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Quantum Mechanics

Non-relativistic Theory

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Fundamental Concepts of Quantum Mechanics

§ 1. The Uncertainty Principle

When classical mechanics and electrodynamics are applied in an attempt to explain atomic phenomena, they give results sharply at variance with experiment. This is already seen most clearly in the contradiction produced by applying ordinary electrodynamics to a model of the atom in which electrons move around the nucleus in classical orbits. Since any accelerated charge radiates, electrons in such orbits would be obliged to emit electromagnetic waves without interruption. Through radiation they would lose energy and eventually fall into the nucleus. Classical electrodynamics would therefore render the atom unstable—a conclusion entirely at odds with reality.

So profound a conflict between theory and experiment shows that a theory applicable to atomic phenomena—phenomena involving particles of very small mass in very small regions of space—requires a fundamental alteration of the basic classical ideas and laws.

A convenient point of departure for identifying these changes is the experimentally observed phenomenon known as electron diffraction¹. When a uniform beam of electrons passes through a crystal, the transmitted beam displays a succession of intensity maxima and minima entirely analogous to the diffraction pattern observed for electromagnetic waves. Under certain conditions, therefore, the behavior of material particles—electrons—shows features characteristic of wave processes.

The depth of this phenomenon's conflict with ordinary ideas of motion is best shown by the following thought experiment, an idealization of electron diffraction by a crystal. Imagine a barrier impermeable to electrons, with two slits cut in it. If we observe an electron beam² passing through one slit while the other is closed, we obtain some intensity distribution on a solid screen behind the slit. Opening the second slit and closing the first gives another distribution. With both slits open at once, ordinary reasoning would lead us to expect simply the superposition of those two distributions: each electron, moving on its own trajectory, passes through one slit without affecting electrons passing through the other. Electron diffraction shows instead that the actual result is a diffraction pattern which, because of interference, is by no means the sum of the patterns from the separate slits. Plainly, this result cannot be reconciled in any way with the idea that electrons move along trajectories.

The mechanics governing atomic phenomena—the so-called *quantum* or *wave mechanics*—must therefore rest on concepts of motion fundamentally different from those of classical mechanics. Quantum mechanics has no concept of a particle trajectory.

¹In fact, electron diffraction was discovered after quantum mechanics had been created. Our presentation, however, does not follow the historical sequence in which the theory developed; it is arranged instead to show as clearly as possible how the fundamental principles of quantum mechanics are connected with observed phenomena.

²The beam is assumed to be so dilute that interactions among its particles play no part.

This fact constitutes the content of the so-called *uncertainty principle*, a fundamental principle of quantum mechanics, due to *Heisenberg* (*W. Heisenberg*, 1927)³. Since it rejects the customary ideas of classical mechanics, the uncertainty principle may be said to have negative content. By itself it is naturally quite insufficient as the basis for constructing a new mechanics of particles. Such a theory must of course rest on certain positive propositions, which will be considered below (§ 2). Before those propositions can be formulated, however, the character of the problems posed to quantum mechanics must first be made clear.

For this purpose, let us first consider the special relation between quantum and classical mechanics.

Ordinarily, a more general theory can be formulated as a logically closed system independently of the less general theory that arises as its limiting case. Relativistic mechanics, for example, can be constructed from its own fundamental principles without any reference to Newtonian mechanics. The fundamental propositions of quantum mechanics, on the other hand, cannot in principle be formulated without invoking classical mechanics.

The absence of a definite trajectory for an electron⁴ also deprives the electron, considered by itself, of any other dynamical characteristics⁵. It is therefore clear that no logically closed mechanics could be constructed for a system consisting solely of quantum objects. A quantitative description of electron motion requires the presence of physical objects which obey classical mechanics to sufficient accuracy. When an electron interacts with a “classical object,” the state of that object is, in general, altered. Both the character and the magnitude of this alteration are governed by the electron’s state and can therefore serve as its quantitative characteristic.

In this connection, the “classical object” is usually called an “apparatus,” and its interaction with the electron is spoken of as a “measurement.” It must be emphasized, however, that this does not mean a process of “measurement” involving a physicist as observer. In quantum mechanics, measurement means any interaction between a classical and a quantum object, occurring apart from and independently of any observer. The recognition of the profound role of the measurement concept in quantum mechanics is due to *Bohr* (*N. Bohr*).

We have defined an apparatus as a physical object which obeys classical mechanics to sufficient accuracy; a body of sufficiently great mass is one example. It should not be supposed, however, that an apparatus must be macroscopic. Under suitable conditions an object known to be microscopic can also play the part of an apparatus, since the meaning of “with sufficient accuracy” depends on the specific problem under consideration. In a Wilson chamber, for instance, one tracks the electron by its mist trail, whose thickness far exceeds atomic dimensions; at such a level of accuracy in locating the trajectory, the

³It is noteworthy that the complete mathematical apparatus of quantum mechanics had already been constructed by *W. Heisenberg* and *E. Schrödinger* in 1925–1926—prior to formulation of the uncertainty principle, which lays bare the physical content of that apparatus.

⁴In this and the next section, for brevity, we speak of an electron while meaning any quantum object in general: a particle or system of particles which obeys quantum but not classical mechanics.

⁵Here we mean quantities characterizing the electron’s motion, not those characterizing the electron as a particle (charge and mass), which are parameters.

electron is a fully classical object.

Quantum mechanics thus occupies a very distinctive place among physical theories: it contains classical mechanics as its limiting case and, at the same time, requires that limiting case for its own foundation.

We can now formulate the problem addressed by quantum mechanics. A typical problem is to predict the outcome of a repeated measurement from the known outcomes of earlier measurements. We shall also see below that, compared with classical mechanics, quantum mechanics generally restricts the set of values that various physical quantities (energy, for example) may assume, that is, the values that may be found by measuring the quantity in question. The formalism of quantum mechanics must make it possible to determine these allowed values.

In quantum mechanics the measurement process has an essential peculiarity: it always acts upon the electron being measured, and for a given measurement accuracy this action cannot in principle be made arbitrarily weak. The more accurate the measurement, the stronger its action; only in measurements of very low accuracy can the action on the measured object be weak. This property of measurements is logically connected with the fact that the dynamical characteristics of an electron arise only through the measurement itself. It is clear that, if the action of the measurement process on the object could be made arbitrarily weak, the quantity being measured would have a definite value by itself, irrespective of the measurement.

Of the various types of measurement, the principal one is the measurement of an electron's coordinates. Within the domain of applicability of quantum mechanics, the electron's coordinates can always be measured⁶ to any accuracy.

Suppose that successive measurements of the coordinates of an electron are made at specified time intervals Δt . As a rule, these results will fail to follow any smooth curve. Quite the opposite: the greater the measurement precision, the more abrupt and erratic the succession of results becomes, consistent with the absence of a path concept for the electron. A more or less smooth path is obtained only when the electron's coordinates are measured with a low degree of accuracy, for example, by the condensation of droplets of vapour in a Wilson chamber.

If instead, while keeping the measurement accuracy unchanged, the intervals Δt between measurements are reduced, adjacent measurements will, of course, give nearby coordinate values. Yet although the results of a series of successive measurements will lie in a small region of space, within that region they will be distributed quite irregularly and will by no means fit any smooth curve. In particular, as Δt tends to zero, the results of nearby measurements do not tend at all to lie on one straight line.

This last observation makes plain that a classical particle velocity does not exist in quantum mechanics: the ratio of the coordinate change between two time instants to the separation Δt tends to no well-defined limit. As we shall see later, however, quantum mechanics does admit a meaningful definition of a particle's velocity at a given instant, one that reduces to the classical velocity upon passing to classical mechanics.

In classical mechanics a particle possesses, at each instant, definite coordinates and

⁶We emphasize once again that by a "measurement made" we mean the interaction of the electron with a classical "apparatus"; this in no way presupposes the presence of an outside observer.

velocity; in quantum mechanics the situation is entirely different. If a measurement has given an electron definite coordinates, it then has no definite velocity at all. Conversely, an electron having a definite velocity cannot have a definite position in space. Indeed, the simultaneous existence of coordinates and velocity at any instant would imply the presence of a definite velocity, which the electron does not have. Thus, in quantum mechanics the electron's coordinates and velocity are quantities that cannot be measured exactly at the same time, that is, cannot simultaneously have definite values. One may say that the electron's coordinates and velocity are quantities which do not exist simultaneously. A quantitative relation governing the possibility of measuring coordinates and velocity inexactly at the same instant will be derived below. In classical mechanics, the state of a physical system is completely described by specifying all its coordinates and velocities at a given instant; from these initial data the equations of motion fully determine the system's behaviour at all future instants. In quantum mechanics such a description is impossible in principle, since the coordinates and their corresponding velocities do not exist simultaneously. Hence the state of a quantum system is described by fewer quantities than in classical mechanics, that is, its description is less detailed than a classical one.

From this follows a very important consequence regarding the character of predictions in quantum mechanics. Whereas a classical description is sufficient to predict with complete precision the future motion of a mechanical system, the less detailed quantum-mechanical description manifestly cannot be adequate for this purpose. Consequently, even when an electron is in a state described as completely as possible, its behaviour at subsequent instants is in principle not unique. Quantum mechanics therefore cannot make strictly definite predictions about the electron's future behaviour. For a specified initial state of an electron, a subsequent measurement may give different results. The task of quantum mechanics is only to determine the probability that one or another result will be obtained in that measurement. In some cases, of course, the probability of a particular measurement result may equal unity, that is, become a certainty, so that the result of the measurement is unique.

Every measurement process in quantum mechanics falls into one of two categories. The first, which includes the majority of measurements, comprises those that yield no determinate result with certainty for any state of the system. The second comprises measurements such that for every possible result there exists a state in which that result is obtained with certainty. It is precisely these latter measurements, which may be called *predictable*, that have the principal role in quantum mechanics. The quantitative characteristics of a state determined by such measurements are what quantum mechanics calls physical quantities. If in some state a measurement gives a unique result with certainty, we shall say that the corresponding physical quantity has a definite value in that state. Below we shall everywhere understand the expression "physical quantity" in precisely the sense stated here.

We shall repeatedly find below that by no means every collection of physical quantities in quantum mechanics can be measured simultaneously, that is, can have definite values simultaneously (we have already discussed one example: the velocity and coordinates of an electron).

Sets of physical quantities with the following property have an important role in

quantum mechanics: the quantities can be measured simultaneously and, when they simultaneously have definite values, no other physical quantity (unless it is a function of them) can have a definite value in that state. We shall refer to such sets of physical quantities as *complete sets*.

Every description of an electron's state arises from some measurement. We now state what a complete description of a state means in quantum mechanics. Completely described states arise from the simultaneous measurement of a complete set of physical quantities. From the results of such a measurement one can, in particular, determine the probabilities of the results of any subsequent measurement, independently of everything that happened to the electron before the first measurement.

Below, everywhere except in § 14, by states of a quantum system we shall mean states described in precisely this complete manner.

§2. The superposition principle

The radical change in physical ideas about motion that quantum mechanics requires in comparison with classical mechanics naturally calls for an equally radical change in the theory's mathematical formalism. In this connection the first question that arises concerns the means of describing a state in quantum mechanics.

Let q stand for the totality of coordinates of a quantum system, and dq for the product of their differentials (it is called a volume element of the system's *configuration space*); for one particle, dq coincides with the volume element dV of ordinary space.

The mathematical formalism of quantum mechanics rests on the assertion that a system's state may be described by a certain (generally complex) function of the coordinates $\Psi(q)$, and that the squared modulus of this function determines the probability distribution of the coordinate values: $|\Psi|^2 dq$ is the probability that a measurement made on the system will find coordinate values in the element dq of configuration space. The function Ψ is called the system's *wave function*⁷.

Knowledge of the wave function makes it possible in principle to calculate the probabilities of the various outcomes of any measurement in general (not necessarily a coordinate measurement). Every one of these probabilities is given by an expression bilinear in Ψ and Ψ^* . In full generality, such an expression has the form

$$\iint \Psi(q) \Psi^*(q') \varphi(q, q') dq dq', \quad (2.1)$$

where the function $\varphi(q, q')$ depends on the kind and outcome of the measurement, and the integrations extend over the entire configuration space. The probability $\Psi\Psi^*$ itself for the different coordinate values is also an expression of this type⁸.

As time passes, the system's state, and with it the wave function, generally changes. In this sense the wave function may also be regarded as a function of time. If the wave function is known at some initial instant, then, by the very meaning of a complete description of a state, it is thereby determined in principle at every future instant. The

⁷It was first introduced into quantum mechanics by *Schrödinger* (*E. Schrödinger*, 1926).

⁸It is obtained from (2.1) by taking $\varphi(q, q') = \delta(q - q_0) \delta(q' - q_0)$, where δ denotes the so-called δ -function defined below, in § 5; q_0 denotes the coordinate value whose probability is sought.

actual time dependence of the wave function is specified by equations to be derived below.

By definition, the sum of the probabilities of all possible values of the system's coordinates must equal unity. The integral of $|\Psi|^2$ over the whole configuration space must therefore equal unity:

$$\int |\Psi|^2 dq = 1. \quad (2.2)$$

This equality constitutes what is known as the *normalization condition* on wave functions. When the integral of $|\Psi|^2$ is convergent, by choosing a suitable constant coefficient, the function Ψ can always be normalized, as it is said. We shall see below, however, that the integral of $|\Psi|^2$ may diverge, in which case Ψ cannot be normalized by condition (2.2). Here $|\Psi|^2$ naturally does not yield the absolute coordinate probabilities; the ratio of $|\Psi|^2$ between two distinct points of configuration space, however, does determine the relative probability of the coordinate values.

Since all quantities with direct physical meaning that are calculated using the wave function have the form (2.1), in which Ψ occurs multiplied by Ψ^* , it is clear that a normalized wave function is defined only up to a constant *phase factor* of the form $e^{i\alpha}$, where α is any real number. This ambiguity is fundamental and cannot be removed; it is, however, immaterial, since it has no effect on any physical result.

The positive content of quantum mechanics is founded on a number of assertions about the properties of the wave function, as follows.

Suppose that in the state with wave function $\Psi_1(q)$ a certain measurement leads with certainty to a definite outcome, outcome 1, while in the state $\Psi_2(q)$ it leads to outcome 2. It is then postulated that every linear combination of Ψ_1 and Ψ_2 , that is, every function of the form $c_1\Psi_1 + c_2\Psi_2$ (c_1, c_2 being constants), describes a state in which the same measurement yields either outcome 1 or outcome 2. Moreover, if the time dependence of the states is known, being given in one case by the function $\Psi_1(q, t)$ and in the other by $\Psi_2(q, t)$, then any linear combination of them also gives a possible time dependence of a state.

These assertions constitute the so-called *superposition principle for states*, the fundamental positive principle of quantum mechanics. It follows from this principle, in particular, that every equation satisfied by wave functions must be linear in Ψ .

Consider a system made up of two parts, and suppose its state is specified in such a way that each part is described completely⁹. One can then assert that the coordinate probabilities for q_1 in the first part are independent of those for the coordinates q_2 of the second, and therefore that the probability distribution for the whole system must equal the product of the probability distributions for its parts. It follows that the system's wave function $\Psi_{12}(q_1, q_2)$ may be written as a product of the wave functions $\Psi_1(q_1)$ and $\Psi_2(q_2)$ of its parts:

$$\Psi_{12}(q_1, q_2) = \Psi_1(q_1)\Psi_2(q_2). \quad (2.3)$$

If the two parts do not interact with each other, this relation between the wave functions

⁹This, of course, also gives a complete description of the state of the system as a whole. We emphasize, however, that the converse statement is by no means valid: in general, a complete description of the state of the system as a whole does not yet describe the states of its separate parts completely (see also § 14).

of the system and of its parts is preserved at future instants as well:

$$\Psi_{12}(q_1, q_2, t) = \Psi_1(q_1, t)\Psi_2(q_2, t). \quad (2.4)$$

§3. Operators

Consider some physical quantity f that characterizes the state of a quantum system. Strictly speaking, in what follows we ought to speak not of a single quantity but of an entire complete set of them at once. However, none of the reasoning is thereby essentially altered, and for the sake of brevity and simplicity we speak below of only a single physical quantity.

Those values which a given physical quantity may assume are termed its *eigenvalues* in quantum mechanics, and their aggregate is referred to as the *eigenvalue spectrum* of that quantity. In classical mechanics, quantities range, in general, over a continuous series of values. In quantum mechanics too there exist physical quantities (for example, coordinates) whose eigenvalues fill a continuous range; in such cases one speaks of a *continuous spectrum* of eigenvalues. Alongside these quantities, however, quantum mechanics also possesses quantities whose eigenvalues constitute a discrete set; one then speaks of a *discrete spectrum*.

For simplicity we shall first assume that the quantity f under consideration has a discrete spectrum; the case of a continuous spectrum is considered in §5. We denote the eigenvalues of f by f_n , where the index n runs through the values $0, 1, 2, 3, \dots$. We denote the wave function of the system in the state in which f has the value f_n by Ψ_n . The wave functions Ψ_n bear the name *eigenfunctions* of the physical quantity f . Every such function is taken to be normalized:

$$\int |\Psi_n|^2 dq = 1. \quad (3.1)$$

If the system is in some arbitrary state with wave function Ψ , then a measurement of f performed on it will yield as a result one of the eigenvalues f_n . By the superposition principle, the wave function Ψ must be a linear combination of those eigenfunctions Ψ_n that correspond to the values f_n which can be found with non-zero probability in a measurement performed on the system in the state under consideration. In the general case of an arbitrary state, Ψ may accordingly be written as the series

$$\Psi = \sum_n a_n \Psi_n, \quad (3.2)$$

where the summation is over all n , and a_n are certain constant coefficients.

Thus we arrive at the conclusion that every wave function can be, as one says, expanded in the eigenfunctions of any physical quantity. A system of functions in terms of which such an expansion can be carried out is spoken of as a *complete system of functions*.

The expansion (3.2) makes it possible to determine the probabilities of finding (by means of measurements) one or another value f_n of f for a system in the state with

wave function Ψ . Indeed, according to what was said in the preceding section, these probabilities must be determined by certain expressions bilinear in Ψ and Ψ^* and must therefore be bilinear in a_n and a_n^* . Moreover, positivity of these expressions is obviously required. As for the probability associated with a given f_n : it must become unity for a system in the state $\Psi = \Psi_n$ and must vanish whenever expansion (3.2) of Ψ lacks a term containing the corresponding Ψ_n . The only essentially positive quantity satisfying this condition is the squared modulus of the coefficient a_n . Thus we arrive at the result that the squared modulus $|a_n|^2$ of every expansion coefficient in (3.2) gives the probability of the respective value f_n of f in the state with wave function Ψ . The sum of the probabilities of all possible values f_n must equal unity; in other words, the relation

$$\sum_n |a_n|^2 = 1. \quad (3.3)$$

must hold.

If Ψ were not normalized, then the relation (3.3) would also not hold. The sum $\sum_n |a_n|^2$ would then have to be determined by some expression bilinear in Ψ and Ψ^* and reducing to unity for normalized Ψ . The only such expression is the integral $\int \Psi \Psi^* dq$. Thus the equality

$$\sum_n a_n a_n^* = \int \Psi \Psi^* dq. \quad (3.4)$$

must hold.

On the other hand, multiplying the expansion $\Psi^* = \sum_n a_n^* \Psi_n^*$ of the complex conjugate function Ψ^* by Ψ and integrating, we obtain

$$\int \Psi \Psi^* dq = \sum_n a_n^* \int \Psi_n^* \Psi dq.$$

Comparing this with (3.4), we have

$$\sum_n a_n a_n^* = \sum_n a_n^* \int \Psi_n^* \Psi dq,$$

from which we find the following formula determining the coefficients a_n of the expansion of Ψ in the eigenfunctions Ψ_n :

$$a_n = \int \Psi \Psi_n^* dq. \quad (3.5)$$

Substituting (3.2) here, we obtain

$$a_n = \sum_m a_m \int \Psi_m \Psi_n^* dq,$$

whence it is seen that the eigenfunctions must satisfy the conditions

$$\int \Psi_m \Psi_n^* dq = \delta_{nm}, \quad (3.6)$$

where $\delta_{nm} = 1$ for $n = m$ and $\delta_{nm} = 0$ for $n \neq m$. The fact that the integrals of the products $\Psi_m \Psi_n^*$ with $m \neq n$ vanish is spoken of as the mutual *orthogonality* of the functions Ψ_n . The collection of eigenfunctions Ψ_n thus constitutes a complete system of normalized and mutually orthogonal (or, as one says for short, *orthonormal*) functions.

We introduce the concept of the *mean value* $\bar{f}$ of a quantity f in a given state. Following the standard definition, $\bar{f}$ is the sum of every eigenvalue f_n of the quantity in question, weighted by the respective probability $|a_n|^2$:

$$\bar{f} = \sum_n f_n |a_n|^2. \quad (3.7)$$

We write $\bar{f}$ in the form of an expression that involves not the expansion coefficients of Ψ but the function itself. Since (3.7) contains the products $a_n a_n^*$, it is clear that the desired expression must be bilinear in Ψ and Ψ^* . We introduce a certain mathematical *operator*, which we denote by $\hat{f}$ ¹⁰, and define as follows. Let $(\hat{f}\Psi)$ denote the result of the action of the operator $\hat{f}$ on the function Ψ . We define $\hat{f}$ by requiring that the integral of $(\hat{f}\Psi)$ multiplied by the conjugate Ψ^* yield $\bar{f}$:

$$\bar{f} = \int \Psi^* (\hat{f}\Psi) dq. \quad (3.8)$$

It is easy to see that in the general case the operator $\hat{f}$ is some linear integral operator.¹¹ Indeed, using the expression (3.5) for a_n , the definition (3.7) of the mean value can be cast as

$$\bar{f} = \sum_n f_n a_n a_n^* = \int \Psi^* \left(\sum_n a_n f_n \Psi_n \right) dq.$$

Comparing with (3.8), we see that the result of the action of the operator $\hat{f}$ on the function Ψ has the form

$$(\hat{f}\Psi) = \sum_n a_n f_n \Psi_n. \quad (3.9)$$

Substituting here the expression (3.5) for a_n , we find that $\hat{f}$ is an integral operator of the form

$$(\hat{f}\Psi) = \int K(q, q') \Psi(q') dq', \quad (3.10)$$

where the function $K(q, q')$ (the so-called *kernel* of the operator) is

$$K(q, q') = \sum_n f_n \Psi_n^*(q') \Psi_n(q). \quad (3.11)$$

Thus, to each physical quantity in quantum mechanics there is assigned a definite linear operator.

¹⁰We shall everywhere denote operators by letters with a hat.

¹¹A linear operator is one possessing the properties: $\hat{f}(\Psi_1 + \Psi_2) = \hat{f}\Psi_1 + \hat{f}\Psi_2$, $\hat{f}(a\Psi) = a\hat{f}\Psi$, where Ψ_1, Ψ_2 are arbitrary functions and a is an arbitrary constant.

From (3.9) it is seen that if Ψ is one of the eigenfunctions Ψ_n (so that all a_n except one are zero), then upon the action of the operator $\hat{f}$ on it, this function is simply multiplied by the eigenvalue f_n ¹²:

$$\hat{f}\Psi_n = f_n\Psi_n. \quad (3.12)$$

The eigenfunctions of a physical quantity f are therefore solutions of the equation

$$\hat{f}\Psi = f\Psi,$$

in this equation f denotes a constant; the eigenvalues are those particular values of f that yield solutions obeying the necessary conditions. As will be seen later, the explicit form of operators for the various physical quantities can be established on direct physical grounds, whereupon this operator property enables one to obtain eigenfunctions and eigenvalues by solving the equations $\hat{f}\Psi = f\Psi$.

For a real physical quantity, both the eigenvalues themselves and their mean values in any state are real. This fact places a definite constraint on the properties of the associated operators. Setting the expression (3.8) equal to its own complex conjugate gives

$$\int \Psi^*(\hat{f}\Psi) dq = \int \Psi(\hat{f}^*\Psi^*) dq, \quad (3.13)$$

where $\hat{f}^*$ denotes the operator complex conjugate to $\hat{f}$ ¹³. For an arbitrary linear operator such a relation does not, in general, hold, so that it represents a certain restriction imposed on the possible form of the operators $\hat{f}$. For an arbitrary operator $\hat{f}$ one can define, as one says, the *transposed* operator $\tilde{f}$, defined so that

$$\int \Phi(\hat{f}\Psi) dq = \int \Psi(\tilde{f}\Phi) dq, \quad (3.14)$$

where Ψ, Φ are two different functions. Choosing as Φ the conjugate function Ψ^* , comparison with (3.13) reveals that

$$\tilde{f} = \hat{f}^*. \quad (3.15)$$

An operator that satisfies this condition is called *Hermitian*¹⁴. Operators representing real physical quantities within the mathematical apparatus of quantum mechanics must therefore be Hermitian.

One may also formally consider complex physical quantities, i.e. quantities possessing complex eigenvalues. Suppose f is one such quantity; one may then form the complex-conjugate quantity f^* , whose eigenvalues are the complex conjugates of the eigenvalues of f . The operator corresponding to the quantity f^* will be denoted by $\hat{f}^+$. It is called the *adjoint* operator to $\hat{f}$ and must, in general, be distinguished from the complex-conjugate

¹²Below, wherever it cannot lead to misunderstanding, we shall omit the parentheses in the expression $(\hat{f}\Psi)$, the operator being understood to act on the expression written after it.

¹³By definition, if for the operator $\hat{f}$ we have $\hat{f}\psi = \varphi$, then the complex-conjugate operator $\hat{f}^*$ is the operator for which $\hat{f}^*\psi^* = \varphi^*$.

¹⁴For a linear integral operator of the form (3.10), Hermiticity requires that the kernel of the operator satisfy $K(q, q') = K^*(q', q)$.

operator $\hat{f}^*$. Indeed, by definition of the operator $\hat{f}^+$, the mean value of f^* in some state Ψ is

$$\overline{f^*} = \int \Psi^* \hat{f}^+ \Psi dq.$$

On the other hand, we have

$$(\overline{f})^* = \left[\int \Psi^* \hat{f} \Psi dq \right]^* = \int \Psi \hat{f}^* \Psi^* dq = \int \Psi^* \tilde{f}^* \Psi dq.$$

Equating the two expressions, we find that

$$\hat{f}^+ = \tilde{f}^*, \quad (3.16)$$

from which it is clear that $\hat{f}^+$ does not, in general, coincide with $\hat{f}^*$. The condition (3.15) can now be written in the form

$$\hat{f} = \hat{f}^+, \quad (3.17)$$

i.e. the operator of a real physical quantity coincides with its adjoint (Hermitian operators are also called *self-adjoint*).

Let us show how one can directly prove the mutual orthogonality of the eigenfunctions of a Hermitian operator corresponding to different eigenvalues. Let f_n, f_m be two different eigenvalues of the real quantity f , and Ψ_n, Ψ_m the corresponding eigenfunctions:

$$\hat{f}\Psi_n = f_n\Psi_n, \quad \hat{f}\Psi_m = f_m\Psi_m.$$

Multiplying both sides of the first of these equalities by Ψ_m^* , and the complex conjugate of the second equality by Ψ_n , and subtracting these products termwise from each other, we obtain

$$\Psi_m^* \hat{f}\Psi_n - \Psi_n \hat{f}^* \Psi_m^* = (f_n - f_m) \Psi_n \Psi_m^*.$$

Integrate both sides of this equality over dq . Since $\hat{f}^* = \tilde{f}$, by virtue of (3.14) the integral of the left-hand side of the equality vanishes, so that we obtain

$$(f_n - f_m) \int \Psi_n \Psi_m^* dq = 0,$$

whence, in view of $f_n \neq f_m$, the required orthogonality property of the functions Ψ_n and Ψ_m follows.

We have been speaking here all along of only a single physical quantity f , whereas we ought to speak, as was noted at the beginning of the section, of a complete system of simultaneously measurable physical quantities. Then we would find that to each of these quantities $f, g, \dots$ there corresponds its own operator $\hat{f}, \hat{g}, \dots$. The eigenfunctions Ψ_n correspond to states in which all the quantities under consideration have definite values, i.e. correspond to definite sets of eigenvalues $f_n, g_n, \dots$, and are joint solutions of the system of equations

$$\hat{f}\Psi = f\Psi, \quad \hat{g}\Psi = g\Psi, \dots$$

§4. Addition and Multiplication of Operators

If $\hat{f}$ and $\hat{g}$ are the operators corresponding to two physical quantities f and g , then $\hat{f} + \hat{g}$ corresponds to their sum $f + g$. In quantum mechanics, however, the meaning of adding different physical quantities differs essentially according as the quantities are or are not measurable simultaneously. If f and g are simultaneously measurable, the operators $\hat{f}$ and $\hat{g}$ have common eigenfunctions. These are also eigenfunctions of $\hat{f} + \hat{g}$, whose eigenvalues are the sums $f_n + g_n$.

If f and g cannot have definite values simultaneously, their sum $f + g$ has a more restricted meaning. All that can be stated is that its mean value in an arbitrary state equals the sum of the separate mean values of the two terms:

$$\overline{f + g} = \bar{f} + \bar{g}. \quad (4.1)$$

The eigenvalues and eigenfunctions of $\hat{f} + \hat{g}$, on the other hand, will in general have no relation to those of f and g . Clearly, when $\hat{f}$ and $\hat{g}$ are Hermitian, so is $\hat{f} + \hat{g}$; its eigenvalues are therefore real and are the eigenvalues of the new quantity $f + g$ defined in this way.

We note the following theorem. Let f_0 and g_0 be the least eigenvalues of f and g , and let $(f + g)_0$ denote the corresponding quantity for $f + g$. Then

$$(f + g)_0 \geq f_0 + g_0. \quad (4.2)$$

The equality sign applies if f and g are simultaneously measurable. The proof follows from the evident fact that a quantity's mean value is always at least its least eigenvalue. In the state in which $(f + g)$ has the value $(f + g)_0$, we have $\overline{(f + g)} = (f + g)_0$; on the other hand, $\overline{(f + g)} = \bar{f} + \bar{g} \geq f_0 + g_0$, which gives (4.2).

Suppose again that f and g are simultaneously measurable. Besides their sum, we may introduce their product as a quantity whose eigenvalues are the products of the eigenvalues of f and g . The corresponding operator is readily seen to act by applying one operator to a function and then the other. Mathematically it is written as the product of $\hat{f}$ and $\hat{g}$. Indeed, if Ψ_n are common eigenfunctions of $\hat{f}$ and $\hat{g}$, then

$$\hat{f}\hat{g}\Psi_n = \hat{f}(\hat{g}\Psi_n) = \hat{f}g_n\Psi_n = g_n\hat{f}\Psi_n = g_nf_n\Psi_n$$

(the symbol $\hat{f}\hat{g}$ denotes the operator which acts on a function Ψ first by applying $\hat{g}$ to Ψ , and then by applying $\hat{f}$ to the function $\hat{g}\Psi$). We could equally well have used $\hat{g}\hat{f}$, whose factors occur in the opposite order. Plainly, the actions of these two operators on the functions Ψ_n give the same result. Since every wave function Ψ is expandable in the functions Ψ_n , it follows that $\hat{f}\hat{g}$ and $\hat{g}\hat{f}$ yield the same result when acting on an arbitrary function. This fact may be written symbolically as $\hat{f}\hat{g} = \hat{g}\hat{f}$, or

$$\hat{f}\hat{g} - \hat{g}\hat{f} = 0. \quad (4.3)$$

Two such operators $\hat{f}$ and $\hat{g}$ are said to *commute* with one another. We have therefore reached an important result: if two quantities f and g can have definite values simultaneously, their operators commute.

The converse theorem can also be proved (see § 11): when $\hat{f}$ and $\hat{g}$ commute, all their eigenfunctions can be chosen in common, which physically means that the corresponding physical quantities are simultaneously measurable. Thus commutation of the operators is a necessary and sufficient condition for simultaneous measurability of physical quantities.

Raising an operator to a power is a special case of multiplying operators. It follows from what has been said that the eigenvalues of $\hat{f}^p$ (where p is an integer) are the eigenvalues of $\hat{f}$ raised to the same p th power. More generally, any function of an operator, $\varphi(\hat{f})$, may be defined as the operator whose eigenvalues are the same function $\varphi(f)$ of the eigenvalues of $\hat{f}$. When $\varphi(f)$ has a Taylor-series expansion, this expansion reduces the action of $\varphi(\hat{f})$ to the actions of the various powers $\hat{f}^p$.

In particular, $\hat{f}^{-1}$ is called the operator *inverse* to $\hat{f}$. Clearly, successive application of $\hat{f}$ and $\hat{f}^{-1}$ to an arbitrary function leaves it unchanged; that is, $\hat{f}\hat{f}^{-1} = \hat{f}^{-1}\hat{f} = 1$. When f and g are not simultaneously measurable, their product has no such direct meaning. This is already apparent because $\hat{f}\hat{g}$ is then not Hermitian and consequently cannot correspond to a real physical quantity. Indeed, by the definition of the transposed operator,

$$\int \Psi \hat{f} \hat{g} \Phi dq = \int \Psi \hat{f} (\hat{g} \Phi) dq = \int (\hat{g} \Phi) (\tilde{f} \Psi) dq.$$

Here $\tilde{f}$ acts only on Ψ , while $\hat{g}$ acts on Φ ; the integrand is therefore simply the product of the two functions $\hat{g} \Phi$ and $\tilde{f} \Psi$. Applying the definition of the transposed operator once more gives

$$\int \Psi \hat{f} \hat{g} \Phi dq = \int (\tilde{f} \Psi) (\hat{g} \Phi) dq = \int \Phi \tilde{g} \tilde{f} \Psi dq.$$

We have thus obtained an integral in which Ψ and Φ have exchanged places relative to the original one. In other words, $\tilde{g} \tilde{f}$ is the operator transposed to $\hat{f} \hat{g}$, so we may write

$$\tilde{f} \tilde{g} = \tilde{g} \tilde{f}, \quad (4.4)$$

that is, the transpose of the product $\hat{f} \hat{g}$ is the product of the transposed factors written in reverse order. Taking the complex conjugate of both sides of (4.4), we find

$$(\hat{f} \hat{g})^+ = \hat{g}^+ \hat{f}^+. \quad (4.5)$$

If both $\hat{f}$ and $\hat{g}$ are Hermitian, then $(\hat{f} \hat{g})^+ = \hat{g} \hat{f}$. Hence $\hat{f} \hat{g}$ is Hermitian only when the factors $\hat{f}$ and $\hat{g}$ commute.

From the products $\hat{f} \hat{g}$ and $\hat{g} \hat{f}$ of two noncommuting Hermitian operators one can nevertheless form another Hermitian operator, their *symmetrized product*

$$\frac{1}{2}(\hat{f} \hat{g} + \hat{g} \hat{f}). \quad (4.6)$$

It is also readily verified that the difference $\hat{f} \hat{g} - \hat{g} \hat{f}$ is an “anti-Hermitian” operator (that is, its transpose is equal to the negative of its complex conjugate). Multiplication by i makes it Hermitian; hence

$$i(\hat{f} \hat{g} - \hat{g} \hat{f}) \quad (4.7)$$

is likewise a Hermitian operator.

For brevity, later on we shall sometimes use the notation

$$\{\hat{f}, \hat{g}\} = \hat{f}\hat{g} - \hat{g}\hat{f} \quad (4.8)$$

for the so-called *commutator* of the operators. It is easy to verify the relation

$$\{\hat{f}\hat{g}, \hat{h}\} = \{\hat{f}, \hat{h}\}\hat{g} + \hat{f}\{\hat{g}, \hat{h}\}. \quad (4.9)$$

Notice that $\{\hat{f}, \hat{h}\} = 0$ and $\{\hat{g}, \hat{h}\} = 0$ do not in general imply that $\hat{f}$ and $\hat{g}$ commute with each other.

§5. Continuous spectrum

All the relations obtained in §3, §4 for the properties of discrete-spectrum eigenfunctions extend without difficulty to a continuous spectrum of eigenvalues.

Let f be a physical quantity possessing a continuous spectrum. We shall denote its eigenvalues simply by the same letter f without an index; the corresponding eigenfunctions will be written Ψ_f . In the same way that an arbitrary wave function Ψ admits the series expansion (3.2) in eigenfunctions of a discrete-spectrum quantity, it can also be expanded — this time as an integral — over the complete system of eigenfunctions of a continuous-spectrum quantity. Such an expansion takes the form

$$\Psi(q) = \int a_f \Psi_f(q) df, \quad (5.1)$$

where the integration is taken over the whole domain of values that the quantity f can assume.

More complicated than in the discrete-spectrum case is the question of normalizing the continuous-spectrum eigenfunctions. As we shall see below, the requirement that the integral of the squared modulus equal unity cannot be satisfied here. Instead, let us set ourselves the aim of normalizing the functions Ψ_f in such a way that $|a_f|^2 df$ represents the probability for the physical quantity under consideration, in the state described by the wave function Ψ , to have a value in the given interval between f and $f + df$. Since the sum of the probabilities of all possible values of f must be equal to unity, we have

$$\int |a_f|^2 df = 1 \quad (5.2)$$

(analogously to relation (3.3) for the discrete spectrum).

Proceeding exactly as we did in deriving formula (3.5), and using the same arguments, we write, on one hand,

$$\int \Psi \Psi^* dq = \int |a_f|^2 df,$$

and, on other hand,

$$\int \Psi \Psi^* dq = \iint a_f^* \Psi_f^* \Psi df dq.$$

By comparing the two expressions we find the formula that determines the expansion coefficients,

$$a_f = \int \Psi(q) \Psi_f^*(q) dq, \quad (5.3)$$

exactly analogous to (3.5).

To derive the normalization condition we now substitute (5.1) into (5.3):

$$a_f = \int a_{f'} \left(\int \Psi_{f'} \Psi_f^* dq \right) df'.$$

This relation must hold for arbitrary a_f and must therefore be satisfied identically. For this to be so it is first of all necessary that the coefficient of $a_{f'}$ under the integral sign (i.e. the integral $\int \Psi_{f'} \Psi_f^* dq$) vanish for all $f' \neq f$. For $f' = f$ this coefficient must become infinite (otherwise the integral over df' would simply be equal to zero). Thus the integral $\int \Psi_{f'} \Psi_f^* dq$ is a function of the difference $f - f'$ which vanishes for non-zero values of the argument and becomes infinite when the argument is zero. We denote this function by $\delta(f' - f)$:

$$\int \Psi_{f'} \Psi_f^* dq = \delta(f' - f). \quad (5.4)$$

The manner in which the function $\delta(f' - f)$ becomes infinite at $f' - f = 0$ is determined by the requirement that there should be

$$\int \delta(f' - f) a_{f'} df' = a_f.$$

Clearly, for this to hold it is necessary that

$$\int \delta(f' - f) df' = 1.$$

A function defined in this way is called the *delta-function*¹⁵. Let us write out once more the formulas that define it. We have

$$\delta(x) = 0 \quad \text{at } x \neq 0, \quad \delta(0) = \infty, \quad (5.5)$$

and moreover in such a way that

$$\int_{-\infty}^{+\infty} \delta(x) dx = 1. \quad (5.6)$$

As the limits of integration one may write any other pair between which the point $x = 0$ lies. If $f(x)$ is a function continuous at $x = 0$, then

$$\int_{-\infty}^{+\infty} \delta(x) f(x) dx = f(0). \quad (5.7)$$

In more general form this formula can be written as

$$\int \delta(x - a) f(x) dx = f(a), \quad (5.8)$$

where the region of integration includes the point $x = a$, and $f(x)$ is continuous at $x = a$. It is also obvious that the delta-function is even, i.e.

$$\delta(-x) = \delta(x). \quad (5.9)$$

¹⁵The delta-function was introduced into theoretical physics by Dirac (P. A. M. Dirac).

Finally, writing

$$\int_{-\infty}^{\infty} \delta(\alpha x) dx = \int_{-\infty}^{\infty} \delta(y) \frac{dy}{|\alpha|} = \frac{1}{|\alpha|},$$

we arrive at the conclusion that

$$\delta(\alpha x) = \frac{1}{|\alpha|} \delta(x), \quad (5.10)$$

where α is any constant. Formula (5.4) expresses the normalization rule for the eigenfunctions of the continuous spectrum; it replaces condition (3.6) of the discrete spectrum. We see that the functions Ψ_f and $\Psi_{f'}$ with $f \neq f'$ remain orthogonal to each other. The integrals of the squares $|\Psi_f|^2$ of the functions of the continuous spectrum, however, diverge.

The functions $\Psi_f(q)$ satisfy one further relation, similar to (5.4). To derive it we substitute (5.3) into (5.1), which gives

$$\Psi(q) = \int \Psi(q') \left(\int \Psi_f^*(q') \Psi_f(q) df \right) dq',$$

whence we immediately conclude that one must have

$$\int \Psi_f^*(q') \Psi_f(q) df = \delta(q - q'). \quad (5.11)$$

An analogous relation can, of course, be derived for the discrete spectrum, where it takes the form

$$\sum_n \Psi_n^*(q') \Psi_n(q) = \delta(q' - q). \quad (5.12)$$

On comparing the pair of formulas (5.1) and (5.4) with the pair (5.3) and (5.11), we see that, on the one hand, the functions $\Psi_f(q)$ effect the expansion of the function $\Psi(q)$ with the expansion coefficients a_f , while, on the other hand, formula (5.3) can be regarded as a completely analogous expansion of the function $a_f \equiv a(f)$ in the functions $\Psi_f^*(q)$, with $\Psi(q)$ playing the role of the expansion coefficients. The function $a(f)$, like $\Psi(q)$, determines the state of the system completely; it is spoken of as the wave function in the f -representation (and the function $\Psi(q)$ as the wave function in the q -representation). Just as $|\Psi(q)|^2$ determines the probability for the system to have coordinates in a given interval dq , so $|a(f)|^2$ determines the probability of values of the quantity f in a given interval df . The functions $\Psi_f(q)$, for their part, are, on the one hand, the eigenfunctions of the quantity f in the q -representation, while their complex conjugates $\Psi_f^*(q)$ serve as eigenfunctions of the coordinate q in the f -representation.

Suppose $\varphi(f)$ is a function of the quantity f , with φ and f standing in one-to-one correspondence. Every function $\Psi_f(q)$ may then equally be viewed as an eigenfunction of the quantity φ . In doing so, however, the normalization of these functions must be changed. Indeed, the eigenfunctions $\Psi_\varphi(q)$ of the quantity φ must be normalized by the condition

$$\int \Psi_{\varphi(f')} \Psi_{\varphi(f)}^* dq = \delta[\varphi(f') - \varphi(f)],$$

whereas the functions Ψ_f are normalized by condition (5.4). The argument of the delta-function vanishes at $f' = f$. For f' close to f we have

$$\varphi(f') - \varphi(f) = \frac{d\varphi(f)}{df}(f' - f).$$

Making use of expression (5.10), one can write¹⁶

$$\delta[\varphi(f') - \varphi(f)] = \frac{1}{|d\varphi(f)/df|} \delta(f' - f). \quad (5.13)$$

Comparison of (5.13) with (5.4) now shows that the functions Ψ_φ and Ψ_f are related to each other by the relation

$$\Psi_{\varphi(f)} = \frac{1}{\sqrt{|d\varphi(f)/df|}} \Psi_f. \quad (5.14)$$

There exist physical quantities which, over some domain of their values, possess a discrete spectrum, and over another possess a continuous one. For the eigenfunctions of such a quantity, of course, all the same relations that were derived in this and the preceding sections continue to hold. It need only be noted that the complete system of functions is formed by the collection of the eigenfunctions of both spectra together. Therefore the expansion of an arbitrary wave function in the eigenfunctions of such a quantity has the form

$$\Psi(q) = \sum_n a_n \Psi_n(q) + \int a_f \Psi_f(q) df, \quad (5.15)$$

where the summation runs over the discrete spectrum, and the integration over the entire continuous spectrum.

An example of a quantity possessing a continuous spectrum is the coordinate q itself. One readily sees that the associated operator reduces to multiplication by q . Indeed, since the square $|\Psi(q)|^2$ governs the probability of different coordinate values, the mean value of the coordinate is

$$\bar{q} = \int q |\Psi|^2 dq = \int \Psi^* q \Psi dq.$$

Comparing this expression with the definition of operators according to (3.8), we see that¹⁷

$$\hat{q} = q. \quad (5.16)$$

The eigenfunctions of this operator must be determined, according to the general rule, by the equation $q\Psi_{q_0} = q_0\Psi_{q_0}$, where q_0 is used temporarily to denote concrete values of

¹⁶In general, if $\varphi(x)$ is a single-valued function (its inverse, however, may be many-valued), then the formula

$$\delta[\varphi(x)] = \sum_i \frac{1}{|\varphi'(\alpha_i)|} \delta(x - \alpha_i), \quad (5.13a)$$

holds, where α_i are the roots of the equation $\varphi(x) = 0$.

¹⁷In what follows we shall agree, for simplicity of notation, to write operators that reduce to multiplication by some quantity everywhere simply as that quantity itself.

the coordinate, in distinction from the variable q . Since this equality can be satisfied either when $\Psi_{q_0} = 0$ or when $q = q_0$, it is clear that the normalized eigenfunctions are¹⁸

$$\Psi_{q_0} = \delta(q - q_0). \quad (5.17)$$

§6. The Limiting Transition

Classical mechanics is contained in quantum mechanics as a limiting case. We must determine how this limiting transition takes place.

In quantum mechanics a wave function describes the electron, governing the various possible values of its coordinate; thus far we know only that this function solves a certain linear partial differential equation. Classical mechanics, by contrast, treats the electron as a material particle moving along a trajectory fixed completely by the equations of motion. A relation analogous in a certain sense to that between quantum and classical mechanics occurs in electrodynamics between wave optics and geometrical optics. Wave optics describes electromagnetic waves by electric- and magnetic-field vectors satisfying a particular system of linear differential equations (Maxwell's equations). Geometrical optics, however, describes the propagation of light along definite paths—rays. This analogy leads one to conclude that the quantum-to-classical limiting transition is analogous to that from wave optics to geometrical optics.

We now recall how this latter transition is carried out mathematically (see II, § 53). Let u be any component of the field in an electromagnetic wave. It may be written $u = ae^{i\varphi}$, with real amplitude a and phase φ (the phase is called the eikonal in geometrical optics). The geometrical-optics limit corresponds to short wavelengths; mathematically, this means that φ changes greatly over short distances. In particular, its absolute magnitude may be regarded as large.

Correspondingly, assume that the classical limiting case in quantum mechanics is represented by wave functions of the form $\Psi = ae^{i\varphi}$, where a varies slowly and φ takes large values. In mechanics, as is known, a particle trajectory can be found from a variational principle according to which the so-called action S of the mechanical system must be a minimum (the principle of least action). In geometrical optics the ray path is instead fixed by Fermat's principle: the "optical path length" of the ray, that is, the difference between its phases at the end and at the beginning of the path, must be a minimum.

This analogy allows us to conclude that, in the classical limiting case, the phase φ of the wave function must be proportional to the mechanical action S of the physical system under consideration: $S = \text{const} \cdot \varphi$. The proportionality factor is called *Planck's constant* and is denoted by $\hbar$ ¹⁹. Since φ is dimensionless, it has the dimensions of action, and its value is

$$\hbar = 1,055 \cdot 10^{-27} \text{ erg} \cdot \text{s}.$$

¹⁸The expansion coefficients of an arbitrary function Ψ in these eigenfunctions are $a_{q_0} = \int \Psi(q) \delta(q - q_0) dq = \Psi(q_0)$. The probability of values of the coordinate in a given interval dq_0 is $|a_{q_0}|^2 dq_0 = |\Psi(q_0)|^2 dq_0$, as it should be.

¹⁹It was introduced into physics by *Planck* (*M. Planck*, 1900). The constant $\hbar$ used throughout this book is, strictly speaking, Planck's constant h divided by 2π (Dirac's notation).

Thus the wave function of an “almost classical” (or, as it is called, *quasiclassical*) physical system has the form

$$\Psi = ae^{iS/\hbar}. \quad (6.1)$$

Planck’s constant has a fundamental part in every quantum phenomenon. Its relative magnitude (in comparison with other quantities having the same dimensions) determines the “degree of quantumness” of a given physical system. Passage from quantum to classical mechanics corresponds to a large phase and can formally be described by taking the limit $\hbar \rightarrow 0$ (just as the passage from wave to geometrical optics corresponds to the zero-wavelength limit, $\lambda \rightarrow 0$).

We have established the limiting form of the wave function, but must still ask how it is related to classical motion along a trajectory. In general, motion described by a wave function does not at all turn into motion along one fixed trajectory. Its connection with classical motion is this: if the wave function, and hence the probability distribution of the coordinates, is specified at an initial instant, the distribution thereafter “moves” as required by the laws of classical mechanics (see the end of § 17 for details).

To obtain motion along a definite trajectory one must begin with a wave function of a special kind, appreciably different from zero only within a very small region of space (a so-called *wave packet*); the dimensions of this region may be made to tend to zero together with $\hbar$. One can then state that, in the quasiclassical case, the wave packet moves through space along the particle’s classical trajectory.

Finally, in the limit, quantum-mechanical operators must reduce simply to multiplication by their corresponding physical quantities.

§ 7. The Wave Function and Measurements

Let us return to the measurement process, whose properties were discussed qualitatively in § 1, and show how those properties are connected with the mathematical apparatus of quantum mechanics.

Consider a system made up of two parts: a classical apparatus and an electron (treated as a quantum object). Measurement consists in bringing these two parts into interaction. The apparatus consequently passes from its initial state into some other state, and from this change we draw conclusions about the state of the electron. States of the apparatus are distinguished by the values of a physical quantity characterizing it (or quantities)—the apparatus “reading.” Denote this quantity provisionally by g , and its eigenvalues by g_n . Because the apparatus is classical, these values generally range over a continuum; solely to simplify the notation in the formulas below, however, we shall suppose that the spectrum is discrete. The states of the apparatus are described by quasiclassical wave functions, which we denote by $\Phi_n(\xi)$; the index n corresponds to the reading g_n , and ξ stands collectively for its coordinates. That the apparatus is classical manifests itself in the circumstance that at each instant one can assert with certainty that it occupies one of the known states Φ_n , with some definite value of g . Such a statement would of course be false for a quantum system.

Let $\Phi_0(\xi)$ be the wave function of the apparatus in its initial state (before measurement), and let $\Psi(q)$ be an arbitrary normalized initial wave function of the electron (q denotes its coordinates). These functions describe apparatus and electron independently,

so the initial wave function of the complete system is the product

$$\Psi(q)\Phi_0(\xi). \quad (7.1)$$

The apparatus and electron then interact. In principle, the equations of quantum mechanics allow the time variation of the system's wave function to be followed. After the measurement it will plainly no longer be a product of functions of ξ and q . Expanding it in the eigenfunctions Φ_n of the apparatus (which form a complete system of functions), we obtain a sum of the form

$$\sum_n A_n(q)\Phi_n(\xi), \quad (7.2)$$

where the $A_n(q)$ are certain functions of q .

At this point the "classical nature" of the apparatus enters, together with the double role of classical mechanics as both a limiting case and a basis of quantum mechanics. As already noted, because the apparatus is classical, the quantity g (its "reading") has a definite value at every instant. It follows that after measurement the actual state of apparatus plus electron is described not by the whole sum (7.2), but by only the one term corresponding to the apparatus reading g_n :

$$A_n(q)\Phi_n(\xi). \quad (7.3)$$

Consequently, $A_n(q)$ is proportional to the electron's wave function after measurement. It is not itself that wave function, as is already evident from the fact that $A_n(q)$ is not normalized. It contains both information about the properties of the state produced and the probability, determined by the system's initial state, that the apparatus will give its n th reading.

By the linearity of the equations of quantum mechanics, the relation between $A_n(q)$ and the initial electron wave function $\Psi(q)$ is in general given by a linear integral operator,

$$A_n(q) = \int K_n(q, q')\Psi(q') dq', \quad (7.4)$$

whose kernel $K_n(q, q')$ characterizes the measurement process in question.

We suppose that the measurement under consideration yields a complete description of the electron's state. In other words (see § 1), in the state produced, the probabilities for all quantities must be independent of the electron's preceding state (before measurement). Mathematically, this means that the form of the functions $A_n(q)$ must be fixed by the measurement process itself and must not depend on the initial electron wave function $\Psi(q)$.

The A_n must therefore have the form

$$A_n(q) = a_n\varphi_n(q), \quad (7.5)$$

where the φ_n are certain functions that we shall take to be normalized, while only the constants a_n depend on the initial state $\Psi(q)$. In the integral relation (7.4), this corresponds to a kernel $K_n(q, q')$ which factors into functions of q and q' separately:

$$K_n(q, q') = \varphi_n(q)\Psi_n^*(q'). \quad (7.6)$$

The linear dependence of the constants a_n on $\Psi(q)$ is then expressed by formulas of the form

$$a_n = \int \Psi(q) \Psi_n^*(q) dq, \quad (7.7)$$

where the $\Psi_n(q)$ are certain functions fixed by the measurement process.

The functions $\varphi_n(q)$ are the normalized wave functions of the electron after measurement. We thus see how the theory's mathematical formalism represents the possibility of producing, by a measurement, an electron state described by a definite wave function. When a measurement is made on an electron having a specified wave function $\Psi(q)$, the constants a_n have a simple physical meaning: by the general rules, $|a_n|^2$ is the probability that the measurement gives the n th result. The probabilities of all results sum to unity:

$$\sum_n |a_n|^2 = 1. \quad (7.8)$$

For an arbitrary normalized function $\Psi(q)$, the validity of (7.7) and (7.8) is equivalent (compare § 3) to the statement that $\Psi(q)$ can be expanded in the functions $\Psi_n(q)$. Hence the functions $\Psi_n(q)$ form a complete normalized and mutually orthogonal set.

If the electron's initial wave function coincides with one of the functions $\Psi_n(q)$, the corresponding constant a_n is plainly unity and all the others vanish. Put differently, if the measurement is performed on an electron in the state $\Psi_n(q)$, it will with certainty give a definite (n th) result.

Taken together, these properties of the functions $\Psi_n(q)$ demonstrate that they are eigenfunctions of a certain physical quantity that characterizes the electron (call it f), and the measurement being considered may be called a measurement of that quantity.

It is essential that, in general, the functions $\Psi_n(q)$ do not coincide with the functions $\varphi_n(q)$ (indeed, the latter need not even be mutually orthogonal or form an eigenfunction system of any operator). This fact expresses first of all the non-repeatability of measurement results in quantum mechanics. If the electron was in the state $\Psi_n(q)$, a measurement of f made upon it will reveal the value f_n with certainty. After the measurement, however, the electron is in a different state $\varphi_n(q)$, in which f need no longer have any definite value. Thus, if a second measurement were made immediately after the first, the value found for f would not coincide with the result of the first measurement²⁰. To predict (in the sense of calculating the probability of) the result of a repeated measurement when the first result is known, one takes from the first measurement the wave function $\varphi_n(q)$ of the state it produced, and from the second the wave function $\Psi_m(q)$ of the state whose probability is sought. More explicitly, the equations of quantum mechanics determine a wave function $\varphi_n(q, t)$ which equals $\varphi_n(q)$ at the time of the first measurement. If the second measurement is made at time t , the probability of its m th result is the squared modulus of the integral $\int \varphi_n(q, t) \Psi_m^*(q) dq$.

We see that measurement in quantum mechanics is "two-faced": its roles with respect to past and future do not coincide. Toward the past it "verifies" the probabilities of the

²⁰There is, however, an important exception to the non-repeatability of measurements: coordinate is the one quantity whose measurement is repeatable. Two measurements of an electron's coordinate made at a sufficiently short interval must give nearby values; otherwise the electron would have infinite velocity. The mathematical origin of this exception lies in the commutativity of the coordinate with the electron-apparatus interaction energy operator, which (in the nonrelativistic theory) depends solely on coordinates.

various possible results predicted from the state created by the preceding measurement. Toward the future it creates a new state (see also § 44). A profound irreversibility is therefore inherent in the measurement process itself.

This irreversibility has important consequences of principle. As will be seen later (see the end of § 18), the fundamental equations of quantum mechanics by themselves are symmetric under reversal of the sign of time; in this respect quantum mechanics does not differ from classical mechanics. The irreversibility of measurement, however, makes the two directions of time physically inequivalent in quantum phenomena, and thus produces a distinction between future and past.

Energy and Momentum

§8. The Hamiltonian

In quantum mechanics the wave function Ψ fully specifies the state of a physical system. This means that prescribing Ψ at some instant not only describes all the properties of the system at that instant, but also determines its behavior at every later instant—of course, only to the degree of completeness that quantum mechanics permits at all. Mathematically, this means that at each instant the value of the time derivative $\partial\Psi/\partial t$ must be determined by the value of Ψ itself at that same instant, and, by the superposition principle, this dependence must be linear. In the most general form we may write

$$i\hbar \frac{\partial\Psi}{\partial t} = \hat{H}\Psi, \quad (8.1)$$

where $\hat{H}$ is a certain linear operator; the factor $i\hbar$ has been introduced here for a reason that will become clear below.

Since the integral $\int \Psi^*\Psi dq$ is constant in time, we have

$$\frac{d}{dt} \int |\Psi|^2 dq = \int \frac{\partial\Psi^*}{\partial t} \Psi dq + \int \Psi^* \frac{\partial\Psi}{\partial t} dq = 0.$$

Substituting (8.1) here and applying the definition of the transposed operator in the first integral, we obtain (after suppressing the common factor $i/\hbar$)

$$\begin{aligned} \int \Psi \hat{H}^* \Psi^* dq - \int \Psi^* \hat{H} \Psi dq &= \int \Psi^* \tilde{H}^* \Psi dq - \int \Psi^* \hat{H} \Psi dq \\ &= \int \Psi^* (\tilde{H}^* - \hat{H}) \Psi dq = 0. \end{aligned}$$

Because this equality must hold for an arbitrary function Ψ , it follows that identically $\hat{H}^+ = \hat{H}$; the operator $\hat{H}$ is therefore Hermitian. Let us determine the physical quantity to which it corresponds. We use the limiting expression (6.1) for the wave function and write

$$\frac{\partial\Psi}{\partial t} = \frac{i}{\hbar} \frac{\partial S}{\partial t} \Psi$$

(the slowly varying amplitude a need not be differentiated). Comparing this equality with definition (8.1), we see that in the limiting case $\hat{H}$ reduces to multiplication by $-\partial S/\partial t$. Thus this last quantity is the physical quantity into which the Hermitian operator $\hat{H}$ passes.

But $-\partial S/\partial t$ is precisely the Hamilton function H of the mechanical system. Hence $\hat{H}$ is the operator corresponding in quantum mechanics to the Hamilton function. This operator is termed the *Hamiltonian operator*, or simply the *Hamiltonian* of the system.

Once the form of the Hamiltonian is known, equation (8.1) fixes the wave functions of the given physical system. This fundamental equation of quantum mechanics is called the *wave equation*.

§9. Time Differentiation of Operators

In quantum mechanics, the concept of the time derivative of a physical quantity cannot be defined with the sense that it carries in classical mechanics. Indeed, the definition of the derivative in classical mechanics is connected with considering the values of the quantity at two close but different instants. But in quantum mechanics a quantity that has a definite value at some instant has no definite value at all at subsequent instants; this was discussed in more detail in § 1.

For this reason the concept of the time derivative has to be given a different definition in quantum mechanics. A natural course is to define the derivative $\dot{f}$ of a quantity f as that quantity whose mean value equals the time derivative of the mean value $\bar{f}$. Thus, by definition, we have

$$\bar{\dot{f}} = \dot{\bar{f}}. \quad (9.1)$$

Taking this definition as the starting point, it is not hard to derive an expression for the quantum-mechanical operator $\hat{f}$ that corresponds to the quantity f :

$$\begin{aligned} \bar{\dot{f}} = \dot{\bar{f}} &= \frac{d}{dt} \int \Psi^* \hat{f} \Psi dq = \\ &= \int \Psi^* \frac{\partial \hat{f}}{\partial t} \Psi dq + \int \frac{\partial \Psi^*}{\partial t} \hat{f} \Psi dq + \int \Psi^* \hat{f} \frac{\partial \Psi}{\partial t} dq. \end{aligned}$$

In this expression $\partial \hat{f} / \partial t$ stands for the operator that results from differentiating the operator $\hat{f}$ with respect to time, on which the latter may depend parametrically. Substituting for the derivatives $\partial \Psi / \partial t$, $\partial \Psi^* / \partial t$ their expressions according to (8.1), we obtain

$$\bar{\dot{f}} = \int \Psi^* \frac{\partial \hat{f}}{\partial t} \Psi dq + \frac{i}{\hbar} \int (\hat{H}^* \Psi^*) \hat{f} \Psi dq - \frac{i}{\hbar} \int \Psi^* \hat{f} (\hat{H} \Psi) dq.$$

Since the operator $\hat{H}$ is Hermitian,

$$\int (\hat{H}^* \Psi^*) (\hat{f} \Psi) dq = \int \Psi^* \hat{H} \hat{f} \Psi dq;$$

thus, we have

$$\bar{\dot{f}} = \int \Psi^* \left(\frac{\partial \hat{f}}{\partial t} + \frac{i}{\hbar} \hat{H} \hat{f} - \frac{i}{\hbar} \hat{f} \hat{H} \right) \Psi dq.$$

Since, on the other hand, by the definition of mean values we must have $\bar{\dot{f}} = \int \Psi^* \hat{\dot{f}} \Psi dq$, it is seen that the expression in parentheses under the integral is the required operator $\hat{\dot{f}}$ ¹:

$$\hat{\dot{f}} = \frac{\partial \hat{f}}{\partial t} + \frac{i}{\hbar} (\hat{H} \hat{f} - \hat{f} \hat{H}). \quad (9.2)$$

¹In classical mechanics, for the total time derivative of a quantity f that is a function of the generalized

If the operator $\hat{f}$ does not depend explicitly on time, then, up to a factor, $\hat{f}$ reduces to the commutator of the operator $\hat{f}$ with the Hamiltonian.

The physical quantities whose operators have no explicit dependence on time and which, moreover, commute with the Hamiltonian—so that $\dot{\hat{f}} = 0$ —form a very important category. Such quantities are called *conserved*. For these, $\dot{\bar{f}} = \dot{\hat{f}} = 0$, i.e. $\bar{f} = \text{const}$. Put differently, the quantity's mean value stays constant in time. One can assert, moreover, that if the quantity f has a definite value in a given state (i.e. if the wave function is an eigenfunction of the operator $\hat{f}$), then at later instants as well it will have a definite value—and the very same one.

§ 10. Stationary states

The Hamiltonian of a closed system (and also of a system in a constant, but not a variable, external field) cannot contain time explicitly. This follows because all instants of time are equivalent for such a physical system. Since, on the other hand, every operator commutes with itself, we conclude that the Hamilton function is conserved for systems not situated in a variable external field. A conserved Hamilton function is called the energy. The meaning of energy conservation in quantum mechanics is that if the energy has a definite value in a given state, this value remains constant in time.

States possessing definite energy values are called the *stationary states* of the system. They are described by wave functions Ψ_n that are eigenfunctions of the Hamiltonian operator, satisfying $\hat{H}\Psi_n = E_n\Psi_n$, where E_n are the energy eigenvalues. Accordingly, the wave equation (8.1) for the function Ψ_n ,

$$i\hbar \frac{\partial \Psi_n}{\partial t} = \hat{H}\Psi_n = E_n\Psi_n,$$

can be integrated directly with respect to time, giving

$$\Psi_n = \exp\left(-\frac{i}{\hbar}E_nt\right)\psi_n(q), \quad (10.1)$$

coordinates q_i and momenta p_i of the system, we have

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_i \left(\frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial p_i} \dot{p}_i \right).$$

Substituting, according to Hamilton's equations, $\dot{q}_i = \partial H / \partial p_i$, $\dot{p}_i = -\partial H / \partial q_i$, we obtain

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + [H, f], \quad [H, f] \equiv \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} \right);$$

$[H, f]$ is the so-called *Poisson bracket* for the quantities f and H (see I, § 42). Comparing with expression (9.2), we see that in the transition to the classical limit the operator $i(\hat{H}\hat{f} - \hat{f}\hat{H})$ becomes zero to the first approximation, as it should, and in the next approximation (in $\hbar$) becomes the quantity $\hbar[H, f]$.

This result is also valid for any two quantities f and g : in the limit the operator $i(\hat{f}\hat{g} - \hat{g}\hat{f})$ becomes the quantity $\hbar[f, g]$, where $[f, g]$ is the Poisson bracket

$$[f, g] \equiv \sum_i \left(\frac{\partial g}{\partial q_i} \frac{\partial f}{\partial p_i} - \frac{\partial g}{\partial p_i} \frac{\partial f}{\partial q_i} \right).$$

This is a consequence of the fact that we can always, formally, conceive of a system whose Hamiltonian coincides with $\hat{g}$.

where ψ_n is a function of the coordinates alone. This determines the time dependence of the wave functions of stationary states.

We shall use the lower-case letter ψ for stationary-state wave functions without the time factor. These functions and the energy eigenvalues themselves are determined by

$$\hat{H}\psi = E\psi. \quad (10.2)$$

Among all stationary states, the one possessing the lowest energy is termed the *normal*, or *ground*, state of the system.

The expansion of any wave function Ψ over the stationary-state wave functions has the form

$$\Psi = \sum_n a_n \exp\left(-\frac{i}{\hbar} E_n(t)\right) \psi_n(q). \quad (10.3)$$

As usual, the squares $|a_n|^2$ give the probabilities of the various energy values of the system.

The coordinate probability distribution in a stationary state is determined by $|\Psi_n|^2 = |\psi_n|^2$ and is therefore independent of time. The same is true of the mean value $\bar{f} = \int \Psi_n^* \hat{f} \Psi_n, dq = \int \psi_n^* \hat{f} \psi_n, dq$ of any physical quantity f whose operator does not depend explicitly on time.

As stated above, the operator of every conserved quantity commutes with the Hamiltonian. Thus every conserved physical quantity can be measured simultaneously with the energy.

Among the stationary states there may be states corresponding to one and the same energy eigenvalue (or *energy level*) but differing in the values of other physical quantities. A level to which several distinct stationary states correspond is said to be *degenerate*. Physically, degeneracy is possible because the energy does not, in general, constitute by itself a complete set of physical quantities. The energy levels of a system are, in general, degenerate if there are two conserved quantities f and g whose operators do not commute. Let ψ be the wave function of a stationary state in which f , as well as the energy, has a definite value. Then $\hat{g}\psi$ cannot coincide with ψ up to a constant factor; otherwise g too would have a definite value, which is impossible because f and g cannot be measured simultaneously. On the other hand, $\hat{g}\psi$ is a Hamiltonian eigenfunction with the same energy E as ψ :

$$\hat{H}(\hat{g}\psi) = \hat{g}\hat{H}\psi = E(\hat{g}\psi).$$

Thus more than one eigenfunction corresponds to E , and the level is degenerate.

Any linear combination of wave functions belonging to the same degenerate energy level is also an eigenfunction with that energy. The choice of eigenfunctions for a degenerate eigenvalue is therefore not unique. Arbitrarily chosen eigenfunctions of a degenerate level are not in general mutually orthogonal, but suitable linear combinations can always be chosen to form a mutually orthogonal and normalized set².

These statements apply not only to energy eigenfunctions but to the eigenfunctions of any operator. Only functions belonging to different eigenvalues are automatically

²This can be done in infinitely many ways: a linear transformation of n functions involves n^2 independent coefficients, but the normalization and orthogonality conditions number only $n(n+1)/2$, which is fewer than n^2 .

orthogonal; functions belonging to the same degenerate eigenvalue are not in general orthogonal.

If $\hat{H} = \hat{H}_1 + \hat{H}_2$, where one part contains only coordinates q_1 and the other only q_2 , the eigenfunctions of $\hat{H}$ can be written as products of eigenfunctions of $\hat{H}_1$ and $\hat{H}_2$, and the energy eigenvalues are sums of the eigenvalues of these operators. The energy spectrum may be discrete or continuous. A stationary state in the discrete spectrum always corresponds to *finite motion*: neither the system nor any part of it recedes to infinity. Indeed, $\int |\Psi|^2, dq$ is finite over all space for a discrete-spectrum eigenfunction. Thus $|\Psi|^2$ decreases sufficiently rapidly and vanishes at infinity. Equivalently, the probability of infinite coordinate values is zero; the system performs finite motion, or is in a *bound* state.

For continuous-spectrum wave functions, $\int |\Psi|^2, dq$ diverges. $|\Psi|^2$ then determines coordinate probabilities only up to proportionality. The divergence is always associated with $|\Psi|^2$ failing to vanish at infinity, or vanishing too slowly. The integral over the region outside any arbitrarily large but finite closed surface therefore still diverges: the system, or some part of it, is at infinity. A superposition of continuous-spectrum stationary states may have a convergent integral and be localized in a finite region, but this region spreads without bound with time, and ultimately the system recedes to infinity.

Indeed, an arbitrary continuous-spectrum superposition is

$$\Psi = \int a_E \exp\left(-\frac{i}{\hbar}Et\right) \psi_E(q) dE,$$

and

$$|\Psi|^2 = \iint a_E a_{E'}^* \exp\left(\frac{i}{\hbar}(E' - E)t\right) \psi_E(q) \psi_{E'}^*(q) dE dE'.$$

If this expression is averaged over an interval T and then T is allowed to tend to infinity, the oscillating factors average to zero, and with them the entire integral. Thus the time-averaged probability of finding the system at any specified point of configuration space vanishes; this is possible only if the motion occupies all infinite space³.

Thus stationary states of the continuous spectrum correspond to infinite motion of the system.

§ 11. Matrices

Assume for convenience that the system in question possesses a discrete energy spectrum (every relation derived below extends directly to the continuous-spectrum case as well). Let $\Psi = \sum_n a_n \Psi_n$ denote the expansion of an arbitrary wave function over the stationary-state wave functions Ψ_n . If this expansion is substituted into the definition (3.8) of the

³For a superposition of discrete-spectrum functions one would instead have

$$\overline{|\Psi|^2} = \sum_{n,m} a_n a_m^* \exp\left\{\frac{i}{\hbar}(E_m - E_n)t\right\} \psi_n \psi_m^* = \sum_n |a_n \psi_n(q)|^2,$$

so the probability density remains finite under time averaging.

mean value of some quantity f , we obtain

$$\bar{f} = \sum_n \sum_m a_n^* a_m f_{nm}(t), \quad (11.1)$$

where $f_{nm}(t)$ denote the integrals

$$f_{nm}(t) = \int \Psi_n^* \hat{f} \Psi_m dq. \quad (11.2)$$

The set of quantities $f_{nm}(t)$ for all possible n, m is called the *matrix* of the quantity $\hat{f}$, and each $f_{nm}(t)$ is spoken of as a *matrix element*, corresponding to the transition from state m to state n ⁴.

The time dependence of the matrix elements $f_{nm}(t)$ is determined (if the operator $\hat{f}$ does not contain t explicitly) by the time dependence of the functions Ψ_n . Substituting for them the expressions (10.1), we find that

$$f_{nm}(t) = f_{nm} e^{i\omega_{nm}t}, \quad (11.3)$$

where

$$\omega_{nm} = \frac{E_n - E_m}{\hbar} \quad (11.4)$$

is the transition frequency between states n and m , and the quantities

$$f_{nm} = \int \psi_n^* \hat{f} \psi_m dq \quad (11.5)$$

constitute the time-independent matrix of the quantity f , which is what one ordinarily uses⁵.

One obtains the matrix elements of the derivative $\dot{f}$ by differentiating those of the quantity f with respect to time; this is an immediate consequence of the relation

$$\dot{\bar{f}} = \dot{\bar{f}} = \sum_n \sum_m a_n^* a_m \dot{f}_{nm}(t). \quad (11.6)$$

In view of (11.3) we thus have for the matrix elements of $\dot{f}$:

$$\dot{f}_{nm}(t) = i\omega_{nm} f_{nm}(t) \quad (11.7)$$

or (cancelling the time factor $\exp(i\omega_{nm}t)$ on both sides) for the time-independent matrix elements:

$$(\dot{f})_{nm} = i\omega_{nm} f_{nm} = \frac{i}{\hbar} (E_n - E_m) f_{nm}. \quad (11.8)$$

For simplicity of notation in the formulas, we derive below all relations for the time-independent matrix elements; exactly the same relations hold also for matrices that depend on time.

⁴The matrix representation of physical quantities was introduced by *Heisenberg* (W. Heisenberg) in 1925, even before Schrödinger's discovery of the wave equation. "Matrix mechanics" was subsequently developed by *Born, Heisenberg and Jordan* (M. Born, P. Jordan).

⁵In connection with the indeterminacy of the phase factor in the normalized wave functions (see § 2), the matrix elements f_{nm} (and $f_{nm}(t)$) are likewise defined only to within factors of the form $\exp[i(\alpha_m - \alpha_n)]$. Here too this indeterminacy does not affect the physical results.

For the matrix elements of the complex conjugate quantity f^* of f , taking into account the definition of the adjoint operator, we obtain

$$(f^*)_{nm} = \int \psi_n^* \hat{f}^+ \psi_m dq = \int \psi_n^* \tilde{f}^* \psi_m dq = \int \psi_m \hat{f}^* \psi_n^* dq,$$

i.e.

$$(f^*)_{nm} = (f_{mn})^*. \quad (11.9)$$

For real physical quantities, which are usually all that we consider, we therefore have

$$f_{nm} = f_{mn}^*. \quad (11.10)$$

Such matrices, like the operators corresponding to them, are called *Hermitian*.

Matrix elements with $n = m$ are called *diagonal*. These elements do not depend on time at all, and from (11.10) it is clear that they are real. The element f_{nn} is the mean value of f in the state Ψ_n .

The rule of *matrix multiplication* is readily derived. As a preliminary step, observe that the formula

$$\hat{f}\psi_n = \sum_m f_{mn}\psi_m. \quad (11.11)$$

holds. This is nothing other than the expansion of the function $\hat{f}\psi_n$ in the functions ψ_m with coefficients determined according to the general rule (3.5). Bearing this formula in mind, we write

$$\hat{f}\hat{g}\psi_n = \hat{f}(\hat{g}\psi_n) = \sum_k g_{kn}\hat{f}\psi_k = \sum_{k,m} g_{kn}f_{mk}\psi_m.$$

Since, on the other hand, we must have $\hat{f}\hat{g}\psi_n = \sum_m (fg)_{mn}\psi_m$, we obtain

$$(fg)_{mn} = \sum_k f_{mk}g_{kn}. \quad (11.12)$$

This rule matches the standard mathematical prescription for multiplying matrices: one multiplies the rows of the first factor by the columns of the second.

Specifying a matrix is equivalent to specifying the operator itself. In particular, it makes it possible in principle to determine the eigenvalues of a given physical quantity and the eigenfunctions corresponding to them.

Let us consider the values of all quantities at some definite instant of time and expand an arbitrary wave function Ψ in the eigenfunctions of the Hamiltonian, i.e. in the time-independent wave functions ψ_m of the stationary states:

$$\Psi = \sum_m c_m \psi_m, \quad (11.13)$$

where the expansion coefficients are denoted by c_m . Let us substitute this expansion into the equation $\hat{f}\Psi = f\Psi$, which determines the eigenvalues and eigenfunctions of the quantity f . We have

$$\sum_m c_m \hat{f}\psi_m = f \sum_m c_m \psi_m.$$

Multiply this equation on both sides by ψ_n^* and integrate over dq . Each of the integrals $\int \psi_n^* \hat{f} \psi_m dq$ on the left-hand side of the equality is the corresponding matrix element f_{nm} . On the right-hand side, all integrals $\int \psi_n^* \psi_m dq$ with $m \neq n$ vanish by virtue of the orthogonality of the functions ψ_m , and $\int \psi_n^* \psi_n dq = 1$ by virtue of their normalization⁶:

$$\sum_m f_{nm} c_m = f c_n, \quad (11.14)$$

or $\sum_m (f_{nm} - f \delta_{nm}) c_m = 0$, where $\delta_{nm} = 1$ for $n = m$ and $\delta_{nm} = 0$ for $n \neq m$.

Thus we have obtained a system of homogeneous linear algebraic equations (with unknowns c_m). As is well known, such a system has nonzero solutions only if the determinant formed from the coefficients in the equations vanishes, i.e. under the condition

$$|f_{nm} - f \delta_{nm}| = 0. \quad (11.15)$$

The roots of this equation (where f plays the role of the unknown) give the possible values of the quantity f . The set of quantities c_m satisfying equations (11.14) upon setting f equal to one of these values yields the corresponding eigenfunction. Suppose that in the definition (11.5) for the matrix elements of f we adopt, as the functions ψ_n , the eigenfunctions of this very quantity; owing to the equation $\hat{f} \psi_n = f_n \psi_n$ we obtain

$$f_{nm} = \int \psi_n^* \hat{f} \psi_m dq = f_m \int \psi_n^* \psi_m dq.$$

Since the functions ψ_m are orthogonal and normalized, it follows that $f_{nm} = 0$ for $n \neq m$ and $f_{mm} = f_m$. Thus only diagonal matrix elements turn out to be nonzero, each of them being equal to the corresponding eigenvalue of the quantity f ; a matrix in which only these elements are nonzero is said to be reduced to *diagonal form*. In particular, in the ordinary representation—where the stationary-state wave functions serve as the ψ_n —the energy matrix is diagonal (along with the matrices of all other physical quantities that have definite values in the stationary states). More broadly, when the matrix of a quantity f is constructed using the eigenfunctions of an operator g , one speaks of it as the matrix of f in the representation in which g is diagonal. Everywhere where it is not otherwise specified, by the matrix of a physical quantity we shall henceforth understand the matrix in the ordinary representation, in which the energy is diagonal. All that has been said above concerning the time dependence of matrices refers, of course, only to this ordinary representation⁷.

The matrix representation of operators makes it possible to prove the theorem cited in § 4: if two operators commute, they possess a common complete system of eigenfunctions.

⁶In accordance with the general rule (§ 5), the set of coefficients c_n of the expansion (11.13) can be regarded as the wave function in the “energy representation” (the variable here being the index n , which enumerates the energy eigenvalues). The matrix f_{nm} then plays the role of the operator $\hat{f}$ in this representation, whose action on the wave function is given by the expression on the left-hand side of equation (11.14). The formula $\bar{f} = \sum_n \sum_m c_n^* (f_{nm} c_m)$ then corresponds to the general formula for the mean value of a quantity expressed through its operator and the wave function of the state in question.

⁷Keeping in mind that the energy matrix is diagonal, one readily verifies that the equality (11.8) is the operator relation (9.2) written in matrix form.

Let $\hat{f}$ and $\hat{g}$ be two such operators. From $\hat{f}\hat{g} = \hat{g}\hat{f}$ and the rule of matrix multiplication (11.12) it follows that

$$\sum_k f_{mk} g_{kn} = \sum_k g_{mk} f_{kn}.$$

Taking as the system of functions ψ_n , in terms of which the matrix elements are computed, the eigenfunctions of the operator $\hat{f}$, we have $f_{mk} = 0$ for $m \neq k$, so that the written equality reduces to $f_m g_{mn} = g_{mn} f_n$, or $g_{mn}(f_m - f_n) = 0$. If all the eigenvalues f_n of the quantity f are distinct, then for all $m \neq n$ we have $f_m - f_n \neq 0$, so that we must have $g_{mn} = 0$. Thus the matrix g_{mn} likewise turns out to be diagonal, i.e. the functions ψ_n are eigenfunctions also of the physical quantity g . If, however, among the values f_n there are equal ones, then the matrix elements g_{mn} corresponding to each such group of functions ψ_n will, in general, be nonzero. However, linear combinations of the functions ψ_n corresponding to one eigenvalue of the quantity f are likewise its eigenfunctions; one can always choose these combinations in such a way as to make the corresponding nondiagonal matrix elements g_{mn} vanish, and thus in this case too we obtain a system of functions that are eigenfunctions simultaneously for the operators $\hat{f}$ and $\hat{g}$.

Let us note a useful formula

$$\left(\frac{\partial H}{\partial \lambda} \right)_{nn} = \frac{\partial E_n}{\partial \lambda}, \quad (11.16)$$

where λ — a parameter on which the Hamiltonian $\hat{H}$ depends (and with it the energy eigenvalues E_n). Indeed, differentiating the equation $(\hat{H} - E_n)\psi_n = 0$ with respect to λ and then multiplying it from the left by ψ_n^* , we obtain

$$\psi_n^*(\hat{H} - E_n) \frac{\partial \psi_n}{\partial \lambda} = \psi_n^* \left(\frac{\partial E_n}{\partial \lambda} - \frac{\partial \hat{H}}{\partial \lambda} \right) \psi_n.$$

On integrating over dq , the left-hand side of this equality vanishes, since

$$\int \psi_n^*(\hat{H} - E_n) \frac{\partial \psi_n}{\partial \lambda} dq = \int \frac{\partial \psi_n}{\partial \lambda} (\hat{H} - E_n)^* \psi_n^* dq$$

by virtue of the Hermiticity of the operator $\hat{H}$. The right-hand side gives the desired equality.

In the modern literature, a system of notation (introduced by *Dirac*) is widely used, in which matrix elements are written as⁸

$$\langle n | f | m \rangle. \quad (11.17)$$

This symbol is composed of the notation for the quantity f and the symbols $|m\rangle$ and $\langle n|$, denoting respectively the initial and final states. If there is a complete set of state functions $|n_1\rangle, |n_2\rangle, \dots$, then the expansion coefficients of the wave function of the state $|m\rangle$ are denoted by $\langle n_i | m \rangle$:

$$\langle n_i | m \rangle = \int \psi_{n_i}^* \psi_m dq. \quad (11.18)$$

⁸We will use both ways of denoting matrix elements in this book. The notation (11.17) is especially convenient when each index has to be written as a combination of several letters.

§ 12. Transformation of matrices

The matrix elements of one and the same physical quantity may be defined with respect to different sets of wave functions. These may be, for example, wave functions of stationary states described by different sets of physical quantities, or wave functions of stationary states of the same system placed in different external fields. The question therefore arises of transforming matrices from one representation to another.

Let $\psi_n(q)$ and $\psi'_n(q)$ ($n = 1, 2, \dots$) be two complete systems of orthonormal functions. They are related by a linear transformation

$$\psi'_n = \sum_m S_{mn} \psi_m, \quad (12.1)$$

which is simply the expansion of the functions ψ'_n in the complete system ψ_m . This transformation can be written in operator form as

$$\psi'_n = \hat{S} \psi_n. \quad (12.2)$$

The operator $\hat{S}$ must satisfy a certain condition if the functions ψ'_n are to be orthonormal whenever the functions ψ_n are. Indeed, substituting (12.2) into $\int \psi'^*_m \psi'_n dq = \delta_{mn}$ and using the definition (3.14) of the transposed operator, we obtain

$$\int (\hat{S} \psi_m)^* \hat{S} \psi_n dq = \int \psi_m^* \tilde{S}^* \hat{S} \psi_n dq = \delta_{mn}.$$

For this equality to hold for every m, n , we must have $\tilde{S}^* \hat{S} = 1$, or

$$\tilde{S}^* \equiv \hat{S}^+ = \hat{S}^{-1}, \quad (12.3)$$

i.e. the inverse operator coincides with the adjoint. Operators having this property are called *unitary*. By virtue of this property the transformation $\psi_n = \hat{S}^{-1} \psi'_n$, inverse to (12.1), is given by

$$\psi_n = \sum_m S_{nm}^* \psi'_m. \quad (12.4)$$

Writing $\hat{S}^+ \hat{S} = 1$ or $\hat{S} \hat{S}^+ = 1$ in matrix form gives the unitarity conditions

$$\sum_l S_{lm}^* S_{ln} = \delta_{mn} \quad (12.5)$$

or

$$\sum_l S_{ml}^* S_{nl} = \delta_{mn}. \quad (12.6)$$

Now consider a physical quantity f and write its matrix elements in the “new” representation, i.e. with respect to the functions ψ'_n . They are given by

$$\int \psi'^*_m \hat{f} \psi'_n dq = \int (\tilde{S}^* \psi_m^*) (\hat{f} \hat{S} \psi_n) dq = \int \psi_m^* \hat{S}^{-1} \hat{f} \hat{S} \psi_n dq.$$

It follows that the matrix of $\hat{f}$ in the new representation coincides with the matrix of the operator

$$\hat{f}' = \hat{S}^{-1} \hat{f} \hat{S} \quad (12.7)$$

in the old representation⁹. The *trace* of a matrix is defined as the sum of its diagonal elements and is written $\text{Tr } f$ ¹⁰:

$$\text{Tr } f = \sum_n f_{nn}. \quad (12.8)$$

First note that the trace of a product of two matrices does not depend on the order of the factors:

$$\text{Tr}(fg) = \text{Tr}(gf). \quad (12.9)$$

Indeed, by the rule for matrix multiplication,

$$\text{Tr}(fg) = \sum_n \sum_k f_{nk} g_{kn} = \sum_k \sum_n g_{kn} f_{nk} = \text{Tr}(gf).$$

Similarly, the trace of a product of several matrices is unchanged by a cyclic permutation of the factors; thus

$$\text{Tr}(fgh) = \text{Tr}(hfg) = \text{Tr}(ghf). \quad (12.10)$$

The most important property of the trace is its independence of the system of functions relative to which the matrix elements are defined. Indeed,

$$\text{Tr } f' = \text{Tr}(S^{-1} f S) = \text{Tr}(S S^{-1} f) = \text{Tr } f. \quad (12.11)$$

Also, a unitary transformation leaves invariant the sum of the squared moduli of the transformed functions. From (12.6),

$$\sum_i |\psi'_i|^2 = \sum_{k,l,i} S_{ki} S_{li}^* \psi_k \psi_l^* = \sum_{k,l} \delta_{kl} \psi_k \psi_l^* = \sum_k |\psi_k|^2. \quad (12.12)$$

Every unitary operator can be represented as

$$\hat{S} = e^{i\hat{R}}, \quad (12.13)$$

where $\hat{R}$ is Hermitian; indeed, $\hat{R}^+ = \hat{R}$ gives $\hat{S}^+ = e^{-i\hat{R}^+} = e^{-i\hat{R}} = \hat{S}^{-1}$.

We also note the expansion

$$\hat{f}' = \hat{S}^{-1} \hat{f} \hat{S} = \hat{f} + \{\hat{f}, i\hat{R}\} + \frac{1}{2} \{\{\hat{f}, i\hat{R}\}, i\hat{R}\} + \dots, \quad (12.14)$$

which is readily verified directly by expanding the factors $\exp(\pm i\hat{R})$ in powers of $\hat{R}$. This expansion can be useful when $\hat{R}$ is proportional to a small parameter, so that (12.14) becomes an expansion in powers of that parameter.

⁹If $\{\hat{f}, \hat{g}\} = -i\hbar\hat{c}$ is the commutation rule for two operators, then after transformation (12.7) we obtain $\{\hat{f}', \hat{g}'\} = -i\hbar\hat{c}'$, so the rule is unchanged. It was noted in the note on p. 46 that c is the quantum analogue of the classical Poisson bracket $[f, g]$. Poisson brackets in classical mechanics, however, are invariant under canonical variable transformations (see Vol. I, § 45). One may therefore say that unitary transformations in quantum mechanics serve as the counterpart of canonical transformations in classical mechanics.

¹⁰The source's notation comes from the German *Spur*, trace. The notation Tr , from the English *trace*, is also used. Consideration of a matrix trace of course assumes convergence of the sum over n .

§ 13. Heisenberg representation of operators

In the mathematical apparatus of quantum mechanics as here presented, the operators corresponding to the various physical quantities act on functions of the coordinates and as such ordinarily contain no explicit dependence on time. The dependence of the mean values of physical quantities on time arises only through the time dependence of the wave function of the state, in accordance with the formula

$$\bar{f}(t) = \int \Psi^*(q, t) \hat{f} \Psi(q, t) dq. \quad (13.1)$$

The apparatus of quantum mechanics can, however, also be formulated in a somewhat different, equivalent form, in which the dependence on time is transferred from the wave functions to the operators. Although in this book we shall not make use of such a representation of operators—the so-called Heisenberg representation, as distinct from the Schrödinger representation—we shall formulate it here, with later applications in relativistic theory in view.

Let us introduce the unitary operator (cf. (12.13))

$$\hat{S} = \exp \left(-\frac{i}{\hbar} \hat{H} t \right), \quad (13.2)$$

where $\hat{H}$ is the Hamiltonian of the system. By definition its eigenfunctions coincide with the eigenfunctions of the operator $\hat{H}$, i.e. with the wave functions $\psi_n(q)$ of the stationary states, and moreover

$$\hat{S} \psi_n(q) = \exp \left(-\frac{i}{\hbar} E_n t \right) \psi_n(q) = \Psi_n(q, t). \quad (13.3)$$

It follows that expansion (10.3) of any wave function Ψ over the stationary-state wave functions can be recast in operator form as

$$\Psi(q, t) = \hat{S} \Psi(q, 0), \quad (13.4)$$

i.e. the action of the operator $\hat{S}$ carries the wave function of the system at some initial instant of time over into the wave function at an arbitrary instant of time. On introducing, in accordance with (12.7), the time-dependent operator

$$\hat{f}(t) = \hat{S}^{-1} \hat{f} \hat{S}, \quad (13.5)$$

we obtain

$$\bar{f}(t) = \int \Psi^*(q, 0) \hat{f}(t) \Psi(q, 0) dq, \quad (13.6)$$

i.e. we bring the formula for the mean value of the quantity f (which serves as the definition of the operators) into a form in which the dependence on time is wholly transferred to the operator.

It is obvious that the matrix elements of the operator (13.5), with respect to the wave functions of stationary states, coincide with the time-dependent matrix elements $f_{nm}(t)$ defined by formula (11.3).

Finally, on differentiating expression (13.5) with respect to time (assuming here that the operators $\hat{f}$ and $\hat{H}$ themselves contain no t), we obtain the equation

$$\frac{\partial \hat{f}(t)}{\partial t} = \frac{i}{\hbar} [\hat{H}\hat{f}(t) - \hat{f}(t)\hat{H}], \quad (13.7)$$

which is analogous to formula (9.2) but has a somewhat different meaning: expression (9.2) constitutes the definition of the operator $\hat{f}$ corresponding to the physical quantity f , whereas the left-hand side of equation (13.7) contains the time derivative of the operator of the quantity f itself.

§ 14. The density matrix

Describing a system by its wave function represents the fullest description that quantum mechanics allows, in the sense indicated at the end of § 1.

States not admitting such a description arise when the system considered is part of a larger closed system. Suppose the closed system as a whole is in a state described by $\Psi(q, x)$, where x denotes the coordinates of the system considered and q the remaining coordinates of the closed system. In general this function does not factor into functions of x alone and q alone, so that the system has no wave function of its own¹¹.

Let f be a physical quantity pertaining to our system. Its operator acts on the x coordinates only, not on q . Its mean value is

$$\bar{f} = \iint \Psi^*(q, x) \hat{f} \Psi(q, x) dq dx. \quad (14.1)$$

Introduce the function

$$\rho(x, x') = \int \Psi(q, x) \Psi^*(q, x') dq, \quad (14.2)$$

where integration is only over q ; it is called the system's *density matrix*. From (14.2),

$$\rho^*(x, x') = \rho(x', x). \quad (14.3)$$

The “diagonal elements”

$$\rho(x, x) = \int |\Psi(q, x)|^2 dq$$

determine the coordinate probability distribution.

In terms of the density matrix,

$$\bar{f} = \int [\hat{f} \rho(x, x')]_{x'=x} dx. \quad (14.4)$$

Here $\hat{f}$ acts only on the variables x in $\rho(x, x')$; after its action has been evaluated, one sets $x' = x$. Thus the density matrix determines the mean of every quantity characterizing

¹¹For $\Psi(q, x)$ to factor in this way at a given instant, the measurement that produced the state must describe completely and separately the system considered and the remainder of the closed system. For the factorization to persist, these two parts must also not interact (see § 2). Neither condition is assumed here.

the system and hence also the probabilities of its possible values. A state having no wave function can therefore be described by a density matrix. The density matrix contains none of the coordinates q that do not belong to the system, although it naturally depends in substance on the state of the closed system as a whole.

The density-matrix description is the most general form of quantum-mechanical description of systems. The wave-function description is the special case $\rho(x, x') = \Psi(x)\Psi^*(x')$. There is an important distinction. For a state possessing a wave function (a *pure* state), there is always a complete system of measurements whose results can be predicted with certainty (mathematically, Ψ is an eigenfunction of some operator). For states possessing only a density matrix (*mixed* states), no complete system of measurements has uniquely predictable results.

Suppose the system is closed, or became closed at some instant. We derive an equation governing the time dependence of its density matrix, analogous to the wave equation. The desired linear differential equation for $\rho(x, x', t)$ must also hold in the special case

$$\rho(x, x', t) = \Psi(x, t)\Psi^*(x', t).$$

Differentiating and using (8.1),

$$\begin{aligned} i\hbar \frac{\partial \rho}{\partial t} &= i\hbar \Psi^*(x', t) \frac{\partial \Psi(x, t)}{\partial t} + i\hbar \Psi(x, t) \frac{\partial \Psi^*(x', t)}{\partial t} \\ &= \Psi^*(x', t) \hat{H} \Psi(x, t) - \Psi(x, t) \hat{H}'^* \Psi^*(x', t). \end{aligned}$$

Here $\hat{H}$ acts on functions of x , while $\hat{H}'$ is the same operator acting on functions of x' . Bringing the other factors under the respective operator signs gives

$$i\hbar \frac{\partial \rho(x, x', t)}{\partial t} = (\hat{H} - \hat{H}'^*) \rho(x, x', t). \quad (14.5)$$

Let $\Psi_n(x, t)$ be the stationary-state wave functions. Expanding the density matrix in these functions gives the double series

$$\rho(x, x', t) = \sum_m \sum_n a_{mn} \psi_n^*(x') \psi_m(x) \exp \left[\frac{i}{\hbar} (E_n - E_m) t \right]. \quad (14.6)$$

This plays the same role for the density matrix as (10.3) does for the wave function. In place of the single set a_n there is the double set a_{mn} , which, like the density matrix, is Hermitian:

$$a_{mn} = a_{nm}^*. \quad (14.7)$$

Substitution in (14.4) gives

$$\bar{f} = \sum_m \sum_n a_{mn} \int \Psi_n^*(x, t) \hat{f} \Psi_m(x, t) dx,$$

or

$$\bar{f} = \sum_m \sum_n a_{mn} f_{nm}(t) = \sum_m \sum_n a_{mn} f_{nm} \exp \left[\frac{i}{\hbar} (E_n - E_m) t \right]. \quad (14.8)$$

This is analogous to (11.1)¹².

The a_{mn} must satisfy inequalities. Since the diagonal elements $\rho(x, x)$ are probabilities, they must be positive. Hence the quadratic form

$$\sum_n \sum_m a_{mn} \xi_m \xi_n^*$$

must be positive definite for arbitrary complex ξ_n . The standard conditions for quadratic forms therefore apply; in particular,

$$a_{nn} > 0, \quad (14.9)$$

and for every a_{mm}, a_{nn}, a_{mn} ,

$$a_{mm}a_{nn} \geq |a_{mn}|^2. \quad (14.10)$$

The pure case, in which the density matrix is a product of functions, corresponds to

$$a_{mn} = a_m a_n^*. \quad (14.11)$$

A simple criterion distinguishes a pure from a mixed state. In the pure case,

$$(a^2)_{mn} = \sum_k a_{mk} a_{kn} = \sum_k a_m a_k^* a_k a_n^* = a_m a_n^* \sum_k |a_k|^2 = a_m a_n^*,$$

and therefore

$$(a^2)_{mn} = a_{mn}. \quad (14.12)$$

Thus the square of the density matrix coincides with the matrix itself.

§ 15. Momentum

Consider a closed system of particles in no external field. Since all positions of the system as a whole are equivalent, its Hamiltonian is unchanged by an arbitrary parallel displacement. It is enough to require this for every infinitesimal displacement; it then holds for finite displacements as well.

Under an infinitesimal displacement $\delta \mathbf{r}$, every radius vector $\mathbf{r}_a$ acquires the same increment. Thus $\mathbf{r}_a \rightarrow \mathbf{r}_a + \delta \mathbf{r}$, and

$$\psi(\mathbf{r}_1 + \delta \mathbf{r}, \mathbf{r}_2 + \delta \mathbf{r}, \dots) = \left(1 + \delta \mathbf{r} \cdot \sum_a \nabla_a \right) \psi(\mathbf{r}_1, \mathbf{r}_2, \dots).$$

Here a labels the particles and ∇_a differentiates with respect to $\mathbf{r}_a$. The expression in parentheses is the operator of an infinitesimal displacement.

If a transformation does not change the Hamiltonian, transforming $\hat{H}\psi$ gives the same result as first transforming ψ and then applying $\hat{H}$. If $\hat{O}$ produces the transformation, then $\hat{O}(\hat{H}\psi) = \hat{H}(\hat{O}\psi)$, so $\hat{O}\hat{H} - \hat{H}\hat{O} = 0$: $\hat{H}$ commutes with $\hat{O}$.

¹²The quantities a_{mn} constitute the density matrix in the energy representation. Description of states by such a matrix was introduced independently by Landau and Bloch (*F. Bloch*) in 1927.

Here $\hat{O}$ is the infinitesimal-displacement operator. The unit operator commutes with every operator, and the constant $\delta \mathbf{r}$ may be taken outside $\hat{H}$; hence

$$\left(\sum_a \nabla_a \right) \hat{H} - \hat{H} \left(\sum_a \nabla_a \right) = 0. \quad (15.1)$$

An explicitly time-independent operator commuting with the Hamiltonian corresponds to a conserved quantity. The quantity whose conservation follows from the homogeneity of space is momentum (cf. Vol. I, § 7). Thus (15.1) expresses momentum conservation; $\sum_a \nabla_a$ corresponds, up to a constant factor, to the total momentum, and each term to the momentum of one particle.

The factor relating the momentum operator to ∇ follows from the classical limit and equals $-i\hbar$:

$$\hat{\mathbf{p}} = -i\hbar \nabla, \quad (15.2)$$

or

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}, \quad \hat{p}_y = -i\hbar \frac{\partial}{\partial y}, \quad \hat{p}_z = -i\hbar \frac{\partial}{\partial z}.$$

Indeed, using (6.1),

$$\hat{\mathbf{p}}\Psi = -i\hbar \nabla \Psi = \Psi \nabla S.$$

Thus classically the operator acts by multiplication by ∇S , which is the classical momentum (see Vol. I, § 43).

The operator (15.2) is Hermitian, as required. For functions $\psi(x)$ and $\varphi(x)$ vanishing at infinity, integration by parts gives

$$\int \varphi \hat{p}_x \psi dx = -i\hbar \int \varphi \frac{\partial \psi}{\partial x} dx = i\hbar \int \psi \frac{\partial \varphi}{\partial x} dx = \int \psi \hat{p}_x^* \varphi dx,$$

which is the Hermiticity condition.

Since differentiations with respect to different variables commute,

$$\hat{p}_x \hat{p}_y - \hat{p}_y \hat{p}_x = 0, \quad \hat{p}_x \hat{p}_z - \hat{p}_z \hat{p}_x = 0, \quad \hat{p}_y \hat{p}_z - \hat{p}_z \hat{p}_y = 0. \quad (15.3)$$

Thus all three momentum components may have definite values simultaneously.

Their eigenfunctions and eigenvalues satisfy

$$-i\hbar \nabla \psi = \mathbf{p} \psi, \quad (15.4)$$

with solutions

$$\psi = \text{const} \cdot e^{i\mathbf{p} \cdot \mathbf{r} / \hbar}. \quad (15.5)$$

The three components completely determine the wave function and form one possible complete set. Their eigenvalues form a continuous spectrum from $-\infty$ to ∞ .

By (5.4), $\int \psi_{\mathbf{p}'}^* \psi_{\mathbf{p}} dV$ over all space should equal $\delta(\mathbf{p}' - \mathbf{p})$ ¹³. For later applications it is more natural to normalize to the delta function of the momentum difference divided by $2\pi\hbar$:

$$\int \psi_{\mathbf{p}'}^* \psi_{\mathbf{p}} dV = \delta \left(\frac{\mathbf{p}' - \mathbf{p}}{2\pi\hbar} \right),$$

¹³For a vector argument, $\delta(\mathbf{a}) = \delta(a_x) \delta(a_y) \delta(a_z)$.

or equivalently

$$\int \psi_{\mathbf{p}'}^* \psi_{\mathbf{p}} dV = (2\pi\hbar)^3 \delta(\mathbf{p}' - \mathbf{p}). \quad (15.6)$$

Each one-dimensional factor obeys, for example, $\delta[(p'_x - p_x)/(2\pi\hbar)] = 2\pi\hbar\delta(p'_x - p_x)$. Integration uses¹⁴

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\alpha\xi} d\xi = \delta(\alpha). \quad (15.7)$$

It follows that $\text{const} = 1$ in (15.5)¹⁵:

$$\psi_{\mathbf{p}} = e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}. \quad (15.8)$$

Expansion of an arbitrary $\psi(\mathbf{r})$ in momentum eigenfunctions is its Fourier integral:

$$\psi(\mathbf{r}) = \int a(\mathbf{p}) \psi_{\mathbf{p}}(\mathbf{r}) \frac{d^3 p}{(2\pi\hbar)^3} = \int a(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} \frac{d^3 p}{(2\pi\hbar)^3}. \quad (15.9)$$

Here $d^3 p = dp_x dp_y dp_z$, and

$$a(\mathbf{p}) = \int \psi(\mathbf{r}) \psi_{\mathbf{p}}^*(\mathbf{r}) dV = \int \psi(\mathbf{r}) e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} dV. \quad (15.10)$$

The function $a(\mathbf{p})$ is the momentum-representation wave function; $|a(\mathbf{p})|^2 d^3 p / (2\pi\hbar)^3$ is the probability that the momentum lies in $d^3 p$. Just as $\hat{\mathbf{p}}$ represents momentum in coordinate space, introduce $\hat{\mathbf{r}}$ for the coordinates in momentum space, defined by

$$\hat{\mathbf{r}} = \int a^*(\mathbf{p}) \hat{\mathbf{r}} a(\mathbf{p}) \frac{d^3 p}{(2\pi\hbar)^3}. \quad (15.11)$$

But also $\hat{\mathbf{r}} = \int \psi^* \mathbf{r} \psi dV$. Substituting (15.9) and integrating by parts,

$$\mathbf{r} \psi(\mathbf{r}) = \int i\hbar e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} \frac{\partial a(\mathbf{p})}{\partial \mathbf{p}} \frac{d^3 p}{(2\pi\hbar)^3},$$

and hence

$$\hat{\mathbf{r}} = \int i\hbar a^*(\mathbf{p}) \frac{\partial a(\mathbf{p})}{\partial \mathbf{p}} \frac{d^3 p}{(2\pi\hbar)^3}.$$

Comparison with (15.11) gives

$$\hat{\mathbf{r}} = i\hbar \frac{\partial}{\partial \mathbf{p}}. \quad (15.12)$$

In this representation momentum is simply multiplication by $\mathbf{p}$.

¹⁴The formula means that its left-hand side has the defining property (5.8) of the delta function. Substitution in (5.8) gives the Fourier formula

$$f(a) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x) e^{i\xi(x-a)} dx \frac{d\xi}{2\pi}.$$

¹⁵With this normalization $|\psi|^2 = 1$: the function is normalized to “one particle per unit volume.” This agreement is not accidental; cf. the note on p. 221.

Finally, express the operator of translation through an arbitrary finite distance $\mathbf{a}$ in terms of $\hat{\mathbf{p}}$. By definition, $\hat{T}_{\mathbf{a}}\psi(\mathbf{r}) = \psi(\mathbf{r} + \mathbf{a})$. Expanding $\psi(\mathbf{r} + \mathbf{a})$ in a Taylor series gives

$$\psi(\mathbf{r} + \mathbf{a}) = \psi(\mathbf{r}) + \mathbf{a} \cdot \frac{\partial \psi(\mathbf{r})}{\partial \mathbf{r}} + \dots$$

or, on introducing $\hat{\mathbf{p}} = -i\hbar\nabla$,

$$\psi(\mathbf{r} + \mathbf{a}) = \left[1 + \frac{i}{\hbar} \mathbf{a} \cdot \hat{\mathbf{p}} + \frac{1}{2} \left(\frac{i}{\hbar} \mathbf{a} \cdot \hat{\mathbf{p}} \right)^2 + \dots \right] \psi(\mathbf{r}).$$

Thus

$$\hat{T}_{\mathbf{a}} = \exp \left(\frac{i}{\hbar} \mathbf{a} \cdot \hat{\mathbf{p}} \right). \quad (15.13)$$

This is the required *finite-displacement operator*.

§ 16. Uncertainty Relations

Let us derive the commutation rules for the momentum and coordinate operators. Successive differentiation with respect to one of the variables x, y, z and multiplication by another of them gives a result independent of the order of these operations. Hence

$$\hat{p}_x y - y \hat{p}_x = 0, \quad \hat{p}_x z - z \hat{p}_x = 0, \quad (16.1)$$

and similarly for $\hat{p}_y$ and $\hat{p}_z$.

To derive the commutation rule for $\hat{p}_x$ and x , write

$$(\hat{p}_x x - x \hat{p}_x)\psi = -i\hbar \frac{\partial}{\partial x}(x\psi) + i\hbar x \frac{\partial \psi}{\partial x} = -i\hbar\psi.$$

We see that the action of $\hat{p}_x x - x \hat{p}_x$ reduces to multiplication of the function by $-i\hbar$; the same is of course true for the commutation of $\hat{p}_y$ with y and of $\hat{p}_z$ with z . Thus¹⁶

$$\hat{p}_x x - x \hat{p}_x = -i\hbar, \quad \hat{p}_y y - y \hat{p}_y = -i\hbar, \quad \hat{p}_z z - z \hat{p}_z = -i\hbar. \quad (16.2)$$

Relations (16.1) and (16.2) can be written together as

$$\hat{p}_i x_k - x_k \hat{p}_i = -i\hbar \delta_{ik}, \quad i, k = x, y, z. \quad (16.3)$$

Before examining the physical meaning of these relations and their consequences, we give two formulas useful below. If $f(\mathbf{r})$ is a function of the coordinates, then

$$\hat{\mathbf{p}}f(\mathbf{r}) - f(\mathbf{r})\hat{\mathbf{p}} = -i\hbar\nabla f. \quad (16.4)$$

Indeed,

$$(\hat{\mathbf{p}}f - f\hat{\mathbf{p}})\psi = -i\hbar[\nabla(f\psi) - f\nabla\psi] = -i\hbar\psi\nabla f.$$

¹⁶These relations, discovered in matrix form by *Heisenberg* in 1925, served as the starting point in the creation of quantum mechanics.

An analogous relation holds for the commutator with a function of the momentum operator:

$$f(\mathbf{p})\hat{\mathbf{r}} - \hat{\mathbf{r}}f(\mathbf{p}) = -i\hbar \frac{\partial f}{\partial \mathbf{p}}. \quad (16.5)$$

It may be derived in the same way as (16.4) by working in the momentum representation and using (15.12) for the coordinate operators.

Relations (16.1) and (16.2) show that a particle coordinate along one axis can have a definite value simultaneously with the momentum components along the other two axes; a coordinate and the momentum component along that same axis do not exist simultaneously. In particular, a particle cannot occupy a definite point in space while also having a definite momentum $\mathbf{p}$.

Let the particle occupy a finite spatial region whose dimensions along the three axes are of order Δx , Δy , Δz . Let its mean momentum be $\mathbf{p}_0$. Mathematically, this means that the wave function has the form $\psi = u(\mathbf{r})e^{i\mathbf{p}_0 \cdot \mathbf{r}/\hbar}$, where $u(\mathbf{r})$ is appreciably different from zero only within the specified region.

Expand ψ in eigenfunctions of the momentum operator, that is, as a Fourier integral. The coefficients $a(\mathbf{p})$ in this expansion are given by (15.10) as integrals of functions of the form $u(\mathbf{r})e^{i(\mathbf{p}_0 - \mathbf{p}) \cdot \mathbf{r}/\hbar}$. For such an integral to be appreciably different from zero, the periods of the oscillating factor $e^{i(\mathbf{p}_0 - \mathbf{p}) \cdot \mathbf{r}/\hbar}$ must not be small compared with the dimensions Δx , Δy , and Δz of the region where $u(\mathbf{r})$ differs from zero. Thus $a(\mathbf{p})$ is appreciably different from zero only for values of $\mathbf{p}$ satisfying $(p_{0x} - p_x)\Delta x/\hbar \lesssim 1, \dots$. Since $|a(\mathbf{p})|^2$ determines the probabilities of the various momentum values, the intervals of p_x, p_y, p_z in which $a(\mathbf{p})$ is nonzero are precisely the intervals in which the particle's momentum components may lie in the state considered. Denoting these intervals by $\Delta p_x, \Delta p_y$, and Δp_z , we therefore have

$$\Delta p_x \Delta x \sim \hbar, \quad \Delta p_y \Delta y \sim \hbar, \quad \Delta p_z \Delta z \sim \hbar. \quad (16.6)$$

These *uncertainty relations* were established by *Heisenberg* in 1927.

We see that the more accurately a particle coordinate is known (the smaller Δx is), the greater is the uncertainty Δp_x in the momentum component along the same axis, and conversely. In particular, if a particle occupies a strictly definite point of space ($\Delta x = \Delta y = \Delta z = 0$), then $\Delta p_x = \Delta p_y = \Delta p_z = \infty$. All momentum values are then equally probable. Conversely, if the particle has a strictly definite momentum $\mathbf{p}$, all its positions in space are equally probable (this also follows directly from the wave function (15.8), whose squared modulus is entirely independent of the coordinates).

If the coordinate and momentum uncertainties are characterized by their root-mean-square fluctuations,

$$\delta x = \sqrt{(x - \bar{x})^2}, \quad \delta p_x = \sqrt{(p_x - \bar{p}_x)^2},$$

an exact bound can be given for the least possible value of their product (*H. Weyl*).

Consider the one-dimensional case, a packet with wave function $\psi(x)$ depending on only one coordinate; for simplicity, suppose that the mean values of x and p_x in this state

are zero. Begin with the evident inequality

$$\int_{-\infty}^{\infty} \left| \alpha x \psi + \frac{d\psi}{dx} \right|^2 dx \geq 0,$$

where α is an arbitrary real constant. In evaluating this integral, note that

$$\int x^2 |\psi|^2 dx = (\delta x)^2,$$

$$\int \left(x \frac{d\psi^*}{dx} \psi + x \psi^* \frac{d\psi}{dx} \right) dx = \int x \frac{d|\psi|^2}{dx} dx = - \int |\psi|^2 dx = -1,$$

$$\int \frac{d\psi^*}{dx} \frac{d\psi}{dx} dx = - \int \psi^* \frac{d^2\psi}{dx^2} dx = \frac{1}{\hbar^2} \int \psi^* \hat{p}_x^2 \psi dx = \frac{1}{\hbar^2} (\delta p_x)^2,$$

and obtain

$$\alpha^2 (\delta x)^2 - \alpha + \frac{(\delta p_x)^2}{\hbar^2} \geq 0.$$

For this quadratic trinomial in α to be nonnegative for every value of α , its discriminant must be negative. It follows that

$$\delta x \delta p_x \geq \hbar/2. \quad (16.7)$$

The least possible value of the product is $\hbar/2$.

This value is attained by wave packets described by functions of the form

$$\psi = \frac{1}{(2\pi)^{1/4} \sqrt{\delta x}} \exp \left(\frac{i}{\hbar} p_0 x - \frac{x^2}{4(\delta x)^2} \right), \quad (16.8)$$

where p_0 and δx are constants. The probabilities of the various coordinate values in such a state are

$$|\psi|^2 = \frac{1}{\sqrt{2\pi} \delta x} \exp \left(-\frac{x^2}{2(\delta x)^2} \right),$$

that is, they have a Gaussian distribution about the origin (mean value $\bar{x} = 0$), with root-mean-square fluctuation δx . The wave function in the momentum representation is

$$a(p_x) = \int_{-\infty}^{\infty} \psi(x) \exp \left(-i \frac{p_x x}{\hbar} \right) dx.$$

Evaluation of the integral gives an expression of the form

$$a(p_x) = \text{const} \cdot \exp \left(-\frac{(\delta x)^2 (p_x - p_0)^2}{\hbar^2} \right).$$

The probability distribution of the momentum values, $|a(p_x)|^2$, is also Gaussian, about the mean $\bar{p}_x = p_0$, and has root-mean-square fluctuation $\delta p_x = \hbar/(2\delta x)$. Thus the product $\delta p_x \delta x$ is exactly $\hbar/2$.

Finally, let us derive one more useful relation. Let f and g be two physical quantities whose operators obey the commutation rule

$$\hat{f}\hat{g} - \hat{g}\hat{f} = -i\hbar\hat{c}, \quad (16.9)$$

where $\hat{c}$ is the operator of a physical quantity c . The factor $\hbar$ has been introduced on the right because in the classical limit ($\hbar \rightarrow 0$) all operators of physical quantities reduce to multiplication by those quantities and commute with one another. Thus, in the “quasiclassical” case, the right-hand side of (16.9) may be taken as zero in the first approximation. In the next approximation, $\hat{c}$ may be replaced by the operator of simple multiplication by c . We then have

$$\hat{f}\hat{g} - \hat{g}\hat{f} = -i\hbar c.$$

This equation is exactly analogous to $\hat{p}_x x - x \hat{p}_x = -i\hbar$, except that the quantity $\hbar c$ occurs in place of the constant $\hbar$ ¹⁷. By analogy with $\Delta x \Delta p_x \sim \hbar$, we therefore conclude that in the quasiclassical case the quantities f and g obey the uncertainty relation

$$\Delta f \Delta g \sim \hbar c. \quad (16.10)$$

In particular, if one of the quantities is the energy ($f \equiv H$) and the operator of the other (g) has no explicit time dependence, then by (9.2) $c = \dot{g}$, and the quasiclassical uncertainty relation is

$$\Delta E \Delta g \sim \hbar \dot{g}. \quad (16.11)$$

¹⁷The classical quantity c is the Poisson bracket of f and g (see the note on p. 46).

The Schrödinger Equation

§ 17. The Schrödinger equation

The Hamiltonian determines the form of a physical system's wave equation and consequently acquires fundamental significance in the entire mathematical apparatus of quantum mechanics.

For a free particle, the Hamiltonian is already established by the general requirements associated with the homogeneity and isotropy of space and with the Galilean relativity principle. In classical mechanics these requirements imply a quadratic dependence of the particle energy on its momentum: $E = p^2/2m$, where the constant m is called the mass of the particle (see I, § 4). In quantum mechanics the same requirements lead to the same relation for the eigenvalues of energy and momentum, quantities which for a free particle are conserved and can be measured simultaneously.

For the relation $E = p^2/2m$ to hold for every eigenvalue of energy and momentum, however, it must also hold for their operators:

$$\hat{H} = \frac{1}{2m} (\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2). \quad (17.1)$$

Substitution of (15.2) here gives the Hamiltonian of a freely moving particle in the form

$$\hat{H} = -\frac{\hbar^2}{2m} \Delta, \quad (17.2)$$

where $\Delta = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$ is the Laplace operator. The Hamiltonian of a system of noninteracting particles is equal to the sum of the Hamiltonians of the individual particles:

$$\hat{H} = -\frac{\hbar^2}{2} \sum_a \frac{\Delta_a}{m_a}, \quad (17.3)$$

where the index a numbers the particles; Δ_a is the Laplace operator in which differentiation is performed with respect to the coordinates of the a th particle. In classical (nonrelativistic) mechanics, particle interaction is described by an additive term in the Hamilton function, the potential energy of interaction $U(\mathbf{r}_1, \mathbf{r}_2, \dots)$, which is a function of the particle coordinates. Particle interaction in quantum mechanics is likewise described by adding the same function to the Hamiltonian of the system¹:

$$\hat{H} = -\frac{\hbar^2}{2} \sum_a \frac{\Delta_a}{m_a} + U(\mathbf{r}_1, \mathbf{r}_2, \dots); \quad (17.4)$$

¹This assertion is not, of course, a logical consequence of the fundamental principles of quantum mechanics and must be regarded as a consequence of experimental data.

the first term may be regarded as the kinetic-energy operator and the second as the potential-energy operator. In particular, the Hamiltonian for one particle in an external field is

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + U(x, y, z) = -\frac{\hbar^2}{2m}\Delta + U(x, y, z), \quad (17.5)$$

where $U(x, y, z)$ is the potential energy of the particle in the external field.

Substitution of expressions (17.2)–(17.5) in the general equation (8.1) yields the wave equations for the corresponding systems. Here we write out the wave equation for a particle in an external field:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m}\Delta \Psi + U(x, y, z)\Psi. \quad (17.6)$$

Equation (10.2), determining the stationary states, assumes the form

$$\frac{\hbar^2}{2m}\Delta \psi + [E - U(x, y, z)]\psi = 0. \quad (17.7)$$

Equations (17.6) and (17.7) were established by *Schrödinger* in 1926 and are known as the *Schrödinger equations*.

For a free particle equation (17.7) becomes

$$\frac{\hbar^2}{2m}\Delta \psi + E\psi = 0. \quad (17.8)$$

This equation admits solutions that remain finite over all space for any positive value of the energy E . In states with definite directions of motion these solutions are the eigenfunctions of the momentum operator, with $E = p^2/2m$. The full (time-dependent) wave functions of these stationary states have the form

$$\Psi = \text{const} \cdot \exp \left[-\frac{i}{\hbar}(Et - \mathbf{p} \cdot \mathbf{r}) \right]. \quad (17.9)$$

Each of these functions, a *plane wave*, describes a state in which the particle possesses definite energy E and momentum $\mathbf{p}$. The frequency of this wave is $E/\hbar$, and its wave vector is $\mathbf{k} = \mathbf{p}/\hbar$; the associated wavelength $\lambda = 2\pi\hbar/p$ is termed the *de Broglie wavelength of the particle*².

The energy spectrum of a freely moving particle is therefore continuous, extending from zero to $+\infty$. Each of these eigenvalues, except only $E = 0$, is degenerate, and the degeneracy is of infinite multiplicity. Indeed, every nonzero value of E corresponds to an infinite set of eigenfunctions (17.9), differing in the directions of the vector $\mathbf{p}$ while its absolute magnitude remains the same.

Let us trace how the limiting transition to classical mechanics occurs in the Schrödinger equation, considering for simplicity just one particle in an external field. Substitution in the Schrödinger equation (17.6) of the limiting expression (6.1), $\Psi = ae^{iS/\hbar}$, for the wave function gives, after the differentiations are performed,

$$a \frac{\partial S}{\partial t} - i\hbar \frac{\partial a}{\partial t} + \frac{a}{2m}(\nabla S)^2 - \frac{i\hbar}{2m}a\Delta S - \frac{i\hbar}{m}\nabla S \cdot \nabla a - \frac{\hbar^2}{2m}\Delta a + Ua = 0.$$

²The concept of a wave associated with a particle was first introduced by *de Broglie* (*L. de Broglie*) in 1924.

This equation contains purely real and purely imaginary terms (recall that S and a are real); equating each group separately to zero gives two equations:

$$\begin{aligned}\frac{\partial S}{\partial t} + \frac{1}{2m}(\nabla S)^2 + U - \frac{\hbar^2}{2ma}\Delta a &= 0, \\ \frac{\partial a}{\partial t} + \frac{a}{2m}\Delta S + \frac{1}{m}\nabla S \cdot \nabla a &= 0.\end{aligned}$$

Neglecting the term containing $\hbar^2$ in the first of these equations, we obtain

$$\frac{\partial S}{\partial t} + \frac{1}{2m}(\nabla S)^2 + U = 0, \quad (17.10)$$

that is, as expected, the classical Hamilton–Jacobi equation for the action S of the particle. We note in passing that, as $\hbar \rightarrow 0$ classical mechanics is accurate up to and including quantities of first order in $\hbar$ (rather than zeroth order).

After multiplication by $2a$, the second equation obtained above can be rewritten as

$$\frac{\partial a^2}{\partial t} + \nabla \cdot \left(a^2 \frac{\nabla S}{m} \right) = 0. \quad (17.11)$$

This equation has a transparent physical meaning: a^2 is the probability density for finding the particle at one or another place in space ($|\Psi|^2 = a^2$); $\nabla S/m = \mathbf{p}/m$ is the classical velocity $\mathbf{v}$ of the particle. Equation (17.11) is therefore nothing other than the continuity equation, showing that the probability density is “transported” according to the laws of classical mechanics, with the classical velocity $\mathbf{v}$ at every point.

PROBLEM

Find the transformation law of the wave function under a Galilean transformation.

SOLUTION. Perform the transformation on the wave function for the free motion of a particle (a plane wave). Since every function Ψ can be expanded in plane waves, this will also determine the transformation law for an arbitrary wave function.

The plane waves in the reference frames K and K' (where K' moves with velocity $\mathbf{V}$ relative to K) are

$$\begin{aligned}\Psi(\mathbf{r}, t) &= \text{const} \cdot \exp[i(\mathbf{p} \cdot \mathbf{r} - Et)/\hbar], \\ \Psi'(\mathbf{r}', t) &= \text{const} \cdot \exp[i(\mathbf{p}' \cdot \mathbf{r}' - E't)/\hbar],\end{aligned}$$

where $\mathbf{r} = \mathbf{r}' + \mathbf{V}t$, while the particle’s momenta and energies in the two frames are related by

$$\mathbf{p} = \mathbf{p}' + m\mathbf{V}, \quad E = E' + \mathbf{V} \cdot \mathbf{p}' + \frac{mV^2}{2}$$

(see I, § 8). Substitution of these expressions in Ψ gives

$$\begin{aligned}\Psi(\mathbf{r}, t) &= \Psi'(\mathbf{r}', t) \exp \left[\frac{i}{\hbar} \left(m\mathbf{V} \cdot \mathbf{r}' + \frac{mV^2}{2}t \right) \right] \\ &= \Psi'(\mathbf{r} - \mathbf{V}t, t) \exp \left[\frac{i}{\hbar} \left(m\mathbf{V} \cdot \mathbf{r} - \frac{mV^2}{2}t \right) \right].\end{aligned} \quad (1)$$

Written this way, the formula no longer contains quantities characterizing the free motion of the particle; it therefore furnishes the sought-after general rule governing how the wave function of an arbitrary state of the particle transforms. For a system of particles, the exponent of the exponential in (1) must contain a summation over all the particles.

§ 18. Fundamental properties of the Schrödinger equation

The conditions to be satisfied by solutions of the Schrödinger equation are quite general in nature. The wave function, to begin with, must be single-valued and continuous throughout all of space. The continuity requirement remains in force even when the field $U(x, y, z)$ itself has surfaces of discontinuity. On any such surface, the wave function together with its derivatives must still be continuous. The latter are not continuous, however, if the potential energy U becomes infinite beyond some surface. A particle cannot enter at all a region of space where $U = \infty$, that is, $\psi = 0$ must hold everywhere in that region. Continuity of ψ requires it to vanish on the boundary of this region; in this case, however, the derivatives of ψ generally undergo a jump.

If the field $U(x, y, z)$ nowhere becomes infinite, the wave function must also be finite throughout space. The same condition must be observed when U becomes infinite at some point, provided it does not do so too rapidly, namely as $1/r^s$ with $s < 2$ (see also § 35).

Let $U_{\min}$ be the minimum value of the function $U(x, y, z)$. Since the particle's Hamiltonian is the sum of two terms, the kinetic- and potential-energy operators $\hat{T}$ and U , the mean energy in an arbitrary state is the sum $\bar{E} = \bar{T} + \bar{U}$. But all eigenvalues of the operator $\hat{T}$ (which coincides with the Hamiltonian of a free particle) are positive, and therefore the mean value $\bar{T} > 0$ as well. Taking account also of the evident inequality $\bar{U} > U_{\min}$, we find that $\bar{E} > U_{\min}$. Since this inequality holds for every state, it plainly holds for all energy eigenvalues as well:

$$E_n > U_{\min}. \quad (18.1)$$

Suppose a particle moves through a force field that goes to zero at large distances; following standard convention, we fix $U(x, y, z)$ so that it too tends to zero at infinity. One readily verifies that the eigenvalue spectrum for negative energies is then discrete: every state with $E < 0$ in a field that vanishes at infinity is a bound state. Indeed, in stationary states of the continuous spectrum, which correspond to infinite motion, the particle is at infinity (see § 10). At sufficiently large distances the presence of the field may be neglected and the particle's motion may be regarded as free; in free motion, however, the energy can only be positive.

Conversely, the positive eigenvalues form a continuous spectrum and correspond to infinite motion; for $E > 0$ the Schrödinger equation generally does not have (in the field under consideration) solutions for which the integral $\int |\psi|^2 dV$ converges³.

We draw attention to the fact that, in quantum mechanics, a particle in finite motion may also be present in regions of space where $E < U$. Although the probability $|\psi|^2$

³From a purely mathematical standpoint, however, it must be qualified that, for certain forms of the function $U(x, y, z)$ (having no physical significance), a discrete set of values may drop out of the continuous spectrum.

of finding the particle tends rapidly to zero deeper inside such a region, it nevertheless differs from zero at every finite distance. This represents a fundamental departure from classical mechanics, where a particle is altogether unable to penetrate a region with $U > E$. In classical mechanics, penetration into this region is impossible because, for $E < U$, the kinetic energy would become negative, meaning that the velocity would be imaginary. In quantum mechanics, too, the kinetic-energy eigenvalues are positive; nevertheless, no contradiction arises here, because if the measurement process localizes the particle at some definite point of space, that same process disturbs the particle's state in such a way that it ceases altogether to have any definite kinetic energy.

If $U(x, y, z) > 0$ throughout space (with $U \rightarrow 0$ at infinity), inequality (18.1) gives $E_n > 0$. On the other hand, since the spectrum must be continuous for $E > 0$, it follows that in the case under consideration the discrete spectrum is absent altogether, that is, only infinite motion of the particle is possible.

Suppose that at some point, chosen as the coordinate origin, U tends to $-\infty$ according to the law

$$U \approx -\alpha/r^s, \quad \alpha > 0. \quad (18.2)$$

Consider a wave function that is finite in a certain small region of radius r_0 around the origin and zero outside it. The uncertainty in the particle's coordinate values in such a wave packet is of order r_0 ; the uncertainty in the momentum value is therefore $\sim \hbar/r_0$. The mean kinetic energy in this state is of order $\hbar^2/mr_0^2$, and the mean potential energy is of order $-\alpha/r_0^s$. Suppose first that $s > 2$. Then, for sufficiently small r_0 , the sum $\hbar^2/mr_0^2 - \alpha/r_0^s$ takes negative values of arbitrarily large absolute magnitude. But if the mean energy can take such values, this means in any event that there exist negative energy eigenvalues of arbitrarily large absolute magnitude. Energy levels with large $|E|$ correspond to motion of the particle in a very small region of space around the origin. The "normal" state will correspond to the particle being located at the origin itself, that is, the particle will "fall" into the point $r = 0$.

If $s < 2$, on the other hand, the energy cannot take negative values of arbitrarily large absolute magnitude. The diagonal spectrum starts from a certain finite negative value, and the particle does not fall to the center. We note that in classical mechanics a particle can in principle fall to the center in any attractive field (that is, for any positive s). The case $s = 2$ will be considered separately in § 35.

Next, let us investigate the character of the energy spectrum as a function of the behaviour of the field at large distances. Suppose that as $r \rightarrow \infty$ the potential energy, while negative, tends to zero according to the power law (18.2) (in this formula r is now large). Consider a wave packet "filling" a spherical shell of large radius r_0 and thickness $\Delta r \ll r_0$. Then again the order of magnitude of the kinetic energy is $\hbar^2/m(\Delta r)^2$, and that of the potential energy is $-\alpha/r_0^s$. Let r_0 increase while Δr is increased at the same time (so that Δr grows in proportion to r_0). If $s < 2$, then for sufficiently large r_0 the sum $\hbar^2/m(\Delta r)^2 - \alpha/r_0^s$ becomes negative. It follows that stationary states with negative energy exist in which the particle may be found with appreciable probability at large distances from the origin. But this means that negative energy levels of arbitrarily small absolute magnitude exist (it must be remembered that wave functions decay rapidly in the region of space where $U > E$). Thus, in the case under consideration, the discrete

spectrum contains an infinite set of levels which crowd together towards the level $E = 0$.

If, however, the field falls off at infinity as $-1/r^s$ with $s > 2$, there are no negative levels of arbitrarily small absolute magnitude. The discrete spectrum ends at a level with nonzero absolute magnitude, so that the total number of levels is finite.

The Schrödinger equation for the stationary-state wave functions ψ , together with the conditions placed on its solutions, is real. Its solutions can therefore always be chosen real⁴. The eigenfunctions of nondegenerate energy values automatically prove to be real up to an immaterial phase factor. Indeed, ψ^* satisfies the same equation as ψ and is therefore also an eigenfunction for the same value of the energy; hence, if that value is nondegenerate, ψ and ψ^* must be essentially identical, that is, they may differ at most by a constant factor whose modulus equals unity. Wave functions that correspond to one and the same degenerate level need not be real, but a set of real functions can always be obtained by a suitable choice of their linear combinations.

The full (time-dependent) wave functions Ψ , however, are determined by an equation whose coefficients contain i . This equation nevertheless retains its form if t is replaced by $-t$ and complex conjugation is performed at the same time⁵. The functions Ψ can therefore always be chosen so that Ψ and Ψ^* differ only in the sign of time.

As is known, the equations of classical mechanics are unchanged under *time reversal*, i.e. upon reversing the sign of time. In quantum mechanics, as we see, symmetry with respect to the two directions of time manifests itself in the invariance of the wave equation under reversal of the sign of t accompanied by the replacement of Ψ with Ψ^* . It must be remembered, however, that here this symmetry applies only to the equations and not to the concept of measurement itself, which has a fundamental role in quantum mechanics (as discussed in detail in § 7).

§ 19. Flux Density

In classical mechanics a particle's velocity $\mathbf{v}$ is related to its momentum by $\mathbf{p} = m\mathbf{v}$. As expected, quantum mechanics has the same relation between the corresponding operators. This follows at once by calculating $\hat{\mathbf{v}} = \hat{\mathbf{r}}\dot{\mathbf{r}}$ from the general rule (9.2) for differentiating operators with respect to time:

$$\hat{\mathbf{v}} = \frac{i}{\hbar} (\hat{H}\mathbf{r} - \mathbf{r}\hat{H}).$$

Using (17.5) for $\hat{H}$ together with (16.5), we obtain

$$\hat{\mathbf{v}} = \hat{\mathbf{p}}/m. \quad (19.1)$$

The same relations plainly hold between the eigenvalues of velocity and momentum, and between their mean values in any state.

Like momentum, the velocity of a particle cannot have a definite value simultaneously with its coordinates. Velocity multiplied by an infinitesimal time element dt , however,

⁴This is not true for systems in a magnetic field.

⁵It is assumed that the potential energy U has no explicit dependence on time: the system is either closed or is in a constant (nonmagnetic) field.

determines the particle's displacement during dt . Thus the nonexistence of velocity simultaneously with coordinates means that if a particle occupies a definite point of space at one instant, it no longer has a definite position at the next, infinitesimally close instant.

We record a useful formula for the operator $\hat{f}$, the time derivative of a quantity $f(\mathbf{r})$ which is a function of the particle's radius vector. Since f commutes with $U(\mathbf{r})$, we find

$$\hat{f} = \frac{i}{\hbar}(\hat{H}f - f\hat{H}) = \frac{i}{2m\hbar}(\hat{\mathbf{p}}^2 f - f\hat{\mathbf{p}}^2),$$

while (16.4) gives

$$\hat{\mathbf{p}}^2 f = \hat{\mathbf{p}}(f\hat{\mathbf{p}} - i\hbar\nabla f), \quad f\hat{\mathbf{p}}^2 = (\hat{\mathbf{p}}f + i\hbar\nabla f)\hat{\mathbf{p}}.$$

Hence the desired expression is

$$\hat{f} = \frac{1}{2m}(\hat{\mathbf{p}}\nabla f + \nabla f \cdot \hat{\mathbf{p}}). \quad (19.2)$$

Next, let us find the acceleration operator. We have

$$\hat{\mathbf{v}} = \frac{i}{\hbar}(\hat{H}\hat{\mathbf{v}} - \hat{\mathbf{v}}\hat{H}) = \frac{i}{m\hbar}(\hat{H}\hat{\mathbf{p}} - \hat{\mathbf{p}}\hat{H}) = \frac{i}{m\hbar}(U\hat{\mathbf{p}} - \hat{\mathbf{p}}U).$$

Using (16.4), we find

$$m\hat{\mathbf{v}} = -\nabla U. \quad (19.3)$$

In form, this operator equation coincides exactly with the equation of motion (Newton's equation) of classical mechanics.

The integral $\int_V |\Psi|^2 dV$ over a finite volume V is the probability of finding the particle in that volume. Let us calculate its time derivative:

$$\frac{d}{dt} \int_V |\Psi|^2 dV = \int_V \left(\Psi \frac{\partial \Psi^*}{\partial t} + \Psi^* \frac{\partial \Psi}{\partial t} \right) dV = \frac{i}{\hbar} \int_V (\Psi \hat{H}^* \Psi^* - \Psi^* \hat{H} \Psi) dV.$$

Substitution of

$$\hat{H} = \hat{H}^* = -\frac{\hbar^2}{2m}\Delta + U(x, y, z)$$

and use of the identity

$$\Psi \Delta \Psi^* - \Psi^* \Delta \Psi = \nabla \cdot (\Psi \nabla \Psi^* - \Psi^* \nabla \Psi)$$

give

$$\frac{d}{dt} \int_V |\Psi|^2 dV = - \int_V \nabla \cdot \mathbf{j} dV,$$

where $\mathbf{j}$ denotes the vector⁶

$$\mathbf{j} = \frac{i\hbar}{2m}(\Psi \nabla \Psi^* - \Psi^* \nabla \Psi) = \frac{1}{2m}(\Psi \hat{\mathbf{p}}^* \Psi^* + \Psi^* \hat{\mathbf{p}} \Psi). \quad (19.4)$$

⁶If ψ is written as $|\psi|e^{i\alpha}$, then

$$\mathbf{j} = \frac{\hbar}{m}|\psi|^2 \nabla \alpha. \quad (19.4a)$$

By Gauss's theorem, the integral of $\nabla \cdot \mathbf{j}$ can be transformed into an integral over the closed surface bounding V :

$$\frac{d}{dt} \int_V |\Psi|^2 dV = - \oint \mathbf{j} d\mathbf{f}. \quad (19.5)$$

It follows that $\mathbf{j}$ may be called the *probability-flux-density vector*, or simply the *flux density*. Its surface integral is the probability that the particle crosses that surface per unit time. The vector $\mathbf{j}$ and the probability density $|\Psi|^2$ obey

$$\frac{\partial |\Psi|^2}{\partial t} + \nabla \cdot \mathbf{j} = 0, \quad (19.6)$$

an equation analogous to the classical continuity equation.

The free-motion wave function—the plane wave (17.9)—can be normalized so as to describe a particle flux of unit density (a flux in which, on average, one particle per unit time passes through a unit area of its cross-section). The function is

$$\Psi = \frac{1}{\sqrt{v}} \exp \left[-\frac{i}{\hbar} (Et - \mathbf{p} \cdot \mathbf{r}) \right], \quad (19.7)$$

where v is the particle's speed. Indeed, substituting this function into (19.4) gives $\mathbf{j} = \mathbf{p}/m\gamma$, the unit vector in the direction of motion. It is useful to show how the mutual orthogonality of wave functions of states with different energies follows directly from the Schrödinger equation. Let ψ_m and ψ_n be two such functions; they satisfy

$$-\frac{\hbar^2}{2m} \Delta \psi_m + U \psi_m = E_m \psi_m,$$

$$-\frac{\hbar^2}{2m} \Delta \psi_n^* + U \psi_n^* = E_n \psi_n^*.$$

Multiply the first equation by ψ_n^* and the second by ψ_m , and subtract them term by term. This gives

$$(E_m - E_n) \psi_m \psi_n^* = \frac{\hbar^2}{2m} (\psi_m \Delta \psi_n^* - \psi_n^* \Delta \psi_m) = \frac{\hbar^2}{2m} \nabla \cdot (\psi_m \nabla \psi_n^* - \psi_n^* \nabla \psi_m).$$

Integrating both sides over all space, the right-hand side vanishes after application of Gauss's theorem, and we obtain

$$(E_m - E_n) \int \psi_m \psi_n^* dV = 0.$$

Since $E_m \neq E_n$ by assumption, the required orthogonality relation follows:

$$\int \psi_m \psi_n^* dV = 0.$$

§ 20. The Variational Principle

The Schrödinger equation in its general form, $\hat{H}\psi = E\psi$, can be obtained from the variational principle

$$\delta \int \psi^* (\hat{H} - E) \psi dq = 0. \quad (20.1)$$

Since ψ is complex, variations with respect to ψ and ψ^* may be made independently. Variation with respect to ψ^* gives

$$\int \delta \psi^* (\hat{H} - E) \psi dq = 0,$$

and the arbitrariness of $\delta \psi^*$ then yields the required equation $\hat{H}\psi = E\psi$. Variation with respect to ψ gives nothing new. Indeed, varying ψ and using the Hermitian character of $\hat{H}$, we have

$$\int \psi^* (\hat{H} - E) \delta \psi dq = \int \delta \psi (\hat{H}^* - E) \psi^* dq = 0,$$

which yields the complex-conjugate equation $\hat{H}^* \psi^* = E \psi^*$.

The variational principle (20.1) requires an unconstrained extremum of the integral. It may be put in another form by treating E as a Lagrange multiplier in the problem of finding a constrained extremum of

$$\delta \int \psi^* \hat{H} \psi dq = 0 \quad (20.2)$$

subject to the supplementary condition

$$\int \psi \psi^* dq = 1. \quad (20.3)$$

The minimum value of the integral (20.2), subject to (20.3), is the first energy eigenvalue, that is, the energy E_0 of the normal state. The function ψ which realizes this minimum is correspondingly the normal-state wave function ψ_0 ⁷. The wave functions ψ_n ($n > 0$) of the succeeding stationary states correspond only to extrema, not to true minima of the integral.

To obtain from the minimum condition for (20.2) the wave function ψ_1 and energy E_1 of the state immediately above the normal state, the trial functions ψ must be restricted not only by the normalization condition (20.3), but also by orthogonality to the normal-state wave function ψ_0 : $\int \psi \psi_0 dq = 0$. In general, suppose that the wave functions $\psi_0, \psi_1, \dots, \psi_{n-1}$ of the first n states are known (the states being ordered by increasing energy). The wave function of the next state minimizes (20.2) under the supplementary conditions

$$\int \psi^2 dq = 1, \quad \int \psi \psi_m dq = 0, \quad m = 0, 1, 2, \dots, n-1. \quad (20.4)$$

⁷For the remainder of this section we take the wave functions ψ to be real, as they can always be chosen when no magnetic field is present.

We give here several general theorems which may be proved from the variational principle⁸.

The normal-state wave function ψ_0 does not vanish (or, as one says, has no nodes) at any finite values of the coordinates⁹. In other words, it has the same sign throughout space. It follows that the wave functions ψ_n ($n > 0$) of the other stationary states, being orthogonal to ψ_0 , necessarily have nodal points (if ψ_n also had a constant sign, the integral $\int \psi_0 \psi_n dq$ could not vanish).

The absence of nodes in ψ_0 also implies that the normal energy level cannot be degenerate. Suppose otherwise, and let ψ_0 and ψ'_0 be two different eigenfunctions belonging to the level E_0 . Every linear combination $c\psi_0 + c'\psi'_0$ would also be an eigenfunction; but by a suitable choice of c and c' , this function can always be made to vanish at any prescribed point of space, producing an eigenfunction with a node.

If motion is confined to a bounded region of space, then $\psi = 0$ on its boundary (see § 18). To determine the energy levels one must use the variational principle to find the minimum of (20.2) with this boundary condition. In this case the theorem that the normal-state wave function has no nodes states that ψ_0 vanishes nowhere inside the region.

Notice that increasing the size of the region of motion lowers every energy level E_n . This follows directly because enlarging the region enlarges the class of trial functions available for minimizing the integral, so its minimum value can only decrease. For discrete-spectrum states of a system of particles, the expression

$$\int \psi \hat{H} \psi dq = \int \left[- \sum_a \frac{\hbar^2}{2m_a} \psi \Delta_a \psi + U \psi^2 \right] dq$$

can be transformed into a form more convenient for actual variation. In the first term of the integrand, write

$$\psi \Delta_a \psi = \nabla_a \cdot (\psi \nabla_a \psi) - (\nabla_a \psi)^2.$$

The integral of $\nabla_a \cdot (\psi \nabla_a \psi)$ over dV_a becomes an integral over a closed surface at infinity. Since discrete-spectrum wave functions vanish sufficiently rapidly at infinity, this integral disappears. Thus

$$\int \psi \hat{H} \psi dq = \int \left[\sum_a \frac{\hbar^2}{2m_a} (\nabla_a \psi)^2 + U \psi^2 \right] dq. \quad (20.5)$$

§ 21. General Properties of One-Dimensional Motion

When a particle's potential energy depends on just one coordinate x , the wave function may be sought as a product of a function of y, z and a function of x alone. The

⁸Proofs of the theorems about zeros of eigenfunctions (see also the next section) may be found in: *M. A. Lavrent'ev and L. A. Lyusternik*, Course of the Calculus of Variations, Moscow: Gostekhizdat, 1950, Ch. IX; *R. Courant and D. Hilbert*, Methods of Mathematical Physics, Moscow: Gostekhizdat, 1951, Vol. I, Ch. VI.

⁹This theorem (and the consequences drawn from it below) is not valid in general for wave functions of systems containing several identical particles; see the end of § 63.

first is determined by the free-motion Schrödinger equation, and the second by the one-dimensional Schrödinger equation

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - U(x)]\psi = 0. \quad (21.1)$$

The problem of motion in a field with potential energy $U(x, y, z) = U_1(x) + U_2(y) + U_3(z)$, separated into a sum of functions each depending on only one coordinate, plainly reduces to the same kind of one-dimensional equations. In §§ 22–24 we shall consider several specific examples of such “one-dimensional” motion. First we establish some of its general properties.

We begin by showing that every discrete-spectrum energy level in a one-dimensional problem is nondegenerate. Suppose the contrary, and let ψ_1 and ψ_2 be two different eigenfunctions belonging to the same energy value. Since both satisfy the same equation (21.1),

$$\frac{\psi_1''}{\psi_1} = \frac{2m}{\hbar^2}(U - E) = \frac{\psi_2''}{\psi_2},$$

or $\psi_2\psi_1'' - \psi_1\psi_2'' = 0$ (a prime denotes differentiation with respect to x). Integration gives

$$\psi_1\psi_2' - \psi_2\psi_1' = \text{const}. \quad (21.2)$$

Since $\psi_1 = \psi_2 = 0$ at infinity, the constant must be zero, and hence

$$\psi_1\psi_2' - \psi_2\psi_1' = 0,$$

or $\psi_1'/\psi_1 = \psi_2'/\psi_2$. A second integration gives $\psi_1 = \text{const} \cdot \psi_2$; the two functions are therefore essentially identical.

For the discrete-spectrum wave functions $\psi_n(x)$, the following result, known as the *oscillation theorem*, can be stated: the function $\psi_n(x)$ corresponding to the $(n + 1)$ th eigenvalue E_n in increasing order vanishes n times at finite x ¹⁰.

Suppose that $U(x)$ tends to finite limits as $x \rightarrow \pm\infty$ (it need not in any way be monotonic). Take $U(+\infty)$ as the zero of energy, so that $U(+\infty) = 0$, and denote $U(-\infty)$ by U_0 , with $U_0 > 0$. The discrete spectrum lies in the range of energies for which the particle cannot escape to infinity. The energy must therefore be less than both limiting values and, in particular, negative:

$$E < 0. \quad (21.3)$$

Of course $E > U_{\min}$ must also hold in every case, so $U(x)$ must possess at least one minimum with $U_{\min} < 0$.

Now consider the positive energies below U_0 :

$$0 < E < U_0. \quad (21.4)$$

Here the spectrum is continuous and the particle motion in the corresponding stationary states is infinite, with the particle escaping toward $x = +\infty$. Every energy eigenvalue in

¹⁰If the particle can occupy only a bounded interval of the x axis, the zeros of $\psi_n(x)$ inside that interval are to be counted.

this part of the spectrum is likewise nondegenerate. To see this, it suffices to observe that the proof given above for the discrete spectrum requires ψ_1 and ψ_2 to vanish at only one of the two infinities (here they vanish as $x \rightarrow -\infty$).

At sufficiently large positive x , $U(x)$ can be neglected in the Schrödinger equation (21.1):

$$\psi'' + \frac{2m}{\hbar^2} E \psi = 0.$$

This equation has real solutions in the form of a standing plane wave,

$$\psi = a \cos(kx + \delta), \quad (21.5)$$

where a and δ are constants, and the “wave vector” is $k = p/\hbar = \sqrt{2mE}/\hbar$. This formula gives the asymptotic form, as $x \rightarrow +\infty$, of the wave functions for the nondegenerate energy levels in the continuous-spectrum interval (21.4). At large negative x , the Schrödinger equation becomes

$$\psi'' - \frac{2m}{\hbar^2} (U_0 - E) \psi = 0.$$

The solution which does not become infinite as $x \rightarrow -\infty$ is

$$\psi = b e^{\kappa x}, \quad \kappa = \frac{1}{\hbar} \sqrt{2m(U_0 - E)}. \quad (21.6)$$

This is the asymptotic wave function as $x \rightarrow -\infty$. The wave function thus decays exponentially into the region in which $E < U_0$.

Finally, when

$$E > U_0, \quad (21.7)$$

the spectrum is continuous and the motion is infinite in both directions. Every level in this part of the spectrum is doubly degenerate. The reason is that the corresponding wave functions are determined by the second-order equation (21.1), and both independent solutions meet the required conditions at infinity (whereas, for example, in the preceding case one solution became infinite as $x \rightarrow -\infty$ and had to be rejected). As $x \rightarrow +\infty$, the asymptotic wave function is

$$\psi = a_1 e^{ikx} + a_2 e^{-ikx}, \quad (21.8)$$

and similarly as $x \rightarrow -\infty$. The term containing e^{ikx} corresponds to a particle moving to the right, and that containing e^{-ikx} to a particle moving to the left.

Suppose $U(x)$ is even, $U(-x) = U(x)$. The Schrödinger equation (21.1) is then unchanged on reversing the sign of the coordinate. Hence, if $\psi(x)$ is a solution, $\psi(-x)$ is also a solution and differs from $\psi(x)$ only by a constant factor: $\psi(-x) = c\psi(x)$. Reversing x once more gives $\psi(x) = c^2\psi(x)$, so $c = \pm 1$. Thus, for a potential energy symmetric about $x = 0$, stationary-state wave functions may be either even, $\psi(-x) = \psi(x)$, or odd, $\psi(-x) = -\psi(x)$ ¹¹. In particular, the ground-state wave function is even: it cannot have a node, whereas an odd function necessarily vanishes at $x = 0$, since $\psi(0) = -\psi(0) = 0$.

¹¹This argument assumes that the stationary state is nondegenerate, that is, that its motion is not infinite in both directions. Otherwise the two wave functions belonging to the given energy level may transform into one another when the sign of x is reversed. Even then, the stationary-state wave functions are not necessarily even or odd, but definite parity can always be achieved by forming suitable linear combinations of the two solutions.

There is a simple method for normalizing one-dimensional-motion wave functions in the continuous spectrum, which determines the normalization coefficient directly from the asymptotic wave function at large $|x|$.

Consider the wave function for motion infinite in one direction ($x \rightarrow +\infty$). The normalization integral diverges as $x \rightarrow \infty$ (at $x \rightarrow -\infty$ the function decays exponentially, so the integral converges rapidly). To determine the normalization constant, one may therefore replace ψ by its asymptotic value for large positive x and take any finite value of x , say zero, as the lower limit of integration; this is equivalent to discarding a finite contribution beside a divergent one. We now show that a wave function normalized by

$$\int \psi_p^* \psi_{p'} dx = \delta \left(\frac{p - p'}{2\pi\hbar} \right) = 2\pi\hbar\delta(p - p') \quad (21.9)$$

(where p is the particle momentum at infinity) must have the asymptotic form (21.5) with $a = 2$:

$$\psi_p \approx 2 \cos(kx + \delta) = e^{i(kx + \delta)} + e^{-i(kx + \delta)}. \quad (21.10)$$

Since we are not seeking to verify mutual orthogonality of functions corresponding to different p , when substituting (21.10) into the normalization integral we take p and p' arbitrarily close and may put $\delta = \delta'$ (in general, δ is a function of p). We then retain in the integrand only the terms which diverge when $p = p'$; equivalently, we drop terms containing factors $e^{\pm i(k+k')x}$. We thus obtain

$$\int \psi_p^* \psi_{p'} dx = \int_0^\infty e^{i(k'-k)x} dx + \int_0^\infty e^{-i(k'-k)x} dx = \int_{-\infty}^\infty e^{i(k'-k)x} dx,$$

which, by (15.7), agrees with (21.9). According to (5.14), conversion to normalization by an energy delta function is effected by multiplying ψ_p by

$$\left(\frac{d(p/2\pi\hbar)}{dE} \right)^{1/2} = \frac{1}{\sqrt{2\pi\hbar v}},$$

where v is the particle's speed at infinity. Thus

$$\psi_E = \frac{1}{\sqrt{2\pi\hbar v}} \psi_p = \frac{1}{\sqrt{2\pi\hbar v}} \left(e^{i(kx + \delta)} + e^{-i(kx + \delta)} \right). \quad (21.11)$$

Notice that each of the two traveling waves into which the standing wave (21.11) separates has flux density $1/(2\pi\hbar)$.

We may therefore state the following rule for normalizing, to an energy delta function, the wave function of motion infinite in one direction: write its asymptotic expression as the sum of two plane waves traveling in opposite directions, and choose the normalization coefficient so that the flux density in the wave traveling toward the origin (or away from it) equals $1/(2\pi\hbar)$.

The analogous argument gives the corresponding rule for wave functions of motion infinite in both directions. Such a wave function is normalized to an energy delta function when the sum of the fluxes in the waves traveling toward the origin from the positive and negative sides of the x axis equals $1/(2\pi\hbar)$.

§ 22. The Potential Well

As a simple example of one-dimensional motion, consider motion in a rectangular *potential well*, that is, in a field with the function $U(x)$ shown in Fig. 1: $U(x) = 0$ for $0 < x < a$, and $U(x) = U_0$ for $x < 0$ and $x > a$. It is evident in advance that the spectrum is discrete when $E < U_0$, whereas for $E > U_0$ there is a continuous spectrum of doubly degenerate levels.

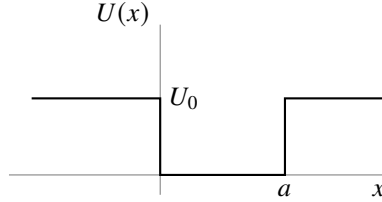


Fig. 1

In the region $0 < x < a$, the Schrödinger equation is

$$\psi'' + \frac{2m}{\hbar^2} E \psi = 0, \quad (22.1)$$

where a prime denotes differentiation with respect to x ; outside the well it is

$$\psi'' + \frac{2m}{\hbar^2} (E - U_0) \psi = 0. \quad (22.2)$$

At $x = 0, a$, the solutions of these equations must pass continuously into one another, with continuous first derivative; at $x = \pm\infty$, the solution of (22.2) must remain finite (and for the discrete spectrum, $E < U_0$, it must vanish).

For $E < U_0$, the solution of (22.2) that vanishes at infinity is $\psi = \text{const} \cdot e^{\mp\kappa x}$, where

$$\kappa = \frac{1}{\hbar} \sqrt{2m(U_0 - E)} \quad (22.3)$$

(the minus and plus signs in the exponent refer respectively to the regions $x > a$ and $x < 0$). The probability $|\psi|^2$ of finding the particle decays exponentially into the region where $E < U(x)$. Instead of requiring ψ and ψ' to be continuous at the boundary of the potential well, it is convenient to require continuity of ψ and of the logarithmic derivative ψ'/ψ . With (22.3), the boundary condition becomes

$$|\psi'|/\psi = \mp\kappa. \quad (22.4)$$

Determining the energy levels for a well of arbitrary depth U_0 is left aside here (see Problem 2); we analyze in full only the limiting case of infinitely high walls ($U_0 \rightarrow \infty$).

When $U_0 = \infty$, motion occurs only on the interval bounded by $x = 0, a$ and, as stated in § 18, the boundary condition at these points is

$$\psi = 0. \quad (22.5)$$

(This condition also follows readily from the general condition (22.4). Indeed, when $U_0 \rightarrow \infty$, we also have $\kappa \rightarrow \infty$, and hence $\psi'/\psi \rightarrow \infty$; since ψ' cannot become infinite, it follows that $\psi = 0$.) Seek a solution of (22.1) inside the well in the form

$$\psi = c \sin(kx + \delta), \quad k = \frac{\sqrt{2mE}}{\hbar}. \quad (22.6)$$

The condition $\psi = 0$ at $x = 0$ gives $\delta = 0$. The same condition at $x = a$ then gives $\sin ka = 0$, whence $ka = n\pi$ (n takes the positive integral values beginning with unity¹²), or

$$E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2, \quad n = 1, 2, 3, \dots \quad (22.7)$$

The energy levels of a particle inside the potential well are thus determined. After normalization, the stationary-state wave functions are

$$\psi_n = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi n}{a} x\right). \quad (22.8)$$

From these results one can write down directly the energy levels of a particle in a rectangular “potential box,” that is, for three-dimensional motion in a field whose potential energy is $U = 0$ for $0 < x < a$, $0 < y < b$, $0 < z < c$, and $U = \infty$ outside this region. Namely, the levels are the sums

$$E_{n_1 n_2 n_3} = \frac{\pi^2 \hbar^2}{2m} \left(\frac{n_1^2}{a^2} + \frac{n_2^2}{b^2} + \frac{n_3^2}{c^2} \right), \quad n_1, n_2, n_3 = 1, 2, 3, \dots, \quad (22.9)$$

and the corresponding wave functions are the products

$$\psi_{n_1 n_2 n_3} = \sqrt{\frac{8}{abc}} \sin\left(\frac{\pi n_1}{a} x\right) \sin\left(\frac{\pi n_2}{b} y\right) \sin\left(\frac{\pi n_3}{c} z\right). \quad (22.10)$$

Notice that, according to (22.7) or (22.9), the energy of the ground state is of order $E_0 \sim \hbar^2/ml^2$, where l denotes the linear dimensions of the region in which the particle moves. This result agrees with the uncertainty relations: for a coordinate uncertainty of order l , the momentum uncertainty, and with it the order of the momentum itself, is $\sim \hbar/l$; the corresponding energy is $\sim (\hbar/l)^2/m$.

PROBLEM

1. Determine the probability distribution of the possible momentum values for the normal state of a particle in an infinitely deep rectangular potential well.

SOLUTION. The coefficients $a(p)$ in the expansion of the function ψ_1 (22.8) in momentum eigenfunctions are

$$a(p) = \int \psi_p^* \psi_1 dx = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{\pi}{a} x\right) \exp\left(-i \frac{px}{\hbar}\right) dx.$$

¹²For $n = 0$, one would obtain identically $\psi = 0$.

Evaluating the integral and taking the square of its modulus gives the required probability distribution

$$|a(p)|^2 \frac{dp}{2\pi\hbar} = \frac{4\pi\hbar^3 a}{(p^2 a^2 - \pi^2 \hbar^2)^2} \cos^2\left(\frac{pa}{2\hbar}\right) dp.$$

PROBLEM

2. Determine the energy levels for the potential well shown in Fig. 2.

SOLUTION. The discrete part of the spectrum is the range $E < U_1$, which is the range we consider. In the region $x < 0$, the wave function is

$$\psi = c_1 e^{\kappa_1 x}, \quad \kappa_1 = \frac{1}{\hbar} \sqrt{2m(U_1 - E)},$$

and in the region $x > a$ it is

$$\psi = c_2 e^{-\kappa_2 x}, \quad \kappa_2 = \frac{1}{\hbar} \sqrt{2m(U_2 - E)}.$$

Inside the well ($0 < x < a$), seek ψ in the form

$$\psi = c \sin(kx + \delta), \quad k = \sqrt{2mE}/\hbar.$$

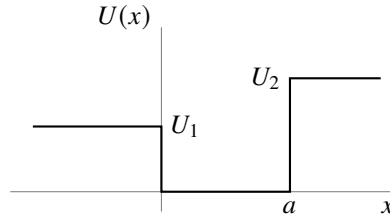


Fig. 2

Continuity of ψ'/ψ at the boundaries of the well gives

$$k \cot \delta = \kappa_1 = \sqrt{\frac{2m}{\hbar^2} U_1 - k^2},$$

$$k \cot(ak + \delta) = -\kappa_2 = -\sqrt{\frac{2m}{\hbar^2} U_2 - k^2},$$

or

$$\sin \delta = \frac{k\hbar}{\sqrt{2mU_1}}, \quad \sin(ka + \delta) = -\frac{k\hbar}{\sqrt{2mU_2}}.$$

Eliminating δ , we obtain the transcendental equation

$$ka = n\pi - \arcsin \frac{k\hbar}{\sqrt{2mU_1}} - \arcsin \frac{k\hbar}{\sqrt{2mU_2}} \quad (1)$$

(where $n = 1, 2, 3, \dots$, and the values of $\arcsin$ are taken between 0 and $\pi/2$), whose roots determine the energy levels $E = k^2 \hbar^2 / 2m$. In general there is one root for each n ; the values of n number the levels in increasing order.

Since the argument of arcsin cannot exceed unity, it is clear that the values of k can lie only in the interval from zero to $\sqrt{2mU_1}/\hbar$. The left-hand side of (1) is a monotonically increasing function of k , while the right-hand side is monotonically decreasing. Consequently, for a root of (1) to exist, its right-hand side at $k = \sqrt{2mU_1}/\hbar$ must be less than its left-hand side. In particular, the inequality

$$a \frac{\sqrt{2mU_1}}{\hbar} \geq \frac{\pi}{2} - \arcsin \sqrt{\frac{U_1}{U_2}}, \quad (2)$$

obtained for $n = 1$, is the condition for the well to contain at least one energy level. Thus, for given $U_1 \neq U_2$, there always exist sufficiently small values of the width a for which the well has no discrete energy level at all. When $U_1 = U_2$, condition (2) is plainly always satisfied.

When $U_1 = U_2 \equiv U_0$ (a symmetric well), equation (1) reduces to

$$\arcsin \frac{\hbar k}{\sqrt{2mU_0}} = \frac{n\pi - ka}{2}. \quad (3)$$

Introducing the variable $\xi = ka/2$, for odd n we obtain

$$\cos \xi = \pm \gamma \xi, \quad \gamma = \frac{\hbar}{a} \sqrt{\frac{2}{mU_0}}, \quad (4)$$

where the roots for which $\tan \xi > 0$ must be selected. For even n we obtain

$$\sin \xi = \pm \gamma \xi, \quad (5)$$

where the roots for which $\tan \xi < 0$ must be selected. The energy levels are found from the roots of these two equations as $E = 2\xi^2 \hbar^2 / ma^2$; for $\gamma \neq 0$ only finitely many levels exist.

Consider as a special case a shallow well, in which $U_0 \ll \hbar^2 / ma^2$, we have $\gamma \gg 1$, and equation (5) has no roots at all. Equation (4), however, has one root (with the upper sign on its right-hand side), equal to $\xi \approx (1/\gamma)(1 - 1/2\gamma^2)$. The well therefore contains only one energy level,

$$E_0 \approx U_0 - \frac{ma^2}{2\hbar^2} U_0^2,$$

lying close to its “top.”

PROBLEM

3. Determine the pressure exerted on the walls of a rectangular “potential box” by a particle inside it.

SOLUTION. On a wall normal to the x axis, the force is given by the mean of the derivative $-\partial H / \partial a$ of the particle’s Hamilton function with respect to the length of the box along the x axis; division of this force by the wall area bc gives the pressure. By (11.16), the required mean value is found by differentiating the energy eigenvalue (22.9). The resulting pressure is

$$p^{(x)} = \frac{\pi^2 \hbar^2}{ma^3 bc} n_1^2.$$

§ 23. The Linear Oscillator

Consider a particle executing small one-dimensional oscillations (the so-called *linear oscillator*). Its potential energy is $m\omega^2 x^2/2$, where in classical mechanics ω is the natural frequency of the oscillations. The oscillator Hamiltonian is consequently

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2 x^2}{2}. \quad (23.1)$$

Since the potential energy tends to infinity at $x = \pm\infty$, the particle can execute only finite motion. Accordingly, the entire energy spectrum of the oscillator is discrete.

Let us determine the oscillator's energy levels by the matrix method¹³. We begin with the equations of motion in the form (19.3), which in this case give

$$\hat{x} + \omega^2 x = 0. \quad (23.2)$$

In matrix form this equation is

$$(\ddot{x})_{mn} + \omega^2 x_{mn} = 0.$$

For the matrix elements of the acceleration, (11.8) gives $(\ddot{x})_{mn} = i\omega_{mn}(\dot{x})_{mn} = -\omega_{mn}^2 x_{mn}$. Hence

$$(\omega_{mn}^2 - \omega^2)x_{mn} = 0.$$

Thus all the matrix elements x_{mn} vanish except those for which $\omega_{mn} = \pm\omega$. Number all stationary states so that the frequencies $\pm\omega$ correspond to the transitions $n \rightarrow n \mp 1$, that is, $\omega_{n,n\mp 1} = \pm\omega$. The only nonzero matrix elements are then $x_{n,n\mp 1}$.

We shall suppose that the wave functions ψ_n have been chosen real. Since x is a real quantity, every matrix element x_{mn} is then real as well. The Hermiticity condition (11.10) now makes the matrix x_{mn} symmetric:

$$x_{mn} = x_{nm}.$$

To calculate the nonzero coordinate matrix elements, use the commutation rule

$$\hat{x} \hat{p} - \hat{p} \hat{x} = -i\frac{\hbar}{m},$$

written in matrix form as

$$(\dot{x}x)_{mn} - (x\dot{x})_{mn} = -\frac{i\hbar}{m}\delta_{mn}.$$

Using the matrix-multiplication rule (11.12), for $m = n$ this gives

$$i \sum_l (\omega_{nl} x_{nl} x_{ln} - x_{nl} \omega_{ln} x_{ln}) = 2i \sum_l \omega_{nl} x_{nl}^2 = -\frac{i\hbar}{m}.$$

Only the terms with $l = n \pm 1$ are nonzero in this sum, and therefore

$$(x_{n+1,n})^2 - (x_{n,n-1})^2 = \frac{\hbar}{2m\omega}. \quad (23.3)$$

¹³This was done by Heisenberg (1925), before Schrödinger's discovery of the wave equation.

This equality shows that the quantities $(x_{n+1,n})^2$ form an arithmetic progression, unbounded above but necessarily bounded below, since it can contain only positive terms. Thus far we have established only the relative ordering of the state labels n , not their absolute values; we may therefore choose arbitrarily the value of n assigned to the oscillator's first—the normal—state. Set it equal to zero. Correspondingly, $x_{0,-1}$ must be regarded as identically zero, and successive application of (23.3) for $n = 0, 1, \dots$ gives

$$(x_{n,n-1})^2 = \frac{n\hbar}{2m\omega}.$$

The final expression for the nonzero coordinate matrix elements is therefore¹⁴

$$x_{n,n-1} = x_{n-1,n} = \sqrt{\frac{n\hbar}{2m\omega}}. \quad (23.4)$$

The matrix of $\hat{H}$ is diagonal, and its matrix elements H_{nn} are the required oscillator energy eigenvalues E_n . To calculate them, write

$$\begin{aligned} H_{nn} = E_n &= \frac{m}{2} [(\dot{x}^2)_{nn} + \omega^2 (x^2)_{nn}] \\ &= \frac{m}{2} \left[\sum_l i\omega_{nl} x_{nl} i\omega_{ln} x_{ln} + \omega^2 \sum_l x_{nl} x_{ln} \right] = \frac{m}{2} \sum_l (\omega^2 + \omega_{nl}^2) x_{ln}^2. \end{aligned}$$

Only the terms $l = n \pm 1$ are nonzero in the sum. Substitution of (23.4) gives

$$E_n = (n + 1/2)\hbar\omega, \quad n = 0, 1, 2, \dots \quad (23.5)$$

Thus the oscillator energy levels are separated by equal intervals $\hbar\omega$. The energy of the normal state ($n = 0$) is $\hbar\omega/2$; we emphasize that it is nonzero.

The result (23.5) may also be obtained by solving the Schrödinger equation. For the oscillator this equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} \left(E - \frac{m\omega^2 x^2}{2} \right) \psi = 0. \quad (23.6)$$

It is convenient here to replace x by a dimensionless variable ξ , defined by

$$\xi = \sqrt{\frac{m\omega}{\hbar}} x. \quad (23.7)$$

We then obtain

$$\psi'' + \left(\frac{2E}{\hbar\omega} - \xi^2 \right) \psi = 0. \quad (23.8)$$

(A prime here denotes differentiation with respect to ξ .)

For large ξ , the term $2E/\hbar\omega$ may be omitted in comparison with ξ^2 ; the equation $\psi'' = \xi^2\psi$ has the asymptotic integrals $\psi = e^{\pm\xi^2/2}$. (Indeed, differentiation of this

¹⁴We choose the undetermined phases α_n (see the note on p. 52) so that every matrix element in (23.4) has a plus sign before the root. A phase convention of this kind can always be arranged when a matrix has nonzero entries solely for transitions connecting states whose labels are adjacent.

function gives $\psi'' = \xi^2 \psi$ when lower-order terms in ξ are neglected.) Since the wave function ψ must remain finite at $\xi = \pm\infty$, the minus sign must be chosen in the exponent. It is therefore natural to make the substitution

$$\psi = e^{-\xi^2/2} \chi(\xi) \quad (23.9)$$

in (23.8). For $\chi(\xi)$ we obtain the equation (introducing the notation $2E/\hbar\omega - 1 = 2n$; since we know in advance that $E > 0$, we have $n > -1/2$)

$$\chi'' - 2\xi\chi' + 2n\chi = 0, \quad (23.10)$$

where χ must be finite for every finite ξ and, at $\xi = \pm\infty$, may become infinite no faster than a finite power of ξ (so that ψ tends to zero).

Such solutions of (23.10) exist only for integer positive values of n , including zero (see § a of the mathematical supplements); this gives the energy eigenvalues (23.5), already known to us. For the different integral values of n , the corresponding solutions of (23.10) are

$$\chi = \text{const} \cdot H_n(\xi),$$

where $H_n(\xi)$ are the so-called Hermite polynomials, polynomials of degree n in ξ defined by

$$H_n(\xi) = (-1)^n e^{\xi^2} \frac{d^n e^{-\xi^2}}{d\xi^n}. \quad (23.11)$$

Choosing the constant so that ψ_n obeys the normalization condition

$$\int_{-\infty}^{+\infty} \psi_n^2(x) dx = 1,$$

we obtain (see (a.7))

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} \exp\left(-\frac{m\omega}{2\hbar}x^2\right) H_n\left(x\sqrt{\frac{m\omega}{\hbar}}\right). \quad (23.12)$$

Thus, the normal-state wave function is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}x^2\right). \quad (23.13)$$

As expected, it has no zeros at finite x .

Evaluation of the integrals $\int_{-\infty}^{+\infty} \psi_n \psi_m \xi d\xi$ determines the coordinate matrix elements; the calculation naturally gives the same values (23.4).

We conclude by demonstrating how the wave functions ψ_n can be computed via the matrix method. In the matrices of the operators $\hat{x} \pm i\omega\hat{x}$, the only nonzero elements are

$$(\hat{x} - i\omega\hat{x})_{n-1,n} = -(\hat{x} + i\omega\hat{x})_{n,n-1} = -i\sqrt{\frac{2\omega\hbar n}{m}}. \quad (23.14)$$

Using the general formula (11.11) and recalling that $\psi_{-1} \equiv 0$, we obtain

$$(\hat{x} - i\omega\hat{x})\psi_0 = 0.$$

Substitution of $\hat{x} = -i(\hbar/m)d/dx$ then gives

$$\frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar}x\psi_0,$$

whose normalized solution is (23.13). Further, since

$$(\hat{x} + i\omega\hat{x})\psi_{n-1} = (\hat{x} + i\omega x)_{n,n-1}\psi_n = i\sqrt{\frac{2\omega\hbar n}{m}}\psi_n,$$

we obtain the recurrence formula

$$\begin{aligned}\psi_n &= \sqrt{\frac{m}{2\omega\hbar n}} \left(-\frac{\hbar}{m} \frac{d}{dx} + \omega x \right) \psi_{n-1} = \frac{1}{\sqrt{2n}} \left(-\frac{d}{d\xi} + \xi \right) \psi_{n-1} \\ &= -\frac{1}{\sqrt{2n}} \exp \frac{\xi^2}{2} \frac{d}{d\xi} \left(\exp \left(-\frac{\xi^2}{2} \right) \psi_{n-1} \right),\end{aligned}$$

whose n -fold application to (23.13) gives (23.12) for the normalized functions ψ_n .

PROBLEM

1. Determine the probability distribution of the possible momentum values for the oscillator.

SOLUTION. Rather than expanding a stationary-state wave function in momentum eigenfunctions, for the oscillator it is simpler to begin directly with the Schrödinger equation in the momentum representation. Substituting the coordinate operator (15.12), $\hat{x} = i\hbar d/dp$, into (23.1), we obtain the Hamiltonian in the momentum representation

$$\hat{H} = \frac{p^2}{2m} - \frac{m\omega^2\hbar^2}{2} \frac{d^2}{dp^2}.$$

The corresponding Schrödinger equation $\hat{H}a(p) = Ea(p)$ for the momentum-representation wave function $a(p)$ is

$$\frac{d^2a(p)}{dp^2} + \frac{2}{m\omega^2\hbar^2} \left(E - \frac{p^2}{2m} \right) a(p) = 0.$$

This equation has exactly the same form as (23.6), so its solutions can be written down directly by analogy with (23.12). The required probability distribution is therefore

$$|a_n(p)|^2 \frac{dp}{2\pi\hbar} = \frac{1}{2^n n! \sqrt{\pi m \omega \hbar}} \exp \left(-\frac{p^2}{m\omega\hbar} \right) H_n^2 \left(\frac{p}{\sqrt{m\omega\hbar}} \right) dp.$$

PROBLEM

2. Use the uncertainty relation (16.7) to determine a lower bound on the possible energy values of the oscillator.

SOLUTION. Noting that $\overline{x^2} = \bar{x}^2 + (\delta x)^2$ and $\overline{p^2} = \bar{p}^2 + (\delta p)^2$, and using (16.7), for the oscillator's mean energy we have

$$\bar{E} = \frac{m\omega^2}{2}\overline{x^2} + \frac{\overline{p^2}}{2m} \geq \frac{m\omega^2}{2}(\delta x)^2 + \frac{1}{2m}(\delta p)^2 \geq \frac{m\omega^2\hbar^2}{8(\delta p)^2} + \frac{(\delta p)^2}{2m}.$$

Finding the minimum of this expression as a function of δp , we obtain the lower bound for the mean values, and hence for all possible energy values: $E \geq \hbar\omega/2$.

PROBLEM

3. Find the wave functions of linear-oscillator states that minimize the uncertainty relation, that is, states in which the root-mean-square fluctuations of coordinate and momentum in the wave packet are related by $\delta p \delta x = \hbar/2$ (E. Schrödinger, 1926)¹⁵.

SOLUTION. The required wave functions must have the form

$$\Psi(x, t) = \frac{1}{(2\pi)^{1/4}(\delta x)^{1/2}} \exp \left\{ \frac{i\bar{p}x}{\hbar} - \frac{(x - \bar{x})^2}{4(\delta x)^2} - i\varphi(t) \right\}. \quad (1)$$

At every instant their coordinate dependence corresponds to (16.8), where $\bar{x} = \bar{x}(t)$ and $\bar{p} = \bar{p}(t) = m\dot{\bar{x}}(t)$ are the mean coordinate and momentum. By (19.3), for the linear oscillator ($U = m\omega^2 x^2/2$) we have $\dot{p} = -m\omega^2 x$, and hence for the mean values $\dot{\bar{p}} = -m\omega^2 \bar{x}$, or

$$\ddot{\bar{x}} + \omega^2 \bar{x} = 0, \quad (2)$$

that is, $\bar{x}(t)$ satisfies the classical equation of motion. The constant factor in (1) is fixed by the normalization condition $\int_{-\infty}^{\infty} |\Psi|^2 dx = 1$; besides this factor, Ψ may contain a phase factor with a time-dependent phase $\varphi(t)$. To find the unknown δx and the function $\varphi(t)$, we substitute (1) into the wave equation

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + \frac{m\omega^2 x^2}{2} \Psi = i\hbar \frac{\partial \Psi}{\partial t}.$$

Taking account of (2), the substitution gives

$$\left(\frac{x^2}{2} - x\bar{x} \right) \left(\frac{m^2\omega^2}{\hbar^2} - \frac{1}{4(\delta x)^4} \right) + \left[\frac{m^2\dot{\bar{x}}^2}{2\hbar^2} - \frac{\bar{x}^2}{8(\delta x)^4} + \frac{1}{4(\delta x)^2} - \frac{m}{\hbar} \dot{\varphi}(t) \right] = 0.$$

It follows that $(\delta x)^2 = \hbar/(2m\omega)$ and then

$$\dot{\varphi} = \frac{m}{2\hbar} (\dot{\bar{x}}^2 - \omega^2 \bar{x}^2) + \frac{\omega}{2}, \quad \varphi = \frac{1}{2\hbar} \bar{p}\bar{x} + \frac{\omega}{2}t.$$

¹⁵These states are called *coherent*.

Thus, finally,

$$\Psi(x, t) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left\{\frac{i\bar{p}x}{\hbar} - \frac{m\omega(x - \bar{x})^2}{2\hbar}\right\} \exp\left\{-\frac{i\omega t}{2} - \frac{i\bar{p}\bar{x}}{2\hbar}\right\}. \quad (3)$$

For $\bar{x} = 0$, $\bar{p} = 0$, this function becomes $\psi_0(x)e^{-i\omega t/2}$, the wave function of the oscillator ground state.

The oscillator's mean energy in a coherent state is

$$\bar{E} = \frac{\bar{p}^2}{2m} + \frac{m\omega^2\bar{x}^2}{2} = \frac{\bar{p}^2}{2m} + \frac{m\omega^2\bar{x}^2}{2} + \frac{\hbar\omega}{2} = \hbar\omega\left(\bar{n} + \frac{1}{2}\right); \quad (4)$$

the quantity $\bar{n}$ introduced here is the mean “number of quanta” $\hbar\omega$ in the state. We see that a coherent state is specified completely by giving a dependence $\bar{x}(t)$ that satisfies the classical equation (2). Its general form can be written

$$\frac{m\omega\bar{x} + i\bar{p}}{\sqrt{2m\hbar\omega}} = ae^{-i\omega t}, \quad |a|^2 = \bar{n}. \quad (5)$$

The function (3) may be expanded in the wave functions of the oscillator's stationary states:

$$\Psi = \sum_{n=0}^{\infty} a_n \Psi_n, \quad \Psi_n(x, t) = \psi_n(x) \exp\left\{-i\left(n + \frac{1}{2}\right)\omega t\right\}.$$

The coefficients of this expansion are¹⁶

$$a_n = \int_{-\infty}^{\infty} \Psi_n^* \Psi dx. \quad (6)$$

It follows that the probability for the oscillator to be in its n th state is

$$w_n = |a_n|^2 = e^{-\bar{n}} \frac{\bar{n}^n}{n!}, \quad (7)$$

that is, it is given by the familiar Poisson distribution.

PROBLEM

4. Determine the energy levels of a particle moving in a field with the potential energy shown in Fig. 3 (Ph. Morse, 1929).

SOLUTION. The spectrum of positive energy eigenvalues is continuous (and the levels are nondegenerate), while the spectrum of negative values is discrete.

The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - Ae^{-2\alpha x} + 2Ae^{-\alpha x})\psi = 0.$$

Introduce the new variable

$$\xi = \frac{2\sqrt{2mA}}{\alpha\hbar} e^{-\alpha x}$$

¹⁶Compare the calculation in Problem 1 of § 41.

(which ranges from zero to $+\infty$), and the notation (we consider the discrete spectrum, so $E < 0$)

$$s = \frac{\sqrt{-2mE}}{\alpha\hbar}, \quad n = \frac{\sqrt{2mA}}{\alpha\hbar} - \left(s + \frac{1}{2}\right). \quad (1)$$

The Schrödinger equation then becomes

$$\psi'' + \frac{1}{\xi}\psi' + \left(-\frac{1}{4} + \frac{n+s+1/2}{\xi} - \frac{s^2}{\xi^2}\right)\psi = 0.$$

As $\xi \rightarrow \infty$, ψ behaves asymptotically as $e^{\pm\xi/2}$, while as $\xi \rightarrow 0$ it is proportional to $\xi^{\pm s}$. Finiteness requires the solution behaving as $e^{-\xi/2}$ at $\xi \rightarrow \infty$ and as ξ^s at $\xi \rightarrow 0$.

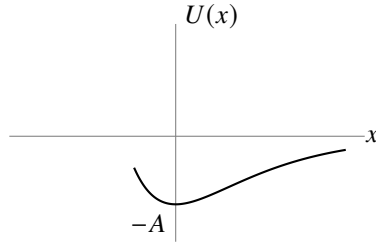


Fig. 3

Make the substitution

$$\psi = e^{-\xi/2}\xi^s w(\xi),$$

which gives for w the equation

$$\xi w'' + (2s+1-\xi)w' + nw = 0, \quad (2)$$

to be solved subject to the conditions that w is finite at $\xi = 0$, and that as $\xi \rightarrow \infty$ it becomes infinite no faster than a finite power of ξ . Equation (2) is the confluent hypergeometric equation (see § d of the mathematical supplements),

$$w = F(-n, 2s+1, \xi).$$

The solution satisfying the required condition is obtained for a nonnegative integer n (and F then reduces to a polynomial). From the definitions (1), the energy levels are consequently

$$-E_n = A \left[1 - \frac{\alpha\hbar}{\sqrt{2mA}} \left(n + \frac{1}{2} \right) \right]^2,$$

where n takes integer positive values beginning at zero and ending at the greatest value for which

$$\frac{\sqrt{2mA}}{\alpha\hbar} > n + \frac{1}{2}$$

(so that the parameter s is positive, in agreement with its definition). Thus the discrete spectrum contains a finite sequence of levels. If

$$\frac{\sqrt{2mA}}{\alpha\hbar} < \frac{1}{2},$$

there is no discrete spectrum at all.

PROBLEM

5. The same problem for $U = -U_0/\cosh^2 \alpha x$ (Fig. 4).

SOLUTION. The positive-energy spectrum is continuous and the negative-energy spectrum is discrete; we consider the latter. The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} \left(E + \frac{U_0}{\cosh^2 \alpha x} \right) \psi = 0.$$

Make the change of variable $\xi = \tanh \alpha x$ and introduce the notation

$$\varepsilon = \frac{\sqrt{-2mE}}{\hbar\alpha}, \quad \frac{2mU_0}{\alpha^2\hbar^2} = s(s+1), \quad s = \frac{1}{2} \left(-1 + \sqrt{1 + \frac{8mU_0}{\alpha^2\hbar^2}} \right).$$

We then obtain

$$\frac{d}{d\xi} \left[(1 - \xi^2) \frac{d\psi}{d\xi} \right] + \left[s(s+1) - \frac{\varepsilon^2}{1 - \xi^2} \right] \psi = 0.$$

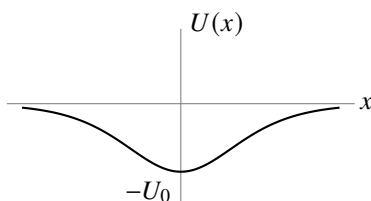


Fig. 4

This is the equation of the generalized Legendre functions. Reduce it to hypergeometric form by the substitution

$$\psi = (1 - \xi^2)^{\varepsilon/2} w(\xi)$$

and the temporary change of variable $\frac{1}{2}(1 - \xi) = u$:

$$u(1-u)w'' + (\varepsilon+1)(1-2u)w' - (\varepsilon-s)(\varepsilon+s+1)w = 0.$$

The solution finite at $\xi = 1$ (that is, at $x = \infty$) is

$$\psi = (1 - \xi^2)^{\varepsilon/2} F[\varepsilon - s, \varepsilon + s + 1, \varepsilon + 1, (1 - \xi)/2].$$

For ψ to remain finite also at $\xi = -1$ (that is, at $x = -\infty$), we must have $\varepsilon - s = -n$, where $n = 0, 1, 2, \dots$ (then F is a polynomial of degree n , finite at $\xi = -1$).

Thus the energy levels are fixed by $s - \varepsilon = n$, which gives

$$E_n = -\frac{\hbar^2\alpha^2}{8m} \left[-(1+2n) + \sqrt{1 + \frac{8mU_0}{\alpha^2\hbar^2}} \right]^2.$$

There is a finite number of levels, determined by the condition $\varepsilon > 0$, that is, $n < s$.

§ 24. Motion in a Uniform Field

Consider the motion of a particle in a uniform external field. Choose the direction of the field as the x axis, and let F be the force exerted by the field on the particle; in an electric field of strength E , this force is $F = eE$, where e is the particle's charge.

The potential energy of the particle in a uniform field has the form $U = -Fx + \text{const}$. Choosing the constant so that $U = 0$ at $x = 0$, we have $U = -Fx$. The Schrödinger equation for the problem is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E + Fx)\psi = 0. \quad (24.1)$$

Since U tends to $+\infty$ as $x \rightarrow -\infty$ and to $-\infty$ as $x \rightarrow \infty$, it is evident in advance that the energy levels form a continuous spectrum filling the entire interval from $-\infty$ to $+\infty$. All these eigenvalues are nondegenerate and correspond to motion that is finite on the side $x = -\infty$ and infinite in the direction $x \rightarrow +\infty$.

In place of the coordinate x , introduce the dimensionless variable

$$\xi = \left(x + \frac{E}{F}\right) \left(\frac{2mF}{\hbar^2}\right)^{1/3}. \quad (24.2)$$

Equation (24.1) then becomes

$$\psi'' + \xi\psi = 0. \quad (24.3)$$

This equation contains no energy parameter at all. Consequently, once a solution satisfying the required finiteness conditions has been found, it provides an eigenfunction for every value of the energy.

The solution of (24.3) that is finite for all x is (see § b of the mathematical supplements)

$$\psi(\xi) = A\Phi(-\xi), \quad (24.4)$$

where

$$\Phi(\xi) = \frac{1}{\sqrt{\pi}} \int_0^\infty \cos\left(\frac{u^3}{3} + u\xi\right) du$$

is the so-called Airy function, and A is a normalization factor to be determined below.

As $\xi \rightarrow -\infty$, the function $\psi(\xi)$ tends exponentially to zero. The asymptotic expression determining $\psi(\xi)$ for negative ξ of large absolute magnitude is (see (b.4))

$$\psi(\xi) \approx \frac{A}{2|\xi|^{1/4}} \exp\left(-\frac{2}{3}|\xi|^{3/2}\right). \quad (24.5)$$

For large positive ξ , the asymptotic expression for $\psi(\xi)$ is instead (see (b.5)¹⁷):

$$\psi(\xi) = \frac{A}{\xi^{1/4}} \sin\left(\frac{2}{3}\xi^{3/2} + \frac{\pi}{4}\right). \quad (24.6)$$

¹⁷We note in anticipation that the asymptotic expressions (24.5) and (24.6) correspond precisely to the quasiclassical forms of the wave function in classically forbidden and allowed regions, respectively (§ 47).

Following the general rule (5.4) for normalizing eigenfunctions of a continuous spectrum, put the functions (24.4) into a form normalized to a delta function of the energy:

$$\int_{-\infty}^{+\infty} \psi(\xi) \psi(\xi') dx = \delta(E' - E). \quad (24.7)$$

In § 21 a simple method was given for determining the normalization coefficient from the asymptotic form of the wave functions. Following that method, write (24.6) as a sum of two traveling waves:

$$\psi(\xi) \approx \frac{A}{2\xi^{1/4}} \left\{ \exp \left[i \left(\frac{2}{3} \xi^{3/2} - \frac{\pi}{4} \right) \right] + \exp \left[-i \left(\frac{2}{3} \xi^{3/2} - \frac{\pi}{4} \right) \right] \right\}.$$

The flux density calculated for each of these two terms is

$$v \left(\frac{A}{2\xi^{1/4}} \right)^2 = \sqrt{\frac{2}{m} (E + Fx)} \left(\frac{A}{2\xi^{1/4}} \right)^2 = A^2 \frac{(2\hbar F)^{1/3}}{4m^{2/3}}.$$

Equating it to $1/(2\pi\hbar)$, we find

$$A = \frac{(2m)^{1/3}}{\pi^{1/2} F^{1/6} \hbar^{2/3}}. \quad (24.8)$$

PROBLEM

Determine the momentum-representation wave functions for a particle in a uniform field.

SOLUTION. The Hamiltonian in the momentum representation is

$$\hat{H} = \frac{p^2}{2m} - i\hbar F \frac{d}{dp},$$

so the Schrödinger equation for the wave function $a(p)$ is

$$-i\hbar F \frac{da}{dp} + \left(\frac{p^2}{2m} - E \right) a = 0.$$

Solving this equation gives the required functions

$$a_E(p) = \frac{1}{\sqrt{2\pi\hbar F}} \exp \left\{ \frac{i}{\hbar F} \left(Ep - \frac{p^3}{6m} \right) \right\}.$$

These functions are normalized by the condition

$$\int_{-\infty}^{+\infty} a_E^*(p) a_{E'}(p) dp = \delta(E' - E).$$

§ 25. Transmission Coefficient

We consider particle motion in a field of the type depicted in Fig. 5: $U(x)$ increases monotonically from one constant limiting value ($U = 0$ as $x \rightarrow -\infty$) to another ($U = U_0$ as $x \rightarrow +\infty$). According to classical mechanics, a particle with energy $E < U_0$ moving in such a field from left to right arrives at the *potential wall*, reflects off it, and proceeds in the reverse direction; for $E > U_0$, however, the particle keeps traveling in its original direction with a reduced velocity. Quantum mechanics introduces a new phenomenon: even when $E > U_0$, the particle may be reflected from the potential wall. In principle, the reflection probability is to be calculated as follows.

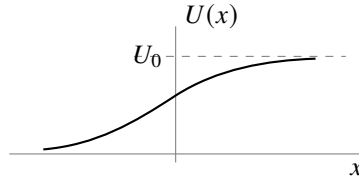


Figure 5

Let the particle move from left to right. At large positive values of x , the wave function must describe a particle that has passed “over the wall” and is moving in the positive direction of the x axis; that is, it must have the asymptotic form

$$\text{as } x \rightarrow \infty : \quad \psi \approx A e^{ik_2 x}, \quad k_2 = \frac{1}{\hbar} \sqrt{2m(E - U_0)} \quad (25.1)$$

(A is a constant). Having found a solution of the Schrödinger equation that satisfies this limiting condition, we calculate its asymptotic expression as $x \rightarrow -\infty$. This is a linear combination of two solutions of the free-motion equation and therefore has the form

$$\text{as } x \rightarrow -\infty : \quad \psi \approx e^{ik_1 x} + B e^{-ik_1 x}, \quad k_1 = \frac{1}{\hbar} \sqrt{2mE}. \quad (25.2)$$

The first term describes the particle impinging on the wall (ψ is taken normalized so that this term has unit coefficient); the second term depicts the particle reflected from the wall. The flux density is proportional to k_1 in the incident wave, to $k_1|B|^2$ in the reflected wave, and to $k_2|A|^2$ in the transmitted wave. Define the particle’s *transmission coefficient* D as the ratio of the flux density in the transmitted wave to that in the incident wave:

$$D = \frac{k_2}{k_1} |A|^2. \quad (25.3)$$

Similarly, the *reflection coefficient* R may be defined as the ratio of the reflected flux density to the incident flux density; evidently $R = 1 - D$:

$$R = |B|^2 = 1 - \frac{k_2}{k_1} |A|^2 \quad (25.4)$$

(this relation between A and B follows automatically from the constancy of the flux along the x axis).

If a particle moves from left to right with energy $E < U_0$, then k_2 is purely imaginary and the wave function decreases exponentially as $x \rightarrow +\infty$. The reflected flux equals the incident flux; that is, the particle is totally reflected from the potential wall. We stress, however, that the probability for the particle to be located in the region $E < U$ remains nonzero even in this case, though it decreases rapidly with growing x .

For the general case of an arbitrary stationary state (energy $E > U_0$), the asymptotic form of the wave function both as $x \rightarrow -\infty$ and as $x \rightarrow +\infty$ is a sum of two waves propagating in the two directions along the x axis:

$$\begin{aligned}\psi &= A_1 e^{ik_1 x} + B_1 e^{-ik_1 x} & \text{as } x \rightarrow -\infty, \\ \psi &= A_2 e^{ik_2 x} + B_2 e^{-ik_2 x} & \text{as } x \rightarrow +\infty.\end{aligned}\tag{25.5}$$

Since both expressions are asymptotic forms of one and the same solution of a linear differential equation, the coefficients A_1 , B_1 and A_2 , B_2 are linearly related. Let $A_2 = \alpha A_1 + \beta B_1$, where α and β are constants (in general complex) that depend on the particular form of the field $U(x)$. The analogous relation for B_2 may then be written by using the fact that the Schrödinger equation is real. By this fact, if ψ is a solution of the given Schrödinger equation, the complex-conjugate function ψ^* is also a solution of the same equation.

The asymptotic form

$$\begin{aligned}\psi^* &= A_1^* e^{-ik_1 x} + B_1^* e^{ik_1 x} & \text{as } x \rightarrow -\infty, \\ \psi^* &= A_2^* e^{-ik_2 x} + B_2^* e^{ik_2 x} & \text{as } x \rightarrow +\infty\end{aligned}$$

differs from (25.5) only in the notation of the constant coefficients; hence $B_2^* = \alpha B_1^* + \beta A_1^*$, or $B_2 = \alpha^* B_1 + \beta^* A_1$. Thus the coefficients in (25.5) are related by equations of the form

$$A_2 = \alpha A_1 + \beta B_1, \quad B_2 = \beta^* A_1 + \alpha^* B_1.\tag{25.6}$$

The condition that the flux be constant along the x axis gives the following relation for the coefficients in (25.5):

$$k_1(|A_1|^2 - |B_1|^2) = k_2(|A_2|^2 - |B_2|^2).$$

On expressing A_2 and B_2 here in terms of A_1 and B_1 according to (25.6), we obtain

$$|\alpha|^2 - |\beta|^2 = \frac{k_1}{k_2}.\tag{25.7}$$

The relations (25.6) can be used to show that the reflection coefficients are identical (at a given energy $E > U_0$) for particles propagating in the positive or negative direction along the x axis. Indeed, the first case is obtained by setting $B_2 = 0$ in the functions (25.5); then $B_1/A_1 = -\beta^*/\alpha^*$. In the second case we put $A_1 = 0$, whereupon $A_2/B_2 = \beta/\alpha^*$. The corresponding reflection coefficients are

$$R_1 = \left| \frac{B_1}{A_1} \right|^2 = \left| \frac{\beta^*}{\alpha^*} \right|^2, \quad R_2 = \left| \frac{A_2}{B_2} \right|^2 = \left| \frac{\beta}{\alpha^*} \right|^2,$$

and it is clear that $R_1 = R_2$.

The quantities $B_1/A_1 = -\beta^*/\alpha^*$ and $A_2/B_2 = \beta/\alpha^*$ are naturally called the *reflection amplitudes* for motion in the positive and negative directions, respectively. These amplitudes have equal moduli but may differ by a phase factor.

PROBLEM

1. For a particle with energy $E > U_0$, calculate the coefficient of reflection from a rectangular potential wall (Fig. 6).

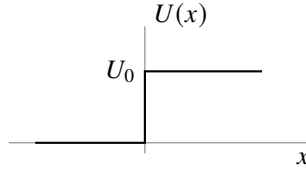


Figure 6

SOLUTION. Throughout the region $x > 0$ the wave function has the form (25.1), and in the region $x < 0$ it has the form (25.2). The constants A and B are determined by the conditions that ψ and $d\psi/dx$ be continuous at $x = 0$:

$$1 + B = A, \quad k_1(1 - B) = k_2A,$$

whence

$$A = \frac{2k_1}{k_1 + k_2}, \quad B = \frac{k_1 - k_2}{k_1 + k_2}.$$

The reflection coefficient is (25.4) ¹⁸

$$R = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2 = \left(\frac{p_1 - p_2}{p_1 + p_2} \right)^2.$$

At $E = U_0$ ($k_2 = 0$), R becomes unity; as $E \rightarrow \infty$, it tends to zero as $R = (U_0/4E)^2$.

PROBLEM

2. Determine the transmission coefficient of a particle through a rectangular potential barrier (Fig. 7).

SOLUTION. Let $E > U_0$, and let the incident particle move from left to right. The wave function then has the following forms in the different regions:

$$\begin{aligned} \psi &= e^{ik_1x} + Ae^{-ik_1x} && \text{for } x < 0, \\ \psi &= Be^{ik_2x} + B'e^{-ik_2x} && \text{for } 0 < x < a, \\ \psi &= Ce^{ik_1x} && \text{for } x > a. \end{aligned}$$

(On the side $x > a$ there must be only the transmitted wave, propagating in the positive direction of the x axis.) The constants A , B , B' , and C are determined by the conditions

¹⁸In the limiting case of classical mechanics the reflection coefficient must go to zero. However, the resulting expression does not contain a quantum constant at all. This seeming contradiction is resolved as follows. The classical limit corresponds to the situation where the particle's de Broglie wavelength $\lambda \sim \hbar/p$ is small in comparison with the characteristic dimensions of the problem, that is, with the distances over which the field $U(x)$ changes appreciably. In the schematic example under consideration this distance is zero (at $x = 0$), so that the limiting transition cannot be made.

that ψ and $d\psi/dx$ be continuous at the points $x = 0, a$. The transmission coefficient is $D = k_1|C|^2/k_1 = |C|^2$. Calculation gives

$$D = \frac{4k_1^2 k_2^2}{(k_1^2 - k_2^2)^2 \sin^2 ak_2 + 4k_1^2 k_2^2}.$$

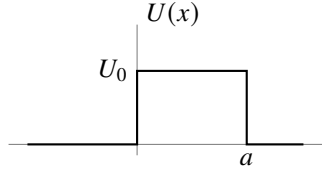


Figure 7

When $E < U_0$, k_2 is purely imaginary; the corresponding expression for D is obtained by replacing k_2 by $i\kappa_2$, where $\hbar\kappa_2 = \sqrt{2m(U_0 - E)}$:

$$D = \frac{4k_1^2 \kappa_2^2}{(k_1^2 + \kappa_2^2)^2 \sinh^2(a\kappa_2) + 4k_1^2 \kappa_2^2}.$$

PROBLEM

3. Determine the reflection coefficient of a particle from the potential wall defined by

$$U(x) = U_0/(1 + e^{-\alpha x})$$

(see Fig. 5); the particle energy is $E > U_0$.

SOLUTION. The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} \left(E - \frac{U_0}{1 + e^{-\alpha x}} \right) \psi = 0.$$

We must find the solution that, as $x \rightarrow +\infty$, has the form

$$\psi = \text{const} \cdot e^{ik_2 x}.$$

Introduce the new variable

$$\xi = -e^{-\alpha x}$$

(which ranges from $-\infty$ to 0), and seek a solution in the form

$$\psi = \xi^{-ik_2/\alpha} w(\xi),$$

where $w(\xi)$ tends to a constant as $\xi \rightarrow 0$ (that is, as $x \rightarrow \infty$). For $w(\xi)$ we obtain an equation of hypergeometric type,

$$\xi(1-\xi)w'' + \left(1 - \frac{2i}{\alpha}k_2\right)(1-\xi)w' + \frac{1}{\alpha^2}(k_2^2 - k_1^2)w = 0,$$

whose solution is the hypergeometric function

$$w = F \left[-\frac{i}{\alpha}(k_1 - k_2), -\frac{i}{\alpha}(k_1 + k_2), -\frac{2i}{\alpha}k_2 + 1, \xi \right]$$

(a constant factor is omitted). As $\xi \rightarrow 0$, this function tends to 1 and thus satisfies the stated condition.

The asymptotic form of ψ as $\xi \rightarrow -\infty$ (that is, as $x \rightarrow -\infty$) is ¹⁹

$$\begin{aligned} \psi &\approx \xi^{-ik_2/\alpha} \left[C_1(-\xi)^{i(k_2-k_1)/\alpha} + C_2(-\xi)^{i(k_1+k_2)/\alpha} \right] \\ &= (-1)^{ik_2/\alpha} \left[C_1 e^{ik_1 x} + C_2 e^{-ik_1 x} \right], \end{aligned}$$

where

$$\begin{aligned} C_1 &= \frac{\Gamma(-(2i/\alpha)k_1)\Gamma(-(2i/\alpha)k_2 + 1)}{\Gamma(-(i/\alpha)(k_1 + k_2))\Gamma(-(i/\alpha)(k_1 + k_2) + 1)}, \\ C_2 &= \frac{\Gamma((2i/\alpha)k_1)\Gamma(-(2i/\alpha)k_2 + 1)}{\Gamma((i/\alpha)(k_1 - k_2))\Gamma((i/\alpha)(k_1 - k_2) + 1)}. \end{aligned}$$

The required reflection coefficient is $R = |C_2/C_1|^2$; calculation using the familiar formula

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x}$$

gives

$$R = \left\{ \frac{\sinh[(\pi/\alpha)(k_1 - k_2)]}{\sinh[(\pi/\alpha)(k_1 + k_2)]} \right\}^2.$$

At $E = U_0$ ($k_2 = 0$), R becomes unity, while as $E \rightarrow \infty$ it tends to zero according to

$$R = \left(\frac{\pi U_0}{\alpha \hbar} \right)^2 \frac{2m}{E} \exp \left(-\frac{4\pi}{\alpha \hbar} \sqrt{2mE} \right).$$

In the limiting transition to classical mechanics, R vanishes, as it should.

PROBLEM

4. Determine the transmission coefficient of a particle through the potential barrier defined by

$$U(x) = \frac{U_0}{\cosh^2(\alpha x)}$$

(Fig. 8); the particle energy is $E < U_0$.

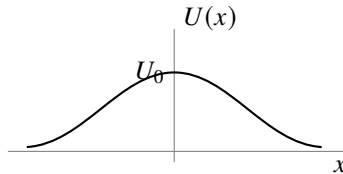


Figure 8

¹⁹See formula (e.6), in each of whose two terms only the first term of the expansion is to be retained; that is, the hypergeometric functions of $1/z$ are to be replaced by unity.

SOLUTION. The Schrödinger equation for this problem is obtained from the example considered in Problem 5 of § 23 by changing the sign of U_0 , with the energy E now taken to be positive.

The same method gives the solution

$$\psi = (1 - \xi^2)^{-ik/2\alpha} F\left(-\frac{ik}{\alpha} - s, -\frac{ik}{\alpha} + s + 1, -\frac{ik}{\alpha} + 1, \frac{1 - \xi}{2}\right), \quad (1)$$

where

$$\xi = \tanh(\alpha x), \quad k = \frac{1}{\hbar} \sqrt{2mE}, \quad s = \frac{1}{2} \left(-1 + \sqrt{1 - \frac{8mU_0}{\alpha^2 \hbar^2}} \right).$$

This solution already satisfies the condition that as $x \rightarrow \infty$ (that is, as $\xi \rightarrow 1$, $(1 - \xi) \approx 2e^{-2\alpha x}$) the wave function contain only the transmitted wave ($\sim e^{ikx}$). The asymptotic expression for the wave function as $x \rightarrow -\infty$ ($\xi \rightarrow -1$) is obtained by transforming the hypergeometric function via formula (e.7):

$$\psi \sim e^{-ikx} \frac{\Gamma(ik/\alpha)\Gamma(1 - ik/\alpha)}{\Gamma(-s)\Gamma(1 + s)} + e^{ikx} \frac{\Gamma(-ik/\alpha)\Gamma(1 - ik/\alpha)}{\Gamma(-ik/\alpha - s)\Gamma(-ik/\alpha + s + 1)}. \quad (2)$$

Taking the squared modulus of the ratio of the coefficients in this function, we obtain the following expression for the transmission coefficient $D = 1 - R$:

$$D = \frac{\sinh^2(\pi k/\alpha)}{\sinh^2(\pi k/\alpha) + \cos^2\left(\frac{\pi}{2} \sqrt{1 - \frac{8mU_0}{\hbar^2 \alpha^2}}\right)} \quad \text{when} \quad \frac{8mU_0}{\hbar^2 \alpha^2} < 1,$$

$$D = \frac{\sinh^2(\pi k/\alpha)}{\sinh^2(\pi k/\alpha) + \cosh^2\left(\frac{\pi}{2} \sqrt{\frac{8mU_0}{\hbar^2 \alpha^2} - 1}\right)} \quad \text{when} \quad \frac{8mU_0}{\hbar^2 \alpha^2} > 1.$$

The first of these formulas also applies when $U_0 < 0$, that is, when the particle passes not over a potential barrier but over a potential well. It is noteworthy that then $D = 1$ if $1 + (8m|U_0|/\hbar^2 \alpha^2) = (2n + 1)^2$; thus, for certain values of the well depth $|U_0|$, particles passing over it are not reflected. This is already clear from expression (2), in which the term containing e^{-ikx} is entirely absent when s is a positive integer.

PROBLEM

5. Determine how the transmission coefficient tends to zero as $E \rightarrow 0$, assuming that the potential energy $U(x)$ decreases rapidly at distances $|x| \gg a$, where a is the characteristic size of the interaction region.

SOLUTION. In the range of distances $k|x| \ll 1$, the energy E may be neglected in the Schrödinger equation. If also $|x| \gg a$, the potential energy may be neglected as well, and the equation reduces to

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} = 0,$$

whose solution we write as

$$\psi = a_1 + b_1x, \quad x < 0, \quad \psi = a_2 + b_2x, \quad x > 0. \quad (1)$$

Solving the equation at distances $|x| \sim a$ determines the relation among a_1 , b_1 and a_2 , b_2 . This relation is linear and has the form

$$a_1 = \rho a_2 + \mu b_2, \quad b_1 = \nu a_2 + \tau b_2. \quad (2)$$

The coefficients ρ , μ , ν , and τ are real and independent of the energy, since the energy no longer enters the equation²⁰. Solution (1) must agree with the first two terms in the expansions of (25.1) and (25.2) in powers of x , whence

$$a_1 = 1 + B, \quad b_1 = ik(1 - B), \quad a_2 = A, \quad b_2 = ikA.$$

Substituting these expressions in (2) and solving the equations for A , we find, at small k , $A \approx 2ik/\nu$, whence

$$D \approx \frac{4k^2}{\nu^2} \sim E.$$

Thus the transmission coefficient tends to zero in proportion to the particle energy. This general relation is, of course, satisfied in the examples considered in Problems 2 and 4.

²⁰By constancy of the flux, they satisfy $\rho\tau - \mu\nu = 1$.

Angular Momentum

§ 26. Angular momentum

In deriving momentum conservation in § 15, we used the homogeneity of space for a closed system of particles. Besides being homogeneous, space is also isotropic: every direction in it is equivalent. The Hamiltonian of a closed system must therefore remain unchanged when the system as a whole is rotated through an arbitrary angle about an arbitrary axis. It is enough to impose this requirement for an arbitrary infinitesimal rotation.

Let $\delta\boldsymbol{\varphi}$ be the vector of an infinitesimal rotation. Its magnitude is the angle of rotation $\delta\varphi$, and its direction is the axis about which the rotation is made. Under such a rotation, the changes $\delta\mathbf{r}_a$ of the particles' radius vectors $\mathbf{r}_a$ are

$$\delta\mathbf{r}_a = \delta\boldsymbol{\varphi} \times \mathbf{r}_a.$$

An arbitrary function $\psi(\mathbf{r}_1, \mathbf{r}_2, \dots)$ is transformed into

$$\begin{aligned} \psi(\mathbf{r}_1 + \delta\mathbf{r}_1, \mathbf{r}_2 + \delta\mathbf{r}_2, \dots) &= \psi(\mathbf{r}_1, \mathbf{r}_2, \dots) + \sum_a \delta\mathbf{r}_a \cdot \nabla_a \psi = \\ &= \psi(\mathbf{r}_1, \mathbf{r}_2, \dots) + \sum_a (\delta\boldsymbol{\varphi} \times \mathbf{r}_a) \cdot \nabla_a \psi = \\ &= \left(1 + \delta\boldsymbol{\varphi} \cdot \sum_a (\mathbf{r}_a \times \nabla_a) \right) \psi(\mathbf{r}_1, \mathbf{r}_2, \dots). \end{aligned}$$

The expression

$$1 + \delta\boldsymbol{\varphi} \cdot \sum_a (\mathbf{r}_a \times \nabla_a)$$

is the operator of an infinitesimal rotation. The fact that such a rotation does not change the system's Hamiltonian is expressed (cf. § 15) by the commutation of the rotation operator with $\hat{H}$. Since $\delta\boldsymbol{\varphi}$ is a constant vector, this condition reduces to

$$\left(\sum_a (\mathbf{r}_a \times \nabla_a) \right) \hat{H} - \hat{H} \left(\sum_a (\mathbf{r}_a \times \nabla_a) \right) = 0, \quad (26.1)$$

which expresses a conservation law.

The quantity conserved in a closed system as a consequence of the isotropy of space is the system's *angular momentum* (cf. Vol. I, § 9). Thus, apart from a constant factor, the operator $\sum_a (\mathbf{r}_a \times \nabla_a)$ must correspond to the total angular momentum of the system's motion, while each term $(\mathbf{r}_a \times \nabla_a)$ must correspond to the angular momentum of one particle.

The proportionality factor must be chosen as $-i\hbar$. The resulting particle angular-momentum operator $-i\hbar(\mathbf{r} \times \nabla) = \mathbf{r} \times \hat{\mathbf{p}}$ then agrees exactly with the classical expression $\mathbf{r} \times \mathbf{p}$. Henceforth we shall always measure angular momentum in units of $\hbar$. We denote the operator of the angular momentum of one particle, defined in this way, by $\hat{\mathbf{L}}$, and that of the entire system by $\hat{\mathbf{L}}$. Thus the particle angular-momentum operator is

$$\hbar \hat{\mathbf{L}} = \mathbf{r} \times \hat{\mathbf{p}} = -i\hbar(\mathbf{r} \times \nabla) \quad (26.2)$$

or, in components,

$$\hbar \hat{L}_x = y\hat{p}_z - z\hat{p}_y, \quad \hbar \hat{L}_y = z\hat{p}_x - x\hat{p}_z, \quad \hbar \hat{L}_z = x\hat{p}_y - y\hat{p}_x.$$

For a system in an external field, angular momentum is not generally conserved. It may nevertheless be conserved when the field has a suitable symmetry. In a centrally symmetric field, for example, all spatial directions from the center are equivalent, so angular momentum about that center is conserved. When the field has axial symmetry, the same reasoning shows that what is conserved is the angular-momentum component directed along the symmetry axis. All these conservation laws of classical mechanics remain valid in quantum mechanics.

If a system's angular momentum is not conserved, it has no definite value in a stationary state. In such cases the mean angular momentum in a given stationary state can sometimes be of interest. Its mean value is readily seen to vanish in every nondegenerate stationary state. Indeed, the energy is unchanged when the sign of time is reversed; and since only one stationary state corresponds to the given energy level, replacing t by $-t$ must leave the state of the system unchanged. The mean values of all quantities, including angular momentum, must therefore also remain unchanged. But angular momentum reverses sign under time reversal, which would give $\bar{\mathbf{L}} = -\bar{\mathbf{L}}$; hence $\bar{\mathbf{L}} = 0$. A direct calculation confirms this: the integral $\int \psi^* \hat{\mathbf{L}} \psi dq$, which by definition gives the mean $\bar{\mathbf{L}}$, can be evaluated using the fact that nondegenerate wave functions are real (see the end of § 18). One then finds that

$$\bar{\mathbf{L}} = -i\hbar \int \psi^* \left(\sum_a (\mathbf{r}_a \times \nabla_a) \right) \psi dq$$

is purely imaginary. Since $\bar{\mathbf{L}}$ must of course be real, it follows that $\bar{\mathbf{L}} = 0$.

We now determine the commutation rules between the angular-momentum operators and the coordinate and momentum operators. From (16.2) we readily obtain

$$\begin{aligned} \{\hat{L}_x, x\} &= 0, & \{\hat{L}_x, y\} &= iz, & \{\hat{L}_x, z\} &= -iy, \\ \{\hat{L}_y, y\} &= 0, & \{\hat{L}_y, z\} &= ix, & \{\hat{L}_y, x\} &= -iz, \\ \{\hat{L}_z, z\} &= 0, & \{\hat{L}_z, x\} &= iy, & \{\hat{L}_z, y\} &= -ix. \end{aligned} \quad (26.3)$$

For example,

$$\hat{L}_x y - y \hat{L}_x = \frac{1}{\hbar} (y \hat{p}_z - z \hat{p}_y) y - y (y \hat{p}_z - z \hat{p}_y) \frac{1}{\hbar} = -\frac{z}{\hbar} \{\hat{p}_y, y\} = iz.$$

All the relations (26.3) may be written in tensor form as

$$\{\hat{l}_i, x_k\} = i e_{ikl} x_l, \quad (26.4)$$

where e_{ikl} is the antisymmetric unit tensor of third rank¹, with summation understood over each pair of repeated “dummy” indices. The angular-momentum and momentum operators satisfy analogous commutation relations:

$$\{\hat{l}_i, \hat{p}_k\} = i e_{ikl} \hat{p}_l. \quad (26.5)$$

These formulas make it easy to obtain the commutation rules among the operators for the components of angular momentum. We have

$$\begin{aligned} \hbar(\hat{l}_x \hat{l}_y - \hat{l}_y \hat{l}_x) &= \hat{l}_x(z\hat{p}_x - x\hat{p}_z) - (z\hat{p}_x - x\hat{p}_z)\hat{l}_x = \\ &= (\hat{l}_x z - z\hat{l}_x)\hat{p}_x - x(\hat{l}_x \hat{p}_z - \hat{p}_z \hat{l}_x) = -iy\hat{p}_x + ix\hat{p}_y = i\hbar \hat{l}_z. \end{aligned}$$

Thus

$$\{\hat{l}_y, \hat{l}_z\} = i\hat{l}_x, \quad \{\hat{l}_z, \hat{l}_x\} = i\hat{l}_y, \quad \{\hat{l}_x, \hat{l}_y\} = i\hat{l}_z \quad (26.6)$$

or

$$\{\hat{l}_i, \hat{l}_k\} = i e_{ikl} \hat{l}_l. \quad (26.7)$$

Exactly the same relations hold for the operators $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ of the system's total angular momentum. Indeed, since the angular-momentum operators of different particles commute with one another, we have, for example,

$$\sum_a \hat{l}_{ay} \sum_a \hat{l}_{az} - \sum_a \hat{l}_{az} \sum_a \hat{l}_{ay} = \sum_a (\hat{l}_{ay} \hat{l}_{az} - \hat{l}_{az} \hat{l}_{ay}) = i \sum_a \hat{l}_{ax}.$$

Therefore

$$\{\hat{L}_y, \hat{L}_z\} = i\hat{L}_x, \quad \{\hat{L}_z, \hat{L}_x\} = i\hat{L}_y, \quad \{\hat{L}_x, \hat{L}_y\} = i\hat{L}_z. \quad (26.8)$$

Relations (26.8) show that the three components of angular momentum cannot have definite values simultaneously (except when all three components vanish at once; see below). In this respect angular momentum differs essentially from momentum, whose three components can be measured simultaneously.

From $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ we form the operator for the square of the magnitude of the angular-momentum vector:

$$\hat{\mathbf{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2. \quad (26.9)$$

¹The antisymmetric unit tensor of third rank e_{ikl} (also called the unit axial tensor) is defined to be antisymmetric in all three indices, with $e_{123} = 1$. Evidently, of its 27 components, only the 6 for which i, k, l form a permutation of the numbers 1, 2, 3 are nonzero. A component is +1 if the permutation i, k, l is obtained from 1, 2, 3 by an even number of pairwise interchanges (transpositions), and is -1 for an odd number. Clearly,

$$e_{ikl} e_{ikm} = 2\delta_{lm}, \quad e_{ikl} e_{ikl} = 6.$$

The components of the vector $\mathbf{C} = \mathbf{A} \times \mathbf{B}$, the vector product of $\mathbf{A}$ and $\mathbf{B}$, can be expressed through e_{ikl} as

$$C_i = e_{ikl} A_k B_l.$$

This operator commutes with each of $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$:

$$\{\hat{\mathbf{L}}^2, \hat{L}_x\} = 0, \quad \{\hat{\mathbf{L}}^2, \hat{L}_y\} = 0, \quad \{\hat{\mathbf{L}}^2, \hat{L}_z\} = 0. \quad (26.10)$$

Indeed, using (26.8), we find, for example,

$$\begin{aligned} \{\hat{L}_x^2, \hat{L}_z\} &= \hat{L}_x \{\hat{L}_x, \hat{L}_z\} + \{\hat{L}_x, \hat{L}_z\} \hat{L}_x = -i(\hat{L}_x \hat{L}_y + \hat{L}_y \hat{L}_x), \\ \{\hat{L}_y^2, \hat{L}_z\} &= i(\hat{L}_x \hat{L}_y + \hat{L}_y \hat{L}_x), \\ \{\hat{L}_z^2, \hat{L}_z\} &= 0. \end{aligned}$$

Adding these equalities gives the last relation in (26.10).

Physically, (26.10) means that the square of the angular momentum (that is, its magnitude) can have a definite value simultaneously with one of its components.

Instead of $\hat{L}_x$ and $\hat{L}_y$, it is often convenient to use their complex combinations

$$\hat{L}_+ = \hat{L}_x + i\hat{L}_y, \quad \hat{L}_- = \hat{L}_x - i\hat{L}_y. \quad (26.11)$$

Direct calculation from (26.8) readily gives the following commutation rules for these combinations:

$$\{\hat{L}_+, \hat{L}_-\} = 2\hat{L}_z, \quad \{\hat{L}_z, \hat{L}_+\} = \hat{L}_+, \quad \{\hat{L}_z, \hat{L}_-\} = -\hat{L}_-. \quad (26.12)$$

It is also easy to verify that

$$\hat{\mathbf{L}}^2 = \hat{L}_+ \hat{L}_- + \hat{L}_z^2 - \hat{L}_z = \hat{L}_- \hat{L}_+ + \hat{L}_z^2 + \hat{L}_z. \quad (26.13)$$

Finally, we give the frequently used expressions for the angular-momentum operator of one particle in spherical coordinates. Introducing these coordinates through the usual relations

$$x = r \sin \theta \cos \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta,$$

a straightforward calculation gives

$$\hat{L}_z = -i \frac{\partial}{\partial \varphi}, \quad (26.14)$$

$$\hat{L}_\pm = e^{\pm i\varphi} \left(\pm \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \varphi} \right). \quad (26.15)$$

Substitution in (26.13) gives the square of the particle's angular-momentum operator in the form

$$\hat{\mathbf{L}}^2 = - \left[\frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) \right]. \quad (26.16)$$

Notice that, apart from a factor, this is the angular part of the Laplace operator.

§ 27. Eigenvalues of angular momentum

In finding the eigenvalues of a particle's angular-momentum projection onto a chosen direction, one may orient the polar axis along that direction and employ the spherical-coordinate form of the operator. Using (26.14), the equation $\hat{l}_z\psi = l_z\psi$ takes the form

$$-i\frac{\partial\psi}{\partial\varphi} = l_z\psi. \quad (27.1)$$

Its solution is

$$\psi = f(r, \theta)e^{il_z\varphi},$$

where $f(r, \theta)$ is an arbitrary function of r and θ . For ψ to be single-valued, it must be periodic in φ , with period 2π . Hence we find²

$$l_z = m, \quad m = 0, \pm 1, \pm 2, \dots \quad (27.2)$$

Thus the eigenvalues l_z are the positive and negative integers, including zero. We denote the factor depending on φ that is characteristic of the eigenfunctions of $\hat{l}_z$ by

$$\Phi_m(\varphi) = \frac{1}{\sqrt{2\pi}}e^{im\varphi}. \quad (27.3)$$

These functions are normalized so that

$$\int_0^{2\pi} \Phi_m^*(\varphi)\Phi_{m'}(\varphi) d\varphi = \delta_{mm'}. \quad (27.4)$$

The eigenvalues of the z component of the total angular momentum of a system are evidently likewise positive and negative integers:

$$L_z = M, \quad M = 0, \pm 1, \pm 2, \dots \quad (27.5)$$

(this follows because $\hat{L}_z$ is the sum of the mutually commuting operators $\hat{l}_z$ of the individual particles).

Since the direction of the z axis has not been distinguished in advance, the same result clearly holds for L_x , L_y , and, in general, for the component of angular momentum along any direction: each can take only integer values. At first sight this result may seem paradoxical, particularly if it is applied to two infinitely close directions. One must bear in mind, however, that the only common eigenfunction of the operators $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ corresponds to the simultaneous values

$$L_x = L_y = L_z = 0;$$

in this case the angular-momentum vector, and therefore its projection on every direction, is zero. If at least one of the eigenvalues L_x , L_y , L_z is nonzero, the corresponding operators have no common eigenfunctions. Put differently, no state exists in which definite nonzero values are possessed simultaneously by two or three angular-momentum

²The customary notation of the eigenvalues of the angular-momentum projection by the letter m —the same letter that also denotes the particle mass—cannot in practice lead to confusion.

components referred to distinct directions; one can therefore only discuss the integer nature of a single component at a time.

Stationary states of a system that differ only in the value of M have the same energy. This follows already from general considerations, since the direction of the z axis has not been distinguished beforehand. Thus the energy levels of a system with conserved nonzero angular momentum are in any event degenerate³.

We now turn to finding the eigenvalues of the square of angular momentum and show how they can be obtained solely from the commutation rules (26.8). Let ψ_M denote the wave functions of stationary states having the same value of $\mathbf{L}^2$, belonging to one degenerate energy level, and differing in the value of M ⁴.

First, since the two directions of the z axis are physically equivalent, every possible positive value $M = |M|$ has a corresponding negative value $M = -|M|$. Let L (a nonnegative integer) denote the largest possible value of $|M|$ for a given $\mathbf{L}^2$. The very existence of such an upper bound follows from the fact that the difference $\hat{\mathbf{L}}^2 - \hat{L}_z^2 = \hat{L}_x^2 + \hat{L}_y^2$ is the operator of the necessarily nonnegative physical quantity $L_x^2 + L_y^2$, and hence its eigenvalues cannot be negative.

Applying the operator $\hat{L}_z \hat{L}_\pm$ to an eigenfunction ψ_M of $\hat{L}_z$, and using the commutation rules (26.12), we obtain

$$\hat{L}_z \hat{L}_\pm \psi_M = (M \pm 1) \hat{L}_\pm \psi_M. \quad (27.6)$$

It follows that $\hat{L}_\pm \psi_M$ is, apart from a normalization constant, an eigenfunction corresponding to the value $M \pm 1$ of L_z :

$$\psi_{M+1} = \text{const} \cdot \hat{L}_+ \psi_M, \quad \psi_{M-1} = \text{const} \cdot \hat{L}_- \psi_M. \quad (27.7)$$

If $M = L$ is put in the first of these equalities, then we must have identically

$$\hat{L}_+ \psi_L = 0, \quad (27.8)$$

since, by definition, there are no states with $M > L$. Applying $\hat{L}_-$ to this equality and using (26.13), we obtain

$$\hat{L}_- \hat{L}_+ \psi_L = (\hat{\mathbf{L}}^2 - \hat{L}_z^2 - \hat{L}_z) \psi_L = 0.$$

But since the ψ_M are common eigenfunctions of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$,

$$\hat{\mathbf{L}}^2 \psi_L = \mathbf{L}^2 \psi_L, \quad \hat{L}_z^2 \psi_L = L^2 \psi_L, \quad \hat{L}_z \psi_L = L \psi_L,$$

³This circumstance is a special case of the general theorem stated in § 10 on the degeneracy of levels in the presence of at least two conserved quantities whose operators do not commute. Here those quantities are the components of angular momentum.

⁴Here it is understood that there is no additional degeneracy by which the same energy corresponds to different values of the square of angular momentum. This is true for the discrete spectrum (apart from the so-called “accidental degeneracy” in the Coulomb field; see § 36), and is, generally speaking, not true for energy levels in the continuous spectrum. Even when there is additional degeneracy, however, the eigenfunctions can always be chosen to represent states with definite values of $\mathbf{L}^2$; from them one can then select states with the same values of E and $\mathbf{L}^2$. The mathematical basis for this is that simultaneous diagonalization of the matrices of mutually commuting operators is always achievable. In analogous cases below, for brevity, we shall speak as if there were no additional degeneracy, with the understanding that, in view of what has just been said, the results obtained do not in fact depend on this assumption.

so the resulting equation gives

$$\mathbf{L}^2 = L(L + 1). \quad (27.9)$$

Formula (27.9) determines the required eigenvalues of the square of angular momentum; L takes all nonnegative integer values. For a given value of L , the angular-momentum component $L_z = M$ can take the values

$$M = L, L - 1, \dots, -L, \quad (27.10)$$

that is, $2L + 1$ distinct values in all. An energy level corresponding to angular momentum L is therefore $(2L + 1)$ -fold degenerate; this is usually called directional degeneracy of angular momentum. A state with zero angular momentum, $L = 0$ (so that all three of its components vanish), is nondegenerate. We note that the wave function of such a state is spherically symmetric. This is already clear from the fact that its change under any infinitesimal rotation, given by $\hat{\mathbf{L}}\psi$, vanishes in this case. For brevity we shall often use the customary phrase “angular momentum L ” of the system, meaning angular momentum whose square equals $L(L + 1)$; the z component of angular momentum is usually called simply its “projection.”

We shall denote the angular momentum of one particle by the lowercase letter l ; thus for it we write (27.9) as

$$\mathbf{l}^2 = l(l + 1). \quad (27.11)$$

Let us calculate the matrix elements of L_x and L_y in the representation in which, together with the energy, $\mathbf{L}^2$ and L_z are diagonal (*M. Born, W. Heisenberg, P. Jordan, 1926*). We note that, since $\hat{L}_x$ and $\hat{L}_y$ commute with the Hamiltonian, their matrices are diagonal with respect to energy; that is, every matrix element for a transition between states of different energies (and different angular momenta L) is zero. It is therefore sufficient to consider matrix elements for transitions within the group of states with different values of M corresponding to a single degenerate energy level.

From (27.7) one sees that the matrix of $\hat{L}_+$ has nonzero elements only for transitions $M - 1 \rightarrow M$, and that of $\hat{L}_-$ only for $M \rightarrow M - 1$. With this in mind, equating the diagonal matrix elements of the two sides of (26.13) gives⁵

$$L(L + 1) = \langle M | L_+ | M - 1 \rangle \langle M - 1 | L_- | M \rangle + M^2 - M.$$

Since the operators $\hat{L}_x$ and $\hat{L}_y$ are Hermitian,

$$\langle M - 1 | L_- | M \rangle = \langle M | L_+ | M - 1 \rangle^*,$$

we rewrite this equality as

$$|\langle M | L_+ | M - 1 \rangle|^2 = L(L + 1) - M(M - 1) = (L - M + 1)(L + M),$$

whence⁶

$$\langle M | L_+ | M - 1 \rangle = \langle M - 1 | L_- | M \rangle = \sqrt{(L + M)(L - M + 1)}. \quad (27.12)$$

⁵In the notation for matrix elements, for brevity, we omit all indices with respect to which they are diagonal (including the index L).

⁶The choice of sign in this formula is consistent with the choice of phase factors in the angular-momentum eigenfunctions.

For the nonzero matrix elements of L_x and L_y themselves, this gives

$$\begin{aligned}\langle M|L_x|M-1\rangle &= \langle M-1|L_x|M\rangle = \frac{1}{2}\sqrt{(L+M)(L-M+1)}, \\ \langle M|L_y|M-1\rangle &= -\langle M-1|L_y|M\rangle = -\frac{i}{2}\sqrt{(L+M)(L-M+1)}.\end{aligned}\quad (27.13)$$

We draw attention to the absence of diagonal elements in the matrices of L_x and L_y . Since a diagonal matrix element gives the mean value of a quantity in the corresponding state, this means that in states with definite values of L_z the mean values are $\bar{L}_x = \bar{L}_y = 0$. Thus, if the projection of angular momentum on some direction in space has a definite value, the whole vector $\bar{\mathbf{L}}$ lies along that same direction.

§ 28. Angular-Momentum Eigenfunctions

Specifying the values of l and m does not determine a particle's wave function completely. Indeed, the expressions for the operators of these quantities in spherical coordinates contain only the angles θ and φ , so their eigenfunctions may include an arbitrary factor depending on r . Here we shall consider only the angular part of the wave function characteristic of angular-momentum eigenfunctions. Denote it by $Y_{lm}(\theta, \varphi)$ and impose the normalization condition

$$\int |Y_{lm}|^2 d\omega = 1$$

($d\omega = \sin\theta d\theta d\varphi$ is the element of solid angle).

The calculation below shows that the problem of finding the common eigenfunctions of $\hat{\mathbf{I}}^2$ and $\hat{I}_z$ permits separation of the variables θ and φ . These functions may therefore be sought in the form

$$Y_{lm} = \Phi_m(\varphi)\Theta_{lm}(\theta), \quad (28.1)$$

where $\Phi_m(\varphi)$ are the eigenfunctions of $\hat{I}_z$ defined by (27.3). Since the functions Φ_m have already been normalized by (27.4), the functions Θ_{lm} must satisfy

$$\int_0^\pi |\Theta_{lm}|^2 \sin\theta d\theta = 1. \quad (28.2)$$

Functions Y_{lm} with different values of l or m are automatically mutually orthogonal:

$$\int_0^{2\pi} \int_0^\pi Y_{l'm'}^* Y_{lm} \sin\theta d\theta d\varphi = \delta_{ll'} \delta_{mm'}, \quad (28.3)$$

because they are eigenfunctions of angular-momentum operators belonging to different eigenvalues. The functions $\Phi_m(\varphi)$ are also separately orthogonal (see (27.4)), since they are eigenfunctions of $\hat{I}_z$ belonging to its different eigenvalues m . The functions $\Theta_{lm}(\theta)$ by themselves, however, are not eigenfunctions of any angular-momentum operator; they are mutually orthogonal for different l , but not for different m .

The most direct way to calculate the required functions is to solve the eigenfunction problem for $\hat{\mathbf{I}}^2$ written in spherical coordinates (formula (26.16)). The equation $\hat{\mathbf{I}}^2\psi = l^2\psi$ reads

$$\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial\psi}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2\psi}{\partial\varphi^2} + l(l+1)\psi = 0.$$

Substitution of ψ in the form (28.1) gives the following equation for Θ_{lm} :

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta_{lm}}{d\theta} \right) - \frac{m^2}{\sin^2 \theta} \Theta_{lm} + l(l+1)\Theta_{lm} = 0. \quad (28.4)$$

This equation is well known from the theory of spherical harmonics. It has solutions satisfying the finiteness and single-valuedness conditions for nonnegative integral values $l \geq |m|$, in agreement with the angular-momentum eigenvalues obtained above by the matrix method. The corresponding solutions are the associated Legendre polynomials $P_l^m(\cos \theta)$ (see § c of the mathematical supplements). Normalizing the solution by (28.2), we obtain⁷

$$\Theta_{lm} = (-1)^m i^l \sqrt{\frac{(2l+1)(l-m)!}{2(l+m)!}} P_l^m(\cos \theta). \quad (28.5)$$

Here $m \geq 0$ is assumed. For negative m , define Θ_{lm} by

$$\Theta_{l,-|m|} = (-1)^m \Theta_{l|m|}, \quad (28.6)$$

that is, for $m < 0$, Θ_{lm} is given by (28.5) with $|m|$ in place of m and with the factor $(-1)^m$ omitted. Thus, from the mathematical point of view, the angular-momentum eigenfunctions are spherical harmonics with a particular normalization. For later reference, we write their complete expression, incorporating all the definitions just given:

$$Y_{lm}(\theta, \varphi) = (-1)^{(m+|m|)/2} i^l \left[\frac{2l+1}{4\pi} \frac{(l-|m|)!}{(l+|m|)!} \right]^{1/2} P_l^{|m|}(\cos \theta) e^{im\varphi}. \quad (28.7)$$

In particular,

$$Y_{l0} = i^l \sqrt{\frac{2l+1}{4\pi}} P_l(\cos \theta). \quad (28.8)$$

Functions differing only in the sign of m are evidently related by

$$(-1)^{l-m} Y_{l,-m} = Y_{lm}^*. \quad (28.9)$$

For $l = 0$ (and consequently $m = 0$), the spherical harmonic reduces to a constant. In other words, the wave functions of particle states with zero angular momentum depend only on r and hence possess complete spherical symmetry, in accordance with the general statement made in § 27.

For a fixed m , the values of l beginning with $|m|$ enumerate, in ascending order, the successive eigenvalues of $\mathbf{I}^2$. The general theorem on zeros of eigenfunctions (§ 21) therefore implies that Θ_{lm} vanishes at $l - |m|$ different values of θ ; in other words, its nodal lines are $l - |m|$ “circles of latitude” on the sphere. For the complete angular

⁷The normalization condition does not, of course, determine the choice of phase factor. The convention used in this book is the one most natural from the standpoint of the general theory of angular-momentum addition; it differs from the convention usually adopted by a factor i^l . The advantages of this choice will be apparent from the notes on pp. 278, 527, and 533.

functions, if real factors $\cos m\varphi$ or $\sin m\varphi$ are chosen instead of $e^{\pm i|m|\varphi}$ ⁸, they also have $|m|$ “meridian circles” as nodal lines. The total number of nodal lines is therefore l .

Finally, let us show how the functions Θ_{lm} can be calculated by the matrix method. The procedure is analogous to the calculation of the oscillator wave functions in § 23. We start from (27.8), $\hat{l}_+ Y_{ll} = 0$. Using (26.15) for the operator $\hat{l}_+$ and substituting

$$Y_{ll} = \frac{1}{\sqrt{2\pi}} e^{il\varphi} \Theta_{ll}(\theta),$$

we obtain for Θ_{ll} the equation

$$\frac{d\Theta_{ll}}{d\theta} - l \cot \theta \cdot \Theta_{ll} = 0,$$

whence $\Theta_{ll} = \text{const} \cdot \sin^l \theta$. Determining the constant from the normalization condition gives

$$\Theta_{ll} = (-i)^l \sqrt{\frac{(2l+1)!}{2}} \frac{1}{2^l l!} \sin^l \theta. \quad (28.10)$$

Next, using (27.12), write

$$\hat{l}_- Y_{l,m+1} = (l_-)_{m,m+1} Y_{lm} = \sqrt{(l-m)(l+m+1)} Y_{lm}.$$

Repeated application of this formula gives

$$\sqrt{\frac{(l-m)!}{(l+m)!}} Y_{lm} = \frac{1}{\sqrt{(2l)!}} \hat{l}_-^{l-m} Y_{ll}.$$

The right-hand side is readily evaluated with the aid of (26.15) for $\hat{l}_-$, according to which

$$\hat{l}_- [f(\theta) e^{im\varphi}] = e^{i(m-1)\varphi} \sin^{1-m} \theta \frac{d}{d \cos \theta} (f \sin^m \theta).$$

Repeated application of this formula gives

$$\hat{l}_-^{l-m} e^{il\varphi} \Theta_{ll} = e^{im\varphi} \sin^{-m} \theta \frac{d^{l-m}}{(d \cos \theta)^{l-m}} (\sin^l \theta \cdot \Theta_{ll}).$$

Finally, using these relations together with (28.10) for Θ_{ll} , we obtain

$$\Theta_{lm}(\theta) = (-i)^l \sqrt{\frac{2l+1}{2}} \frac{(l+m)!}{(l-m)!} \frac{1}{2^l l!} \frac{d^{l-m}}{\sin^m \theta (d \cos \theta)^{l-m}} \sin^{2l} \theta, \quad (28.11)$$

which agrees with (28.5).

⁸Each such function represents a state in which l_z has no definite value, but may take the values $\pm m$ with equal probability.

§ 29. Matrix elements of vectors

Return once more to a closed system of particles⁹. Let f stand for some scalar physical quantity of this system and $\hat{f}$ for its operator. Any scalar is invariant under a rotation of the coordinate system. Hence the scalar operator $\hat{f}$ is unchanged by a rotation; that is, it commutes with the rotation operator. We know, however, that the operator of an infinitesimal rotation coincides, apart from a constant factor, with the angular-momentum operator. Therefore

$$\{\hat{f}, \hat{\mathbf{L}}\} = 0. \quad (29.1)$$

It follows from the commutation of $\hat{f}$ with the angular-momentum operator that, for transitions between states with definite L and M , the matrix of f is diagonal in these indices. Moreover, specifying M determines only the orientation of the system relative to the coordinate axes, while the value of a scalar quantity does not depend on this orientation at all. Consequently, the matrix elements $\langle n'LM|f|nLM\rangle$ do not depend on M (the letter n provisionally denotes the collection of all quantum numbers, other than L and M , that specify the state of the system). A formal proof follows from the commutation of $\hat{f}$ and $\hat{L}_+$:

$$\hat{f}\hat{L}_+ - \hat{L}_+\hat{f} = 0. \quad (29.2)$$

Take the matrix element of this equality for the transition $n, L, M \rightarrow n', L, M+1$. Since the matrix of L_+ has only elements for transitions $n, L, M \rightarrow n, L, M+1$, we find

$$\begin{aligned} \langle n', L, M+1|f|n, L, M+1\rangle \langle n, L, M+1|L_+|n, L, M\rangle &= \\ = \langle n', L, M+1|L_+|n', L, M\rangle \langle n', L, M|f|n, L, M\rangle, \end{aligned}$$

and, since the matrix elements of L_+ do not depend on the index n ,

$$\langle n', L, M+1|f|n, L, M+1\rangle = \langle n', L, M|f|n, L, M\rangle, \quad (29.3)$$

which shows that all the elements $\langle n', L, M|f|n, L, M\rangle$ with different M (and identical remaining indices) are equal.

Applied to the Hamiltonian itself, this result gives the already known independence of the stationary-state energy from M , that is, the $(2L+1)$ -fold degeneracy of the energy levels.

Next, let $\mathbf{A}$ be some vector physical quantity characterizing the closed system. Under a rotation of the coordinate system (in particular, an infinitesimal rotation, that is, under the action of the angular-momentum operator), the components of the vector transform into one another. The commutation of the operators $\hat{L}_i$ with the operators $\hat{A}_i$ must therefore again yield components of the same vector $\hat{A}_i$. Which components occur can be found by noting that, in the special case where $\mathbf{A}$ is the radius vector of a particle, the result must reduce to (26.4). We thus obtain the commutation rules

$$\{\hat{L}_i, \hat{A}_k\} = ie_{ikl}\hat{A}_l. \quad (29.4)$$

⁹Everything derived in this section applies equally to a particle in a centrally symmetric field (more broadly, to any situation in which the system's total angular momentum is conserved).

From these relations one can derive a number of results about the structure of the component matrices of the vector $\mathbf{A}$ (*M. Born, W. Heisenberg, P. Jordan, 1926*). First, they make it possible to find the *selection rules*, which determine the transitions for which matrix elements may be nonzero. We shall not, however, give the corresponding rather cumbersome calculations here, since it will be shown below (§ 107) that these rules are in fact direct consequences of the general transformation properties of vector quantities and can be obtained from them essentially without calculation. Here we merely state the rules without derivation.

Matrix elements of every component of a vector can be nonzero only for transitions in which the angular momentum L changes by no more than one:

$$L \rightarrow L, L \pm 1. \quad (29.5)$$

There is also an additional selection rule forbidding transitions between any two states with $L = 0$. This rule is an evident consequence of the complete spherical symmetry of states with zero angular momentum.

The selection rules for the angular-momentum projection M differ among the components of the vector. Specifically, matrix elements may be nonzero for transitions with the following changes of M :

$$\begin{aligned} M &\rightarrow M + 1 && \text{for } A_+ = A_x + iA_y, \\ M &\rightarrow M - 1 && \text{for } A_- = A_x - iA_y, \\ M &\rightarrow M && \text{for } A_z. \end{aligned} \quad (29.6)$$

It is also possible to determine in general form the dependence of the matrix elements of a vector on M . We again give these important and frequently used formulas without derivation, since they too are special cases of more general relations, applying to arbitrary tensor quantities, which will be derived in § 107. The nonzero matrix elements of A_z are given by

$$\begin{aligned} \langle n'LM | A_z | nLM \rangle &= \frac{M}{\sqrt{L(L+1)(2L+1)}} \langle n'L || A || nL \rangle, \\ \langle n'LM | A_z | n, L-1, M \rangle &= \sqrt{\frac{L^2 - M^2}{L(2L-1)(2L+1)}} \langle n'L || A || n, L-1 \rangle, \\ \langle n', L-1, M | A_z | nLM \rangle &= \sqrt{\frac{L^2 - M^2}{L(2L-1)(2L+1)}} \langle n', L-1 || A || nL \rangle. \end{aligned} \quad (29.7)$$

Here the symbol

$$\langle n'L' || A || nL \rangle$$

denotes a so-called *reduced matrix element*, a quantity independent of the quantum

number M^{10} . They are related by

$$\langle n'L' \| A \| nL \rangle = \langle nL \| A \| n'L' \rangle^*, \quad (29.8)$$

which follows directly from the Hermiticity of $\hat{A}_z$.

The matrix elements of A_- and A_+ can be expressed through the same reduced elements. The nonzero matrix elements of A_- are

$$\begin{aligned} \langle n', L, M-1 | A_- | nLM \rangle &= \sqrt{\frac{(L-M+1)(L+M)}{L(L+1)(2L+1)}} \langle n'L \| A \| nL \rangle, \\ \langle n', L, M-1 | A_- | n, L-1, M \rangle &= \sqrt{\frac{(L-M+1)(L-M)}{L(2L-1)(2L+1)}} \langle n'L \| A \| n, L-1 \rangle, \\ \langle n', L-1, M-1 | A_- | nLM \rangle &= -\sqrt{\frac{(L+M-1)(L+M)}{L(2L-1)(2L+1)}} \langle n', L-1 \| A \| nL \rangle. \end{aligned} \quad (29.9)$$

The matrix elements of A_+ need no separate formulas, since the reality of A_x and A_y gives

$$\langle n'L'M' | A_+ | nLM \rangle = \langle nLM | A_- | n'L'M' \rangle^*. \quad (29.10)$$

Let us write out the formula that relates the matrix elements of the scalar $\mathbf{AB}$ to the reduced matrix elements of the two vectors $\mathbf{A}$ and $\mathbf{B}$. The calculation is conveniently made by writing the operator $\hat{\mathbf{A}}\hat{\mathbf{B}}$ as

$$\hat{\mathbf{A}}\hat{\mathbf{B}} = \frac{1}{2}(\hat{A}_+\hat{B}_- + \hat{A}_-\hat{B}_+) + \hat{A}_z\hat{B}_z. \quad (29.11)$$

The matrix of $\mathbf{AB}$, like that of any scalar, is diagonal in L and M . Calculation using (29.7)–(29.9) gives

$$\langle n'LM | \mathbf{AB} | nLM \rangle = \frac{1}{2L+1} \sum_{n'', L''} \langle n'L \| A \| n''L'' \rangle \langle n''L'' \| B \| nL \rangle, \quad (29.12)$$

where L'' takes the values $L, L \pm 1$.

For reference, we give the reduced matrix elements of the vector $\mathbf{L}$ itself. Comparing (29.9) and (27.12), we find

$$\begin{aligned} \langle L \| L \| L \rangle &= \sqrt{L(L+1)(2L+1)}, \\ \langle L-1 \| L \| L \rangle &= \langle L \| L \| L-1 \rangle = 0. \end{aligned} \quad (29.13)$$

A quantity frequently encountered in applications is the unit vector $\mathbf{n}$ along the radius vector of a particle. Let us find its reduced matrix elements. It is sufficient to calculate,

¹⁰The appearance in (29.7) and (29.9) of denominators depending on L accords with the general notation introduced in § 107. The usefulness of these denominators is seen, in particular, in the simple form taken by (29.12) for the matrix elements of the scalar product of two vectors.

One should regard the notation for the reduced matrix element as an indivisible entity (in contrast to what was said about the matrix element notation (11.17)).

for example, the matrix elements of $n_z = \cos \theta$ at zero angular-momentum projection, $m = 0$. We have

$$\langle l-1, 0 | n_z | l 0 \rangle = \int_0^\pi \Theta_{l-1,0}^* \cos \theta \cdot \Theta_{l0} \sin \theta d\theta$$

with the functions Θ_{l0} from (28.11). Evaluation of the integral gives¹¹

$$\langle l-1, 0 | n_z | l 0 \rangle = \frac{il}{\sqrt{(2l-1)(2l+1)}}.$$

The matrix elements for transitions $l \rightarrow l$ are zero (as they are for any polar vector belonging to a single particle; see below, (30.8)). Comparison with (29.7) now gives

$$\langle l-1 || n || l \rangle = -\langle l || n || l-1 \rangle = i\sqrt{l}, \quad \langle l || n || l \rangle = 0. \quad (29.14)$$

PROBLEM

Average the tensor $n_i n_k - (1/3)\delta_{ik}$ (where $\mathbf{n}$ is the unit vector along the particle's radius vector) over a state with a definite value of $|\mathbf{l}|$ but no preferred orientation (that is, l_z is unspecified).

SOLUTION. The required mean value is an operator that can be expressed only in terms of $\hat{\mathbf{l}}$. We seek it in the form

$$\overline{n_i n_k} - \frac{1}{3}\delta_{ik} = a \left[\hat{l}_i \hat{l}_k + \hat{l}_k \hat{l}_i - \frac{2}{3}\delta_{ik} l(l+1) \right];$$

this is the most general form of a symmetric second-rank tensor with zero trace constructed from the components of $\hat{\mathbf{l}}$. To determine the constant a , multiply this equality on the left by $\hat{l}_i$ and on the right by $\hat{l}_k$, with summation over i and k . Since the vector $\mathbf{n}$ is perpendicular to the vector $\hat{\mathbf{h}}\hat{\mathbf{l}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$, we have $n_i \hat{l}_i = 0$. We replace the product $\hat{l}_i \hat{l}_i \hat{l}_k \hat{l}_k = (\hat{\mathbf{l}}^2)^2$ by its eigenvalue $l^2(l+1)^2$, and transform the product $\hat{l}_i \hat{l}_k \hat{l}_i \hat{l}_k$ using the commutation relations (26.7) as follows:

$$\begin{aligned} \hat{l}_i \hat{l}_k \hat{l}_i \hat{l}_k &= \hat{l}_i \hat{l}_i \hat{l}_k \hat{l}_k - i e_{ikl} \hat{l}_i \hat{l}_l \hat{l}_k \\ &= (\hat{\mathbf{l}}^2)^2 - \frac{i}{2} e_{ikl} \hat{l}_i (\hat{l}_l \hat{l}_k - \hat{l}_k \hat{l}_l) \\ &= (\hat{\mathbf{l}}^2)^2 + \frac{1}{2} e_{ikl} e_{lk m} \hat{l}_i \hat{l}_m = (\hat{\mathbf{l}}^2)^2 - \hat{\mathbf{l}}^2 \\ &= l^2(l+1)^2 - l(l+1) \end{aligned}$$

(we have used $e_{ikl} e_{mkl} = 2\delta_{im}$). Straightforward simplification then gives

$$a = -\frac{1}{(2l-1)(2l+3)}.$$

¹¹The calculation is performed by integrating by parts $(l-1)$ times with respect to $d \cos \theta$. For the general formula for integrals of this kind, see (107.14).

§ 30. Parity of a state

Besides translations and rotations of the coordinate system (invariance under which expresses, respectively, the homogeneity and isotropy of space), there is one more operation that preserves the Hamiltonian of a closed system: the *inversion transformation*, which reverses the sign of every coordinate at once (equivalently, it flips the direction of all axes, turning a right-handed frame into a left-handed one and conversely). Invariance of the Hamiltonian under inversion is the mathematical expression of the mirror symmetry of space¹². In classical mechanics, the invariance of the Hamilton function under inversion gives no new conservation laws. In quantum mechanics, however, the situation is essentially different.

Introduce the inversion operator $\hat{P}$ ¹³, whose action on a wave function $\psi(\mathbf{r})$ consists in reversing the signs of the coordinates:

$$\hat{P}\psi(\mathbf{r}) = \psi(-\mathbf{r}). \quad (30.1)$$

The eigenvalues P of this operator, defined by the equation

$$\hat{P}\psi(\mathbf{r}) = P\psi(\mathbf{r}), \quad (30.2)$$

are easily found. Two successive applications of the inversion operator give the identity: the arguments of the function are left entirely unchanged. In other words, $\hat{P}^2\psi = P^2\psi = \psi$, so that $P^2 = 1$, whence

$$P = \pm 1. \quad (30.3)$$

An eigenfunction of the inversion operator is therefore either left entirely unaltered or acquires an overall sign change. Wave functions (and the corresponding states) of the first type are called *even*, and those of the second type *odd*.

The invariance of the Hamiltonian under inversion (that is, the commutation of $\hat{H}$ and $\hat{P}$) therefore expresses the *law of conservation of parity*: if a state of a closed system has definite parity (if it is even or odd), that parity is conserved in time¹⁴.

The angular-momentum operator is likewise invariant under inversion: inversion reverses the signs both of the coordinates and of the operators for differentiation with respect to them, and hence the operator (26.2) remains unchanged. Put differently, inversion and angular momentum commute, so that a definite parity can coexist with definite values of L and of the projection M . Moreover, all states that differ only in M have the same parity. This is already evident from the independence of the properties of a closed system from its orientation in space; formally, it can be proved from the commutation relation $\hat{L}_+\hat{P} - \hat{P}\hat{L}_+ = 0$ by the same method used to obtain (29.3) from (29.2). There are definite *parity selection rules* for the matrix elements of various physical quantities.

Consider scalar quantities first. Here one must distinguish *true scalars*, which remain entirely unchanged under inversion, from *pseudoscalars*, which change sign

¹²Inversion likewise leaves invariant the Hamiltonian of particles moving in a centrally symmetric field (provided the coordinate origin coincides with the center of that field).

¹³From the English word *parity*.

¹⁴To avoid misunderstanding, we recall that the discussion concerns nonrelativistic theory. Interactions exist in nature, and are treated in relativistic theory, that violate parity conservation.

under inversion (the scalar product of an axial vector and a polar vector is a pseudoscalar). The operator $\hat{f}$ of a true scalar commutes with $\hat{P}$. It follows that if the matrix of P is diagonal, then f is likewise diagonal with respect to parity: matrix elements vanish except for the $g \rightarrow g$ and $u \rightarrow u$ transitions (here g and u label even and odd states). For the operator of a pseudoscalar quantity, on the other hand, $\hat{P}\hat{f} = -\hat{f}\hat{P}$; the operators $\hat{P}$ and $\hat{f}$ “anticommute.” The matrix element of this equality for a transition $g \rightarrow g$ is

$$P_{gg}f_{gg} = -f_{gg}P_{gg},$$

and since $P_{gg} = 1$, we have $f_{gg} = 0$; in the same way one finds $f_{uu} = 0$. Thus the matrix of a pseudoscalar quantity has nonzero elements only for transitions in which the parity changes. The selection rules for matrix elements of scalar quantities are therefore

$$\begin{array}{ll} \text{true scalars:} & g \rightarrow g, \quad u \rightarrow u, \\ \text{pseudoscalars:} & g \rightarrow u, \quad u \rightarrow g. \end{array} \quad (30.4)$$

These rules can also be obtained in another way, directly from the definition of matrix elements. Consider, for example, the integral $f_{ug} = \int \psi_u^* \hat{f} \psi_g dq$, where ψ_g is even and ψ_u is odd. On reversing the signs of all coordinates, the integrand changes sign if f is a true scalar. On the other hand, an integral over all space cannot change when the variables of integration are merely renamed. Hence $f_{ug} = -f_{ug}$, so that $f_{ug} = 0$.

The selection rules for vector quantities are obtained in the same way. One must remember that ordinary, polar vectors reverse sign under inversion, whereas axial vectors remain unchanged (an example is the angular-momentum vector, the cross product $\mathbf{r} \times \mathbf{p}$ of two polar vectors). Taking this into account, we find

$$\begin{array}{ll} \text{polar vectors:} & g \rightarrow u, \quad u \rightarrow g, \\ \text{axial vectors:} & g \rightarrow g, \quad u \rightarrow u. \end{array} \quad (30.5)$$

Let us determine the parity of a one-particle state with angular momentum l . Inversion ($x \rightarrow -x$, $y \rightarrow -y$, $z \rightarrow -z$) consists, in spherical coordinates, of the transformation

$$r \rightarrow r, \quad \theta \rightarrow \pi - \theta, \quad \varphi \rightarrow \varphi + \pi. \quad (30.6)$$

The angular dependence of the particle's wave function is given by the spherical function Y_{lm} , which, apart from a constant immaterial here, has the form $P_l^m(\cos \theta)e^{im\varphi}$. When φ is replaced by $\varphi + \pi$, the factor $e^{im\varphi}$ is multiplied by $(-1)^m$; when θ is replaced by $\pi - \theta$, $P_l^m(\cos \theta)$ becomes $P_l^m(-\cos \theta) = (-1)^{l-m}P_l^m(\cos \theta)$. Thus the entire function is multiplied by $(-1)^l$ (which is independent of m , in agreement with the statement above); hence the parity of a state with a given l is

$$P = (-1)^l. \quad (30.7)$$

Accordingly, even values of l yield even states and odd values yield odd ones.

For a vector physical quantity associated with a single particle, nonzero matrix elements can arise only in transitions with $l \rightarrow l, l \pm 1$ (§ 29). Combining this fact with (30.7) and with the preceding statements concerning parity changes in vector

matrix elements, we conclude that matrix elements of vector quantities belonging to an individual particle are nonzero only for the transitions

$$\begin{aligned} \text{polar vectors:} \quad & l \rightarrow l \pm 1, \\ \text{axial vectors:} \quad & l \rightarrow l. \end{aligned} \tag{30.8}$$

§ 31. Addition of angular momenta

Consider a system consisting of two weakly interacting parts. When the interaction is entirely neglected, the conservation of angular momentum holds for each part independently, so that the angular momenta $\mathbf{L}_1$ and $\mathbf{L}_2$ of the two parts can be summed to give the total angular momentum $\mathbf{L}$ of the system. In the next approximation, once the weak interaction is included, the conservation laws for $\mathbf{L}_1$ and $\mathbf{L}_2$ cease to hold rigorously; nonetheless, the quantum numbers L_1 and L_2 , which fix the squares of those angular momenta, remain “good” and can still serve for an approximate characterization of the system’s state. In a pictorial, classical treatment of the angular momenta, one may say that in this approximation $\mathbf{L}_1$ and $\mathbf{L}_2$ precess about the direction of $\mathbf{L}$ while their magnitudes remain unchanged.

The consideration of such systems raises the question of the *rule for adding angular momenta*. What values of L are possible for given L_1 and L_2 ? The addition rule for the angular-momentum projections is evident: from $\hat{L}_z = \hat{L}_{1z} + \hat{L}_{2z}$ it follows that

$$M = M_1 + M_2. \tag{31.1}$$

There is no equally simple relation for the operators of the squares of the angular momenta, and we derive their “addition rule” as follows.

If the quantities $\mathbf{L}_1^2$, $\mathbf{L}_2^2$, L_{1z} , and L_{2z} are chosen as a complete set of physical quantities¹⁵, each state is specified by the numbers L_1 , L_2 , M_1 , and M_2 . For given L_1 and L_2 , the numbers M_1 and M_2 take, respectively, $2L_1 + 1$ and $2L_2 + 1$ values, so there are altogether $(2L_1 + 1)(2L_2 + 1)$ different states with the same L_1 and L_2 . We denote the wave functions of the states in this description by $\varphi_{L_1 L_2 M_1 M_2}$.

Instead of the four quantities just stated, one can choose the four quantities $\mathbf{L}_1^2$, $\mathbf{L}_2^2$, $\mathbf{L}^2$, and L_z as a complete set. Each state is then characterized by the numbers L_1 , L_2 , L , and M (we denote the corresponding wave functions by $\psi_{L_1 L_2 L M}$). For given L_1 and L_2 there must, of course, still be $(2L_1 + 1)(2L_2 + 1)$ different states; that is, for fixed L_1 and L_2 , the pair L, M can take $(2L_1 + 1)(2L_2 + 1)$ pairs of values. These values can be found by the following argument.

Adding the different allowed values of M_1 and M_2 gives the corresponding values of

¹⁵Together with a number of other quantities which, along with the four stated here, make up a complete set. These remaining quantities play no part in the following argument; for conciseness we do not mention them at all and provisionally call the system of the four stated quantities complete.

M :

M_1	M_2	M
L_1	L_2	$L_1 + L_2$
L_1	$L_2 - 1$	$L_1 + L_2 - 1$
$L_1 - 1$	L_2	
$L_1 - 1$	$L_2 - 1$	$L_1 + L_2 - 2$
L_1	$L_2 - 2$	
$L_1 - 2$	L_2	$L_1 + L_2 - 2$
.....		

The largest possible value of M is evidently $M = L_1 + L_2$, and it corresponds to one state φ (one pair of values M_1, M_2). Consequently, the largest possible M in the states ψ , and hence the largest L , is also $L_1 + L_2$. Next, there are two states φ with $M = L_1 + L_2 - 1$. There must therefore also be two states ψ with this value of M ; one is a state with $L = L_1 + L_2$ (and $M = L - 1$), and the other has $L = L_1 + L_2 - 1$ (with $M = L$). For $M = L_1 + L_2 - 2$ there are three different states φ . This means that, in addition to $L = L_1 + L_2$ and $L = L_1 + L_2 - 1$, the value $L = L_1 + L_2 - 2$ is also possible.

This argument can be continued in the same way as long as decreasing M by one increases by one the number of states with that value of M . It is easy to see that this continues until M reaches $|L_1 - L_2|$. As M is reduced further, the number of states ceases to increase and remains equal to $2L_2 + 1$ (if $L_2 \leq L_1$). Thus $|L_1 - L_2|$ is the smallest possible value of L .

We therefore obtain the result that, for given L_1 and L_2 , L can take the values

$$L = L_1 + L_2, L_1 + L_2 - 1, \dots, |L_1 - L_2|, \quad (31.2)$$

altogether $2L_2 + 1$ different values when $L_2 \leq L_1$. One readily checks that these indeed give $(2L_1 + 1)(2L_2 + 1)$ different pairs of values M, L . It is important to note that, apart from the $2L + 1$ different values of M for a given L , each possible value of L in (31.2) corresponds to just one state.

This result can be represented pictorially by the so-called *vector model*. Introduce two vectors $\mathbf{L}_1$ and $\mathbf{L}_2$ of lengths L_1 and L_2 . The values of L are represented by the integer-valued lengths of vectors $\mathbf{L}$ obtained by vector addition of $\mathbf{L}_1$ and $\mathbf{L}_2$; the largest value, $L_1 + L_2$, is obtained when $\mathbf{L}_1$ and $\mathbf{L}_2$ are parallel, and the smallest, $|L_1 - L_2|$, when they are antiparallel.

In states with definite angular momenta L_1, L_2 , and definite total angular momentum L , the scalar products $\mathbf{L}_1 \mathbf{L}_2$, $\mathbf{L} \mathbf{L}_1$, and $\mathbf{L} \mathbf{L}_2$ also have definite values. These values are readily found. To calculate $\mathbf{L}_1 \mathbf{L}_2$, write $\hat{\mathbf{L}} = \hat{\mathbf{L}}_1 + \hat{\mathbf{L}}_2$, or, on squaring and transposing terms,

$$2\hat{\mathbf{L}}_1 \hat{\mathbf{L}}_2 = \hat{\mathbf{L}}^2 - \hat{\mathbf{L}}_1^2 - \hat{\mathbf{L}}_2^2.$$

Replacing the operators on the right by their eigenvalues gives the eigenvalue of the operator on the left:

$$\mathbf{L}_1 \mathbf{L}_2 = \frac{1}{2} \{L(L + 1) - L_1(L_1 + 1) - L_2(L_2 + 1)\}. \quad (31.3)$$

Similarly, we find

$$\mathbf{L}\mathbf{L}_1 = \frac{1}{2}\{L(L+1) + L_1(L_1+1) - L_2(L_2+1)\}. \quad (31.4)$$

Let us now establish the *rule for combining parities*. For a system that consists of two independent parts, the wave function Ψ equals the product of the individual wave functions Ψ_1 and Ψ_2 . Hence, if the latter two have the same parity (both change sign, or both remain unchanged, when all coordinates change sign), the wave function of the whole system is even. If, on the other hand, Ψ_1 and Ψ_2 have opposite parities, Ψ is odd. These statements can be written as

$$P = P_1 P_2, \quad (31.5)$$

where P is the parity of the whole system, and P_1 and P_2 are the parities of its parts. This rule plainly generalizes directly to a system consisting of any number of noninteracting parts.

In particular, for a system of particles in a centrally symmetric field (with the interaction between the particles treated as weak), the parity of the state of the whole system is

$$P = (-1)^{l_1+l_2+\dots} \quad (31.6)$$

(see (30.7)). Note that the exponent involves the algebraic sum of the particle angular momenta, a quantity that, generally speaking, does not coincide with their “vector sum,” i.e., with the angular momentum L of the system.

If a closed system decays into parts under the action of forces within the system itself, its total angular momentum and parity must be conserved. This may make the decay impossible even when it is energetically possible.

For example, consider an atom in an even state with angular momentum $L = 0$, whose energy would permit it to break up into a free electron and an ion occupying an odd state, also with $L = 0$. It is easy to show that this process is in fact impossible (one calls it *forbidden*). Indeed, by conservation of angular momentum the free electron would also have to possess zero angular momentum and would therefore be in an even state ($P = (-1)^0 = +1$). But the state ion+free electron would then be odd, whereas the initial state of the atom was even.

Motion in a Centrally Symmetric Field

Perturbation Theory